Informatica®

10.5.10

Informatica Administrator for Enterprise Data Catalog

Informatica Informatica Administrator for Enterprise Data Catalog
10.5.10
February 2026

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Publication Date: 2026-02-19

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Preface

Use the *Informatica Administrator for Enterprise Data Catalog* to learn about the concepts, components, and tasks that you need to perform in Informatica Administrator in the context of Enterprise Data Catalog. This reference document includes information about the Informatica domain, managing user accounts in Informatica Administrator, and monitoring statistics of applications and services in the domain.

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CHAPTER 1

Understanding Domains

This chapter includes the following topics:

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- Nodes, 15
- Service Manager, 15
- Application Services, 17
- Informatica Data Usage Policy, 24

Understanding Domains Overview

Informatica has a service-oriented architecture that provides the ability to scale services and share resources across multiple machines. High availability functionality helps minimize service downtime due to unexpected failures or scheduled maintenance in the Informatica environment.

The Informatica domain is the fundamental administrative unit in Informatica. The domain supports the administration of the distributed services. A domain is a collection of nodes and services that you can group in folders based on administration ownership.

A node is the logical representation of a machine in a domain. One node in the domain acts as a gateway to receive service requests from clients and route them to the appropriate service and node. Services and processes run on nodes in a domain. The availability of a service or process on a node depends on how you configure the service and the node.

Services for the domain include the Service Manager and a set of application services:

- Service Manager. A service that runs on each node in the domain to manage all domain operations. The Service Manager performs domain functions such as authentication, authorization, and logging. The Service Manager also starts the application services configured to run on the node.
- Application Services. Services that represent server-based functionality, such as the Model Repository Service and the Data Integration Service. The application services that run on a node depend on the way you configure the services.

The Service Manager and application services control security. The Service Manager manages users and groups that can log in to application clients and authenticates the users who log in to the application clients. The Service Manager and application services authorize user requests from application clients.

Informatica Administrator (the Administrator tool), consolidates the administrative tasks for domain objects such as services, nodes, licenses, and grids. You manage the domain and the security of the domain through the Administrator tool.

If you have the high availability option, you can scale services and eliminate single points of failure for services. Services can continue running despite temporary network or hardware failures.

Nodes

A node is a logical representation of a machine in a domain. During installation, you add the installation machine to the domain as a node. You can add multiple nodes to a domain.

Each node in the domain runs the Service Manager that manages domain functions on that node. The Service Manager also supports the application services that run on the node. The domain functions that the node performs and the services that the node runs depend on the following node configurations:

Node type

The node type determines whether the node can serve as a gateway or worker node and determines the domain functions that the node performs. One gateway node serves as the master gateway node for the domain. The master gateway node receives service requests from clients and routes them to the appropriate service and node. A worker node is any node not configured to serve as a gateway. The first time that you install the Informatica services, you create a gateway node and the Informatica domain. When you install the Informatica services on other machines, you create additional gateway nodes or worker nodes that you join to the domain.

Node role

The node role defines the purpose of the node. A node with the service role can run application services. A node with the compute role can perform computations requested by remote application services. A node with both roles can run application services and locally perform computations for those services. By default, each gateway and worker node has both the service and compute roles enabled. If a node is assigned to a Data Integration Service grid, you might want to update the node role. Enable only the service role to dedicate the node to running the Data Integration Service process. Enable only the compute role to dedicate the node to running Data Integration Service mappings.

You can subscribe to alerts to receive notification about node events such as node failure or a master gateway election. You can also generate and upload node diagnostics to the Configuration Support Manager and review information such as available EBFs and Informatica recommendations.

Service Manager

The Service Manager is a service that manages all domain operations. It runs within Informatica services. It runs as a service on Windows and as a daemon on UNIX. When you start Informatica services, you start the Service Manager.

The Service Manager runs on each node in the domain. If the Service Manager is not running, the node is not available.

The Service Manager runs on all nodes in the domain to support the domain and application services:

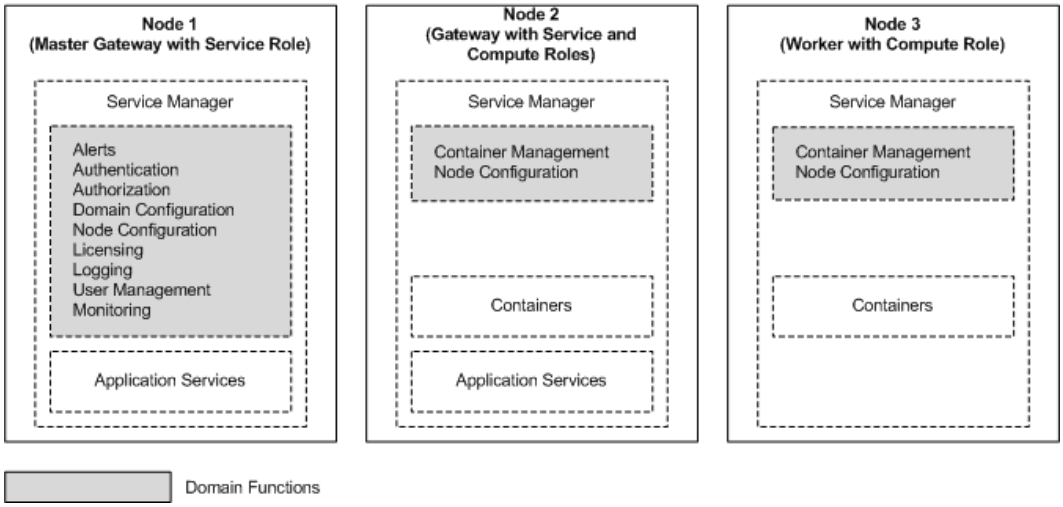
- Domain support. The Service Manager performs functions on each node to support the domain. The functions that the Service Manager performs on a node depend on the type and role of the node. For example, the Service Manager running on the master gateway node performs all domain functions on that node. The Service Manager running on any other type of node performs limited domain functions on that node.
- Application service support. When a node has the service role, the Service Manager starts application services configured to run on that node. It starts and stops services and service processes based on requests from clients. It also directs service requests to application services. The Service Manager uses TCP/IP to communicate with the application services.

The following table describes the domain functions that the Service Manager performs:

Function	Description
Alert Management	The Service Manager sends alerts to subscribed users. You subscribe to alerts to receive notification for node failure and master gateway election on the domain, and for service process failover for services on the domain. When you subscribe to alerts, you receive notification emails. Alert management occurs on the master gateway node.
Authentication	The Service Manager authenticates users who log in to application clients. Authentication occurs on the master gateway node.
Authorization	The Service Manager authorizes user requests for domain objects based on the privileges, roles, and permissions assigned to the user. Requests for domain objects can come from the Administrator tool. Domain authorization occurs on the master gateway node.
Container Management	When a node has the compute role, the Service Manager manages the containers on the node. A container is an allocation of memory and CPU resources. An application service uses the container to remotely perform computations on the node. For example, a Data Integration Service that runs on a grid can remotely run a mapping within a container on a node with the compute role. Container management occurs on any node with the compute role.
Domain Configuration	The Service Manager manages the domain configuration metadata. Domain configuration occurs on the master gateway node.
Node Configuration	The Service Manager manages node configuration metadata in the domain. Node configuration occurs on all nodes in the domain.
Licensing	The Service Manager registers license information and verifies license information when you run application services. Licensing occurs on the master gateway node.
Logging	The Service Manager provides accumulated log events from each service in the domain. To perform the logging function, the Service Manager runs a Log Manager and a Log Agent. The Log Manager runs on the master gateway node. The Log Agent runs on all nodes where Data Integration Service jobs run.

Function	Description
User Management	The Service Manager manages the native and LDAP users and groups that can log in to application clients. It also manages the creation of roles and the assignment of roles and privileges to native and LDAP users and groups. User management occurs on the master gateway node.
Monitoring	The Service Manager persists, updates, retrieves, and publishes run-time statistics for integration objects in the Model repository. The Service Manager stores the monitoring configuration in the domain configuration repository. Monitoring occurs on the master gateway node.

The following image shows where the Service Manager performs domain functions:



Application Services

Application services represent server-based functionality.

Application services include services that you create and system services that are created for you when you create the domain. A system service can have a single instance in the domain.

Application services include the following services:

- Analyst Service
- Catalog Service
- Cluster Service
- Data Integration Service
- Model Repository Service

When you configure an application service, you designate a node to run the service process. When a service process runs, the Service Manager assigns a port number from the range of port numbers assigned to the node.

The service process is the runtime representation of a service running on a node. The service type determines how many service processes can run at a time.

If you have the high availability option, you can run a service on multiple nodes. Designate the primary node to run the service. All other nodes are back-up nodes for the service. If the primary node is not available, the service runs on a back-up node. You can subscribe to alerts to receive notification in the event of a service process failover.

If you do not have the high availability option, configure a service to run on one node. If you assign the service to multiple nodes, then the service will not start.

Analyst Service

The Analyst Service is an application service that runs the Informatica Analyst application in the Informatica domain. The Analyst Service manages the connections between service components and the users that log in to Informatica Analyst. The Analyst Service connects to a Data Integration Service, a Model Repository Service, a Metadata Manager Service, and a Search Service. The Analyst Service also specifies a flat file cache directory and a directory for business glossary export files.

When you configure the Analyst Service, connect it to a Data Integration Service to run profiles, scorecards, and mapping specifications. You can also connect the Analyst Service to a Data Integration Service that runs Human tasks. Connect the Analyst Service to a Model Repository Service to identify a Model repository.

Specify a flat file cache directory to store temporary data from flat files that you upload. Specify a business glossary directory to stores temporary files that you export from the Business Glossary.

Cluster Service

The Informatica Cluster Service is an application service that runs and manages all the associated services that are required to run Enterprise Data Catalog in the Informatica domain. The associated services include Mongo DB, Nomad, Solr, PostgreSQL, and ZooKeeper.

The following table lists the supported methods and algorithms:

Method	Algorithm
Key exchange	- diffie-hellman-group-exchange-sha1 - diffie-hellman-group1-sha1 - diffie-hellman-group14-sha1 - diffie-hellman-group-exchange-sha256 - ecdh-sha2-nistp256 - ecdh-sha2-nistp384 - ecdh-sha2-nistp521
Cipher	- blowfish-cbc - 3des-cbc - aes128-cbc - aes192-cbc - aes256-cbc - aes128-ctr - aes192-ctr - aes256-ctr - 3des-ctr - arcfour - arcfour128 - arcfour256

Method	Algorithm
MAC	- hmac-md5 - hmac-sha1 - hmac-md5-96 - hmac-sha1-96
Host key type	- ssh-dss - ssh-rsa - ecdsa-sha2-nistp256 - ecdsa-sha2-nistp384 - ecdsa-sha2-nistp521

Catalog Service

The Catalog Service is an application service that runs Enterprise Data Catalog in the Informatica domain. The Catalog Service manages the connections between service components and the users that have access to Enterprise Data Catalog and Catalog Administrator.

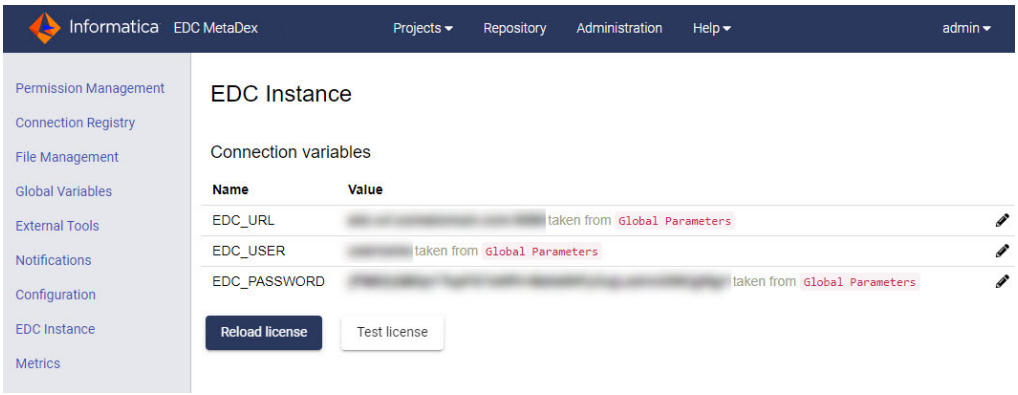
The catalog represents an indexed inventory of all the configured data assets in an enterprise. You can find metadata and statistical information, such as profile statistics, data asset ratings, data domains, and data relationships, in the catalog.

Connect the MetaDex Server to the Catalog Service

Before you run MetaDex configurations, configure the MetaDex server to connect to the Catalog Service.

1. Log in to the MetaDex tool.
2. On the header of the MetaDex tool, click **Administration** > **EDC Instance**.

   The **EDC Instance** page appears with the predefined connection variables and the information on the variable origin, as shown in the following image:



3. Click the **Edit** icon to the right of each variable.

   The **Edit global variable** dialog box appears.

4. Enter the values for the following variables:

Name	Description
EDC_URL	The host name and port number of the Catalog Service in the following format: <host>:<port>
EDC_USER	The user name configured for the catalog administrator user.
EDC_PASSWORD	The password configured for the catalog administrator user. When you add the password, click the **Hide password** icon.

5. Click **Save**.

   The **Edit global variable** dialog box closes.

6. Test and reload the license.

Enable Data Preview and Provisioning

To enable data preview and provisioning, configure custom properties for the Catalog Service in the Informatica Administrator console.

Configure the following custom properties for the Catalog Service:

Name	Value
`LdmCustomOptions.enableDataProvision`	true
`LdmCustomOptions.provision.ics.master.app.url`	Informatica Intelligent Cloud Services Organizations URL

Note: If you do not provide a valid Informatica Intelligent Cloud Services Organizations URL for the `LdmCustomOptions.provision.ics.master.app.url` custom property, Informatica Administrator console considers the following default Informatica Intelligent Cloud Services URL: `https://dm-us.informaticacloud.com/`.

Data Asset Analytics Repository Database Requirements

The Data Asset Analytics repository database stores the analytical information collected from the catalog. The analytical information is displayed as reports and statistical data in the **Data Asset Analytics** tab in Enterprise Data Catalog. You can access the **Data Asset Analytics** tab after you configure Data Asset Analytics in the Catalog Service and assign the required privileges for users to access the tab.

You can configure any of the following databases as the repository database for Data Asset Analytics:

- Oracle
- Microsoft SQL Server
- PostgreSQL

Oracle Database Requirements

Perform the following steps before you configure Oracle as the repository database:

1. Create a user name and password to access the Oracle database using the following command: `CREATE USER <Database schema>`

2. Run the following commands to provide the required user privileges:

```
GRANT CONNECT TO <Database schema>;
GRANT RESOURCE TO <Database schema>;
GRANT CREATE VIEW TO <Database schema>;
GRANT CREATE MATERIALIZED VIEW TO <Database schema>;
GRANT UNLIMITED TABLESPACE TO <Database schema>;
```

   Alternatively, if you do not want to grant the unlimited tablespace privilege, run the command `ALTER USER <Database schema> QUOTA <Size> ON <Tablespace name>;`

Microsoft SQL Server Database Requirements

Perform the following steps before you configure Microsoft SQL Server as the repository database:

1. Use the dbo schema for SQL Server.
2. Run the following commands to create the database, the user name, password and assign the required privileges for the user:

```
USE master
GO
CREATE DATABASE <New database name>;
GO
use <New database name>
CREATE LOGIN <New login name> WITH PASSWORD = '<Password>';
CREATE USER <New user name> FOR LOGIN <New login name> WITH DEFAULT_SCHEMA = [dbo];
GRANT CREATE TABLE TO <New user name>;
GRANT CREATE VIEW TO <New user name>;
GRANT SELECT, ALTER, INSERT, DELETE, UPDATE on schema::dbo to <New user name> ;
GRANT REFERENCES to <New user name>;
```

PostgreSQL Database Requirements

Perform the following steps before you configure PostgreSQL as the repository database:

1. Use the public schema for PostgreSQL.
2. Create the credentials to access the database using the following command: `CREATE USER <New user name> WITH PASSWORD '<Password>';`

   Note: Make sure that you choose a user name that is different from the name of the database schema.
3. Create the database and assign the ownership to the user name that you created. Use the following command to complete this step: `CREATE DATABASE <NEW db NAME> owner=<New user name>;`
4. After you connect to the database, run the command `SHOW search_path`. The command must return the value `<User>, public`.

Data Flow Analytics Repository Database Requirements

Data Flow Analytics connects to the PowerCenter data source to retrieve mappings and performs analyses on the mappings. The statistical information from the analyses appears in the **Data Flow Analytics** tab in the Enterprise Data Catalog tool. Create a database user account and configure the required read privileges for the account. After you create the account, you can configure Data Flow Analytics for the PowerCenter resource using the Catalog Administrator tool.

Use the following guidelines when you set up the database user account for the PowerCenter repository.

Oracle Database Requirements

Perform the following steps before you configure Oracle as the PowerCenter data source for Data Flow Analytics:

1. Create a user name and password to access the database using the `CREATE USER <Database schema> IDENTIFIED BY <database schema>` command.

2. Run the following commands to provide the required privileges to the database user:

```
GRANT CONNECT TO <Database schema>;
GRANT SELECT ON <PowerCenter repository schema>.OPB_WFLOW_RUN TO <Database schema>;
GRANT SELECT ON <PowerCenter repository schema>.OPB_SUBJECT TO <Database schema>;
```

3. Run the following commands to create synonyms for the tables to query the workflow execution details:

```
CREATE synonym <Database schema>.OPB_WFLOW_RUN FOR <PCRS schema>.OPB_WFLOW_RUN;
CREATE synonym <Database schema>.OPB_SUBJECT FOR <PCRS schema>.OPB_SUBJECT;
```

Microsoft SQL Server Database Requirements

Perform the following steps before you configure Microsoft SQL Server as the PowerCenter data source for Data Flow Analytics:

1. Run the following commands to create a user name and password to access the database:

```
USE <PowerCenter repository name>
CREATE LOGIN <New login name> WITH PASSWORD='<Password>';
CREATE USER <New user name> FOR LOGIN <New login name>;
```

2. Run the following commands to provide the required privileges to the database user:

```
GRANT SELECT ON OPB_SUBJECT TO <New user name>;
GRANT SELECT ON OPB_WFLOW_RUN TO <New user name>
```

IBM DB2 Database Requirements

Perform the following steps before you configure IBM DB2 as the PowerCenter data source for Data Flow Analytics:

1. Log in to the machine that hosts the IBM DB2 database as a root user and create a new user account on the machine.
2. Run the following commands to provide the required privileges to the database user:

```
GRANT CONNECT ON <PowerCenter repository name> TO USER <Database user name>;
GRANT SELECT ON TABLE <PowerCenter repository name>."OPB_WFLOW_RUN" TO USER
<Database user name>;
GRANT SELECT ON TABLE <PowerCenter repository name>."OPB_SUBJECT" TO USER <Database
user name>;
```

3. Run the following commands to create aliases for the tables to query the workflow execution details:

```
CREATE ALIAS <Database user name>.OPB_WFLOW_RUN FOR <PowerCenter repository
name>.OPB_WFLOW_RUN;
CREATE ALIAS <Database user name>.OPB_SUBJECT FOR <PowerCenter repository
name>.OPB_SUBJECT;
```

Sybase

Perform the following steps before you configure Sybase as the PowerCenter data source for Data Flow Analytics:

1. Log in to a Sybase client as an administrator and run the following commands to create a user name and password to access the database:

```
sp_addlogin <New login name>, <Password>
use <PowerCenter repository name>
sp_adduser <New login name>, <New user name>
```

2. Run the following commands to provide the required privileges to the database user:

```
GRANT SELECT TO <New user name>
GRANT CREATE VIEW TO <New user name>
```

3. Run the following commands to create views for the tables to query the workflow execution details:

```
CREATE VIEW <Database user name>.OPB_WFLOW_RUN as select * from <PowerCenter
repository schema>.OPB_WFLOW_RUN
CREATE VIEW <Database user name>.OPB_SUBJECT as select * from <PowerCenter
repository schema>.OPB_SUBJECT
```

PostgreSQL

Perform the following steps before you configure PostgreSQL as the PowerCenter data source for Data Flow Analytics:

1. Create a user name and password to access the database using the following command: `CREATE USER <Database user name> WITH PASSWORD '<Database user name>'`
2. Create a role using the `CREATE ROLE <Role name>` command.
3. Run the following commands to provide the required privileges to the user:

```
GRANT <role name> TO <Database user name>;
GRANT CONNECT ON DATABASE "<PowerCenter repository name>" to <Role name>;
GRANT USAGE ON SCHEMA <PowerCenter repository schema> TO <Role name>;
GRANT SELECT ON OPB_WFLOW_RUN TO <Role name>;
GRANT SELECT ON OPB_SUBJECT TO <Role name>;
```

Content Management Service

The Content Management Service is an application service that manages reference data. A reference data object contains a set of data values that Enterprise Data Catalog searches while performing data domain discovery on source data. Data domain discovery finds inferred business semantics based on column data. Examples include Social Security number, phone number, and credit card number.

The Content Management Service uses the Data Integration Service to run mappings to transfer data between reference tables and external data sources.

Data Integration Service

The Data Integration Service is an application service that performs data integration tasks for Informatica Analyst, Informatica Developer, and external clients. Data integration tasks include previewing data and running profiles, SQL data services, web services, and mappings.

When you start a command from the command line or an external client to run SQL data services and mappings in an application, the command sends the request to the Data Integration Service.

You can configure the Data Integration Service to run on the following domain objects:

On nodes

If your license includes high availability, you can configure the service to run on multiple nodes. By default, the service runs on the primary node. If the primary node is not available, the service runs on a back-up node. If the service process fails or the node becomes unavailable, the service fails over to another node. If your license does not include high availability, you can configure the service to run on one node.

On a grid

If your license includes grid, you can configure the service to run on a grid. A grid is an alias assigned to a group of nodes. The Data Integration Service dispatches jobs to available nodes assigned to the grid. When the Data Integration Service runs on a grid, the service remains available if a service process fails or a node becomes unavailable.

Model Repository Service

The Model Repository Service is an application service that manages the Model repository. The Model repository is a relational database that stores the metadata for projects created in Informatica Analyst and Informatica Developer. The Model repository also stores run-time and configuration information for applications that are deployed to a Data Integration Service.

You can configure the Model Repository Service to run on one node. The Model Repository Service is not a highly available service. However, you can run multiple Model Repository Services on the same node. If the Model Repository Service fails, it automatically restarts on the same node.

Informatica Data Usage Policy

Informatica DiscoveryIQ is a product usage tool in the Informatica domain that sends routine reports on data usage and system statistics to Informatica.

Informatica DiscoveryIQ uploads data to Informatica 15 minutes after you install and configure the Informatica domain. Thereafter, the domain sends data to Informatica every few days. Data collection and upload is enabled by default. You can choose to not send any usage statistics to Informatica.

If the network where you install Informatica services needs a proxy server to communicate with the external network, configure the proxy details.

Informatica DiscoveryIQ enables Informatica to provide an environment health check after the analysis of system statistics and domain reports. You can receive best practices and recommendations from Informatica based on the reports to resolve issues on the domain. The usage statistics provide Informatica a proactive insight into product implementation.

Informatica DiscoveryIQ reports the following data to Informatica:

- Operating system details
- CPU information
- Informatica license key serial number
- Gateway information
- Domain options
- Node options
- Application service information

Configuring Informatica DiscoveryIQ Proxy Details

Configure proxy server details if the network on which you install Informatica services use a proxy server to communicate with the external network.

1. In the Administrator tool header area, click **Manage** > **DiscoveryIQ Proxy Details**.
2. Enter the domain, host name, and port number of the proxy server.
3. Enter the user name and password to connect to the proxy server.
4. Click **OK** to save the proxy server details.

Disabling Informatica Data Usage

You can disable the upload of usage data from the Informatica domain in the Administrator tool.

1. In the Administrator tool, click **Help** > **About**.
2. Click **Data Usage Policy**.
3. Clear **Enable Usage Collection**.
4. Click **OK**.

CHAPTER 2

Managing Your Account

This chapter includes the following topics:

- Managing Your Account Overview, 25
- Log In to Informatica Administrator, 25
- Password Management, 26
- Editing Preferences, 27
- Preferences, 27
- Informatica Network Credentials, 28

Managing Your Account Overview

Manage your account to change your password or edit user preferences.

If you have a native user account, you can change your password at any time with the Change Password application. If someone else created your user account, change your password the first time you log in to the Administrator tool.

User preferences control the options that appear in the Administrator tool when you log in. User preferences do not affect the options that appear when another user logs in to the Administrator tool.

You can configure Informatica Network credentials for your account so that you can access the Informatica Knowledge Base from the Administrator tool.

Log In to Informatica Administrator

You must have a user account to log in to the Informatica Administrator web application.

In Microsoft Internet Explorer and Google Chrome, add the URL of the Informatica web application to the list of trusted sites. If you are using Chrome version 41 or later, you must also set the `AuthServerWhitelist` and `AuthNegotiateDelegateWhitelist` policies.

1. Start a Microsoft Internet Explorer or Google Chrome browser.
2. In the **Address** field, enter the URL for the Administrator tool:
   - If the Administrator tool is not configured to use a secure connection, enter the following URL:

     ```
     http://<fully qualified hostname>:<http port>/administrator/
     ```

- If the Administrator tool is configured to use a secure connection, enter the following URL:

  `https://<fully qualified hostname>:<http port>/administrator/`

  Host name and port in the URL represent the host name and port number of the master gateway node. If you configured secure communication for the domain, you must use HTTPS in the URL to ensure that you can access the Administrator tool.

3. Enter the user name, password, and security domain for your user account, and then click **Login**.

   The **Security Domain** field appears when the Informatica domain contains an LDAP security domain. If you do not know the security domain that your user account belongs to, contact the Informatica domain administrator.

   Note: If this is the first time you log in with the user name and password provided by the domain administrator, change your password to maintain security.

Informatica Administrator URL

In the Administrator tool URL, `<host>:<port>` represents the host name of the master gateway node and the Administrator tool port number.

You configure the Administrator tool port when you define the domain. You can define the domain during installation or by running the *infasetup* DefineDomain command line program. If you enter the domain port instead of the Administrator tool port in the URL, the browser is directed to the Administrator tool port.

Note: If the domain fails over to a different master gateway node, the host name in the Administrator tool URL is equal to the host name of the elected master gateway node.

Password Management

You can change the password through the Change Password application.

You can open the Change Password application from the Administrator tool or with the following URL:
`http://<fully qualified host name>:<port>/passwordchange/`

The Service Manager uses the user password associated with a worker node to authenticate the domain user. If you change a user password that is associated with one or more worker nodes, the Service Manager updates the password for each worker node. The Service Manager cannot update nodes that are not running. For nodes that are not running, the Service Manager updates the password when the nodes restart.

Note: For an LDAP user account, change the password in the LDAP directory service.

For a user account in the domain that uses native authentication, if you enable password complexity, use the following guidelines when you create or change a password:

- The length of the password must be at least eight characters.
- It must be a combination of an alphabet character, a numeric character and a non-alphanumeric character, such as:
  `! \ " # $ % & ' ( ) * + , - . / : ; < = > ? @ [ ] ^ _ ` { | } ~`

When you use special characters in a password, the shell sometimes interprets them differently. For example, $ is interpreted as a variable. In this case, use an escape character to escape the special character.

Changing Your Password

Change the password for a native user account at any time. For a user account created by someone else, change the password the first time you log in to the Administrator tool.

1. In the Administrator tool header area, click **Manage** > **Change Password**.

   The Change Password application opens in a new browser window.
2. Enter the current password in the **Password** box, and the new password in the **New Password** and **Confirm Password** boxes.
3. Click **Update**.

Editing Preferences

Edit your preferences to determine the options that appear in the Administrator tool when you log in.

1. In the Administrator tool header area, click **Manage** > **Preferences**.

   The **Preferences** window appears.
2. Click **Edit**.

   The **Edit Preferences** dialog box appears.

Preferences

Your preferences determine the options that appear in the Administrator tool when you log in. Your preferences do not affect the options that appear when another user logs in to the Administrator tool.

The following table describes the options that you can configure for your preferences:

Option	Description
Subscribe for Alerts	Subscribes you to domain and service alerts. You must have a valid email address configured for your user account. Default is No.
Show Custom Properties	Displays custom properties in the contents panel when you click an object in the Navigator. You use custom properties to configure Informatica behavior for special cases or to increase performance. Hide the custom properties to avoid inadvertently changing the values. Use custom properties only if Informatica Global Customer Support instructs you to.

To edit your preferences, click **Manage** > **Preferences** in the Administrator tool header area.

Informatica Network Credentials

You can enter your Informatica Network credentials in the Administrator tool to access the Informatica Knowledge Base from the Administrator tool.

You can also view the search results for an error message in the Informatica Knowledge Base by clicking the error message code in the Administrator tool.

Enter Informatica Network Credentials

Enter your Informatica Network credentials to access the Informatica Knowledge Base from the Administrator tool.

1. Click **Manage** > **Support Portal Credentials**.

   The **Informatica Network Login Credentials** window appears.
2. Enter your Informatica Network credentials and the customer project ID.
3. Click **OK.**

Searching Informatica Knowledge Base

You can search for terms in the Informatica Knowledge Base directly from the Administrator tool.

1. Click **Help** > **Search Knowledge Base**.

   The **Search Knowledge Base** window appears.
2. Enter the term that you want to search in the text box.
3. Click **OK**.

   The search results appear in a different browser window.

CHAPTER 3

Using Informatica Administrator

This chapter includes the following topics:

- Using Informatica Administrator Overview, 29
- Manage Tab, 30
- Manage Tab - Domain View, 31
- ManageTab - Services and Nodes View, 35
- Manage Tab - Connections View, 41
- Manage Tab - Schedules View, 42
- Monitor Tab, 42
- Monitor Tab - Summary Statistics View, 44
- Monitor Tab - Execution Statistics View, 45
- Logs Tab, 51
- Reports Tab, 52
- Security Tab, 52
- Service States, 54
- Process States, 55
- Job States, 56
- Informatica Administrator Accessibility Overview, 57

Using Informatica Administrator Overview

Informatica Administrator is the tool that you use to manage the Informatica domain and Informatica security.

Use the Administrator tool to complete the following types of tasks:

- Domain administrative tasks. Manage logs, domain objects, user permissions, and domain reports. Generate and upload node diagnostics. Monitor Data Integration Service jobs and applications. Domain objects include application services, nodes, grids, folders, database connections, operating system profiles, and licenses
- Domain administrative tasks. Manage logs, domain objects, and user permissions. Monitor Data Integration Service jobs and applications.
- Security administrative tasks. Manage users, groups, roles, and privileges.

The Administrator tool has the following tabs:

- **Manage**. View and edit the properties of the domain and objects within the domain.
- **Monitor**. View the status of profile jobs, scorecard jobs, preview jobs, mapping jobs, SQL data services, web services, and workflows for each Data Integration Service.
- **Logs**. View log events for the domain and services within the domain.
- **Reports**. Run a Web Services Report or License Management Report.
- **Security**. Manage users, groups, roles, and privileges.
- **Cloud**. View information about your Informatica Cloud organization.

The Administrator tool has the following header items:

- **Log out**. Log out of the Administrator tool.
- **Manage**. Manage your account.
- **Help**. Access help for the current tab and determine the Informatica version.

Manage Tab

On the **Manage** tab, you can view and manage the domain and the objects that it contains.

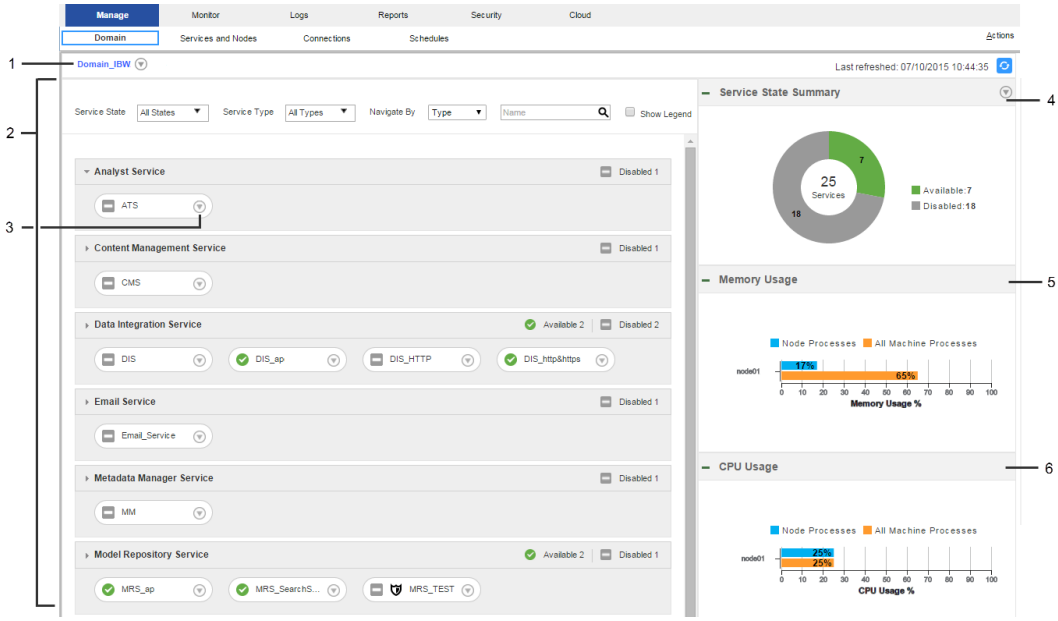
The contents that appear and the tasks you can complete on the **Manage** tab vary based on the view that you select. You can select the following views:

- **Domain**. View and manage domain status, resource consumption, and events.
- **Services and Nodes**. View and manage application services and nodes.
- **Connections**. View and manage connections.
- **Schedules**. View and manage schedules for profiles, scorecards, deployed mappings, and deployed workflows.

Manage Tab - Domain View

The **Domain** view displays an overview of the domain and its contents. You can use the domain view to monitor the domain status, resource consumption, and events. You can also perform domain actions, such as enabling and disabling services.

The following image shows the **Domain** view on the **Manage** tab:



1. Domain Actions menu
2. Contents panel
3. Object Actions menu
4. Service State Summary
5. Memory usage indicator
6. CPU usage indicator

The **Domain** view has the following components:

Domain Actions menu

Use the Domain Actions menu to view more information about the domain or shut down the domain.

Use the Domain Actions menu to perform the following tasks:

- View Properties. Open the Services and Nodes view and display the properties for the domain.
- View History. Open the History view and display domain events from the last day.
- View Logs. Open the Logs tab and display Service Manager log events from the last day.
- View Command History. Open the Command History panel and display the 50 most recent service lifecycle commands that were issued from the Administrator tool.
- Shut Down Domain.

Contents panel

Displays domain objects and their states. Domain objects include services, nodes, and grids.

The following table describes the methods that you can use to view objects in the contents panel:

Method	Description
Service State	Filter services by the following states: - All states - Available - Unavailable - Disabled
Service Type	Filter some or all services in the domain.
Navigate by	Group objects by node, type, or folder.
Search	Search for an object by name. You can use an asterisk (*) as a wildcard character in this field.
Show Legend	View a list of state icons and descriptions.

Object Actions menus

Objects in the contents panel have Actions menus. Use the Actions menus to view information about domain objects or perform common tasks. The information you can view and the tasks that you can perform vary depending on which object you select.

Use the service Actions menu to perform the following tasks:

- View Properties. Open the Services and Nodes view and display the properties for the service.
- View History. Open the History view and display service events from the last day.
- View Logs. Open the Logs tab and display service log events from the last day.
- View Dependencies. Open the Dependency graph and display direct dependencies for the service.
- Recycle Service.
- Enable or Disable Service.

Use the node Actions menu to perform the following tasks:

- View Properties. Open the Services and Nodes view and display properties for the node.
- View History. Open the History view and display node events from the last day.
- View Dependencies. Open the Dependency graph and display direct dependencies for the node.
- Shut Down Node

Use the grid Actions menu to perform the following tasks:

- View Properties. Open the Services and Nodes view and display properties for the grid.
- View Dependencies. Open the Dependency graph and display direct dependencies for the grid.

Service State Summary

Doughnut chart that shows the number and state of the services in the domain. When you click a state in the chart, the contents panel displays services with that state.

Resource usage indicators

Bar charts that compare resource usage of a domain process to resource usage of all processes on the machine. The Domain view contains a memory usage indicator and a CPU usage indicator.

Manage tab Actions menu

Access help for the Domain view.

Details Panel

When you select a domain object, the **Details** panel displays information about the object. The information that you can view varies depending on which object you select.

The following table describes the details that display depending on the object that you select in the contents panel:

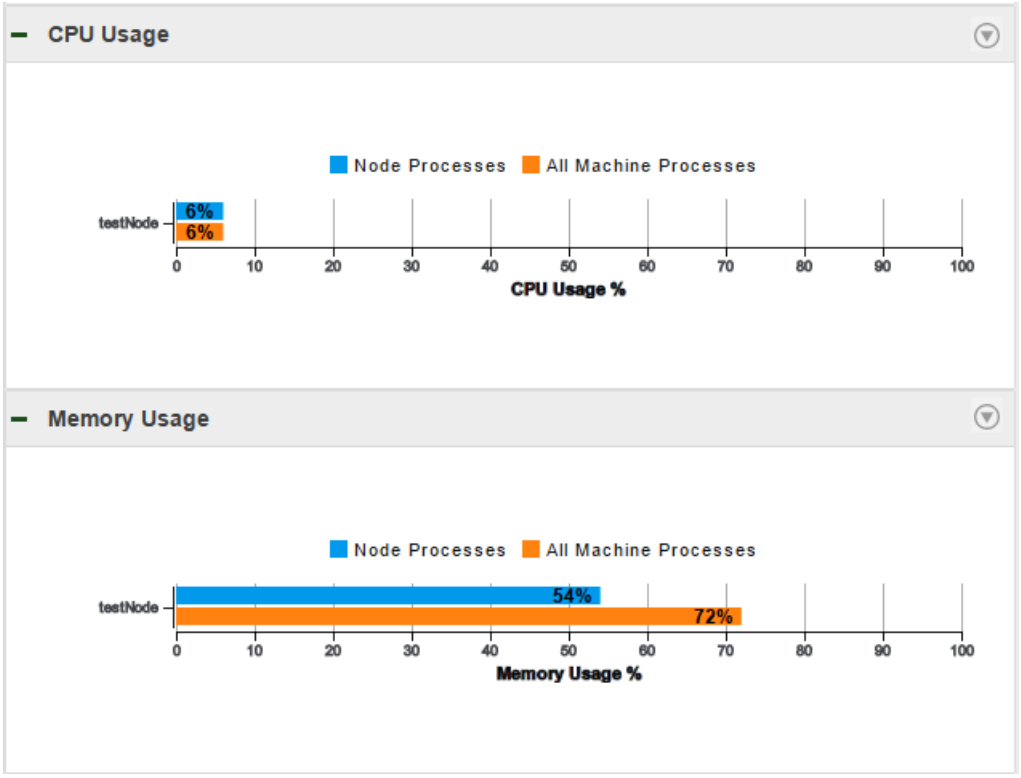
Object	Details Panel Content
Node	Node name and state. Click the node name to view the node properties.
Service	The Details panel displays the following content for a service: - Service name and state. Click the service name to view the service properties. - Node on which the service process runs. Click the node name to view the node properties. - State of the node on which the service process runs. - State of the service process.
Service running on a grid	The Details panel displays the following content for a service that runs on a grid: - Service name and state. Click the service name to view the service properties. - Nodes in the grid. Click a node name to view the node properties. - State of the nodes on which the service processes run. - State of the service processes.
Service in high availability mode	The Details panel displays the following content for a service that is highly available: - Service name and state. Click the service name to view the service properties. - Nodes on which the service is configured to run. Click a node name to view the node properties. - State of the nodes on which the service processes run. - State of the service process on the nodes.
Grid	The Details panel displays the following content for a grid: - Grid name and state. Click the grid name to view the grid properties. - Nodes in the grid. Click a node name to view the node properties. - State of nodes running in the grid.

Resource Usage Indicators

The resource usage indicators are bar charts and graphs that compare resource usage for a domain process to resource usage of all processes on the machine. Select a domain process to compare with all processes. You can view current usage statistics or statistics for the previous 60 minutes.

You can view usage statistics for memory and CPU. Choose whether to view current statistics or to view graphs of usage for the past 60 minutes. Click the selection arrow and choose **Current** or **Last Hour Trend**.

The following image shows the current resource usage in a domain that contains one node:



The information that the graphs display varies based on which domain object you select.

The following table describes the information that you can view when you select the domain or a domain object:

Domain Object	Usage Indicator Content
Domain	The usage indicators display the following content: - Nodes in the domain. - Resource usage of all processes running on each node in the domain. - Resource usage of all processes running on the machine.
Node	The usage indicators display the following content: - The node. - Resource usage of processes running on the node. - Resource usage of all processes running on the machine.
Service	The usage indicators display the following content: - The node on which the service process runs. - Resource usage of the service process that is running on the node. - Resource usage of all processes running on the machine.
Service in high availability mode	The usage indicators display the following content: - The node on which the service process is running. - Resource usage of the service process that is running on the node. - Resource usage of all processes running on the machine.

Domain Object	Usage Indicator Content
Service running on a grid	The usage indicators display the following content: - All nodes on which the service process runs. - Resource usage of the service process that is running on each node. - Resource usage of all processes running on the machine.
Grid	The usage indicators display the following content: - All available nodes in the grid. - Resource usage of all processes running on each node in the domain. - Resource usage of all processes running on the machine.

If a **See More...** link appears in the indicator, you can click it to view the complete list of nodes in the domain. You can sort the list by node name, process usage on the nodes, or process usage on the machine. You can also search the list for a particular node.

ManageTab - Services and Nodes View

The **Services and Nodes** view shows all application services and nodes defined in the domain.

The **Services and Nodes** view has the following components:

Navigator

Appears in the left pane of the **Manage** tab. The Navigator displays the following types of objects:

- Domain. You can view one domain, which is the highest object in the Navigator hierarchy.
- Folders. Use folders to organize domain objects in the Navigator. Select a folder to view information about the folder and the objects in the folder.
- Application services. An application service represents server-based functionality. Select an application service to view information about the service and its processes.
- System services. A system service is an application service that can have a single instance in the domain. Select a system service to view information about the service and its processes.
- Nodes. A node represents a machine in the domain. You configure service processes to run on nodes with the service role.
- Grids. Create a grid to run the Data Integration Service on multiple nodes. Select a grid to view nodes assigned to the grid.
- Licenses. Create a license on the **Manage** tab based on a license key file provided by Informatica. Select a license to view services assigned to the license.

You can search for nodes, services, and grids in the Navigator.

Contents panel

Appears in the right pane of the **Manage** tab and displays information about the domain or domain object that you select in the Navigator.

Actions menu in the Navigator

When you select the domain in the Navigator, you can create a folder, service, node, grid, or license.

When you select a domain object in the Navigator, you can delete the object, move it to a folder, or refresh the object.

Actions menu on the Manage tab

When you select the domain in the Navigator, you can shut down the domain or view logs for the domain.

When you select a node in the Navigator, you can remove a node association, recalculate the CPU profile benchmark, or shut down the node.

When you select a service in the Navigator, you can recycle or disable the service and configure properties for the service.

When you select a license in the Navigator, you can add an incremental key to the license.

Navigator Search

You can search for and filter nodes, application services, and grids in the Navigator.

You can perform the following tasks in the Navigator search section:

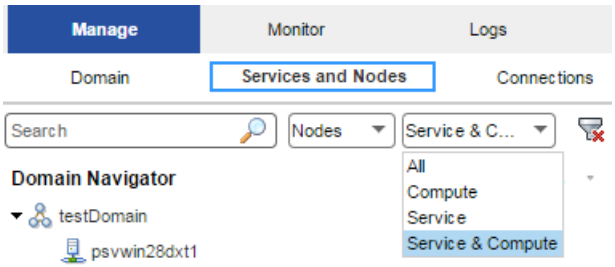
Search by object name.

In the search text box, enter the name or partial name of the object to search for. The Navigator displays the search results.

Filter by object type.

Click **Filters**, and then select the object type to filter by. If you filter by nodes, you can filter further by node role. If you filter by services, you can filter further by service type. The Navigator displays the filtered results.

The following image displays the Navigator search section filtered by nodes with the service and compute roles:



Reset filters.

Click **Reset Filters** to clear any filters or entered search text.

Domain

You can view one domain in the **Services and Nodes** view on the **Manage** tab. It is the highest object in the Navigator hierarchy.

When you select the domain in the Navigator, the contents panel shows the following views and buttons:

- **Properties** view. View or edit domain resilience properties.
- **Resources** view. View available resources for each node in the domain.
- **Permissions** view. View or edit group and user permissions on the domain.

- **Diagnostics** view. View node diagnostics, generate and upload node diagnostics to Customer Support Manager, or edit customer portal login details.
- **Plug-ins** view. View plug-ins registered in the domain.
- **View Domain Logs** button. View logs for the domain and services in the domain.

In the **Actions** menu in the Navigator, you can add a folder, node, grid, application service, or license to the domain.

In the **Actions** menu on the **Manage** tab, you can shut down the domain, view logs, or access help for the current view.

Folders

You can use folders in the domain to organize objects and to manage security.

Folders can contain nodes, services, grids, licenses, and other folders.

When you select a folder in the Navigator, the Navigator opens to display the objects in the folder. The contents panel displays the following information:

- **Properties** view. Displays the name and description of the folder.
- **Permissions** view. View or edit group and user permissions on the folder.

In the **Actions** menu in the Navigator, you can delete the folder, move the folder into another folder, refresh the contents on the **Manage** tab, or access help for the current tab.

Note: The System_Services folder is created for you when you create the domain, and contains all of the system services. A system service is an application service that can have a single instance in the domain. You cannot delete, move, or edit the properties or contents of the System_Services folder.

Application Services

Application services are a group of services that represent Informatica server-based functionality.

In the **Services and Nodes** view on the **Manage** tab, you can create and manage the following application services:

Analyst Service

Runs Informatica Analyst in the Informatica domain. The Analyst Service manages the connections between service components and the users that log in to Informatica Analyst.

The Analyst Service connects to a Data Integration Service, a Model Repository Service, and a Search Service. The Analyst Service also specifies a flat file cache directory and a directory for business glossary export files.

You can create and recycle the Analyst Service in the Informatica domain to access the Analyst tool. You can launch the Analyst tool from the Administrator tool.

When you select an Analyst Service in the Navigator, the contents panel displays the following information:

- **Properties** view. View the state of the service and the URL for the Analyst Service. Manage general, model repository, data integration, metadata manager, flat file cache, business glossary export, logging, and custom properties.
- **Processes** view. View the state of the service process on each node. View and edit service process properties on each assigned node.

- **Permissions** view. View or edit the group and user permissions on the Analyst Service.
- **Actions** menu. Manage the service and repository contents.

Catalog Service

The Catalog Service is an application service that runs Enterprise Data Catalog in the Informatica domain. The Catalog Service manages the connections between service components and the users that have access to Enterprise Data Catalog and Catalog Administrator.

Cluster Service

The Informatica Cluster Service is an application service that runs and manages all the associated services required to run Enterprise Data Catalog.

Data Integration Service

Completes data integration tasks for Informatica Analyst, Informatica Developer, and external clients. When you preview or run data profiles, SQL data services, and mappings in Informatica Analyst or Informatica Developer, the application sends requests to the Data Integration Service to perform the data integration tasks. When you start a command from the command line or an external client to run SQL data services and mappings in an application, the command sends the request to the Data Integration Service.

When you select a Data Integration Service in the Navigator, the contents panel displays the following information:

- **Properties** view. View the state of the service. Manage general, model repository, logging, logical data object and virtual table cache, profiling, data object cache, and custom properties. Set the default deployment option.
- **Processes** view. View the state of the service process on each node. View and edit service process properties on each assigned node.
- **Applications** view. Start and stop applications and SQL data services. Back up applications. Manage application properties.
- **Actions** menu. Manage the service and repository contents.

Model Repository Service

Manages the Model repository. The Model repository stores metadata created by Informatica products, such as Informatica Developer, Informatica Analyst, the Data Integration Service, and Informatica Administrator. The Model repository enables collaboration among the products.

When you select a Model Repository Service in the Navigator, the contents panel displays the following information:

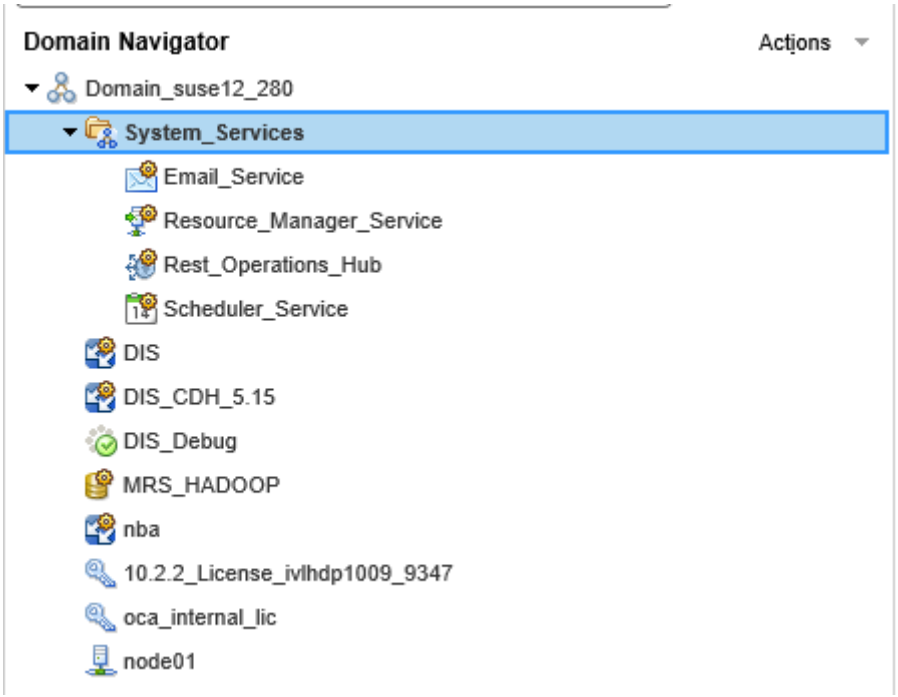
- **Properties** view. View the state of the service. Manage general, repository database, search, and custom properties.
- **Processes** view. View the state of the service process on each node. View and edit service process properties on each assigned node.
- **Actions** menu. Manage the service and repository contents.

System Services

A system service is an application service that can have a single instance in the domain. When you create the domain, the system services are created for you. You can enable, disable, and configure system services.

You can manage system services on the **Services and Nodes** view on the **Manage** tab.

The following image shows the System Services folder in the Domain Navigator:



You can manage the following system services:

Email Service

The Email Service sends email notifications for business glossaries, scorecards, and workflows. Enable the Email Service to allow users to configure email notifications.

When you select the Email Service in the System Services folder in the Navigator, the contents panel displays the following views and buttons:

- **Properties** view. View the state of the service. Manage the Model repository options and email server properties for the service.
- **Processes** view. View the state of the service process for each assigned node.
- **Permissions** view. View or edit the group and user permissions on the service.
- **Actions** menu. Manage the service.

Resource Manager Service

The Resource Manager Service manages computing resources in the domain and dispatches jobs to achieve optimal performance and scalability. The Resource Manager Service collects information about nodes with the compute role. The service matches job requirements with resource availability to identify the best compute node to run the job.

The Resource Manager Service communicates with compute nodes in a Data Integration Service grid. Enable the Resource Manager Service when you configure a Data Integration Service grid to run jobs in separate remote processes.

When you select the Resource Manager Service in the System Services folder in the Navigator, the contents panel displays the following views and buttons:

- **Properties** view. View the state of the service. Manage the log level and the primary and backup nodes for the service.
- **Processes** view. View the state of the service process for each assigned node.

- **Permissions** view. View or edit the group and user permissions on the service.
- **Actions** menu. Manage the service.

REST Operations Hub Service

The REST Operations Hub Service is an application service in the Informatica domain that exposes Informatica product functionality to external clients through REST APIs.

Scheduler Service

The Scheduler Service manages schedules for profiles, scorecards, deployed mappings, and deployed workflows. Enable the Scheduler Service to create, manage, and run schedules.

When you select the Scheduler Service in the System Services folder in the Navigator, the contents panel displays the following views and buttons:

- **Properties** view. View the state of the service. Manage the log level, primary and backup nodes, and Model Repository Service options for the service.
- **Processes** view. View the state of the service process for each assigned node and configure process properties.
- **Permissions** view. View or edit the group and user permissions on the service.
- **Actions** menu. Manage the service.

Nodes

A node is a logical representation of a physical machine in the domain. On the Services and Nodes view on the Manage tab, you assign resources to nodes and configure service processes to run on nodes that have the service role.

When you select a node in the Navigator, the contents panel displays the following information:

- **Properties** view. View the status of the node. View or edit node properties, such as the repository backup directory or range of port numbers for the processes that run on the node.
- **Processes** view. View the status of processes configured to run on the node. Service processes run on nodes that have the service role.
- **Resources** view. View or edit resources assigned to the node.
- **Permissions** view. View or edit group and user permissions on the node.

In the **Actions** menu in the Navigator, you can delete the node, move the node to a folder, refresh the contents on the **Manage** tab, or access help on the current tab.

In the **Actions** menu on the **Manage** tab, you can remove the node association, recalculate the CPU profile benchmark, or shut down the node.

Grids

A grid is an alias assigned to a group of nodes that run Data Integration Service jobs.

When you run a job on a grid, the Integration Service distributes the processing across multiple nodes in the grid. For example, when you run a profile on a grid, the Data Integration Service splits the work into multiple jobs and assigns each job to a node in the grid. You assign nodes to the grid in the **Services and Nodes** view on the **Manage** tab.

When you select a grid in the Navigator, the contents panel displays the following information:

- **Properties** view. View or edit node assignments to a grid.
- **Permissions** view. View or edit group and user permissions on the grid.

In the **Actions** menu in the Navigator, you can delete the grid, move the grid to a folder, refresh the contents on the **Manage** tab, or access help for the current tab.

Important: Profiling might fail with a licensing error if you run data domain discovery using a Data Integration Service on a grid.

Licenses

You create a license object on the **Manage** tab based on a license key file provided by Informatica.

After you create the license, you can assign services to the license.

When you select a license in the Navigator, the contents panel displays the following information:

- **Properties** view. View license properties, such as supported platforms, repositories, and licensed options. You can also edit the license description.
- **Assigned Services** view. View or edit the services assigned to the license.
- **Permissions** view. View or edit user permissions on the license.

In the **Actions** menu in the Navigator, you can delete the license, move the license to a folder, refresh the contents on the **Manage** tab, or access help on the current tab.

In the **Actions** menu on the **Manage** tab, you can add an incremental key to a license.

Manage Tab - Connections View

The **Connections** view shows the domain and all connections in the domain.

The **Connections** view has the following components:

Navigator

Displays the domain and the connections in the domain.

Actions menu in the Navigator

When you select the domain in the Navigator, you can create a connection.

When you select a connection in the Navigator, you can delete the connection.

Contents panel

Displays information about the domain or the connection that you select in the Navigator.

When you select the domain in the Navigator, the contents panel shows all connections in the domain. In the contents panel, you can filter or sort connections, or search for specific connections.

When you select a connection in the Navigator, the contents panel displays information about the connection. The tasks that you can complete for the connection vary depending on which of the following views you select:

- **Properties** view. View or edit connection properties.
- **Pooling** view. View or edit pooling properties for the connection.
- **Permissions** view. View or edit group or user permissions on the connection.

Actions menu on the Manage tab

When you select a connection in the Navigator, you can test the connection.

Manage Tab - Schedules View

Use the **Schedules** view to view and manage schedules for deployed mappings and workflows.

The **Schedules** view has the following components:

Navigator

Displays the domain and schedules in the domain.

All Schedules view

When you select the domain in the Navigator, the **All Schedules** view displays the name, status, and description of the schedules in the domain.

Properties view

When you select a schedule in the Navigator, the **Properties** view displays details about the schedule.

Scheduled Jobs view

When you select the domain in the Navigator, the **Scheduled Jobs** view displays details about the scheduled jobs in the domain.

When you select a schedule in the Navigator, the **Scheduled Jobs** view displays details about the jobs that are running on that schedule.

Actions menu on the Manage tab

The actions that you can perform in the Actions menu vary based on which view you select.

You can perform the following actions using the Actions menu:

- Create a schedule
- Edit a schedule
- Delete a schedule
- Pause a schedule
- Resume a schedule
- Refresh
- Unschedule an object
- Reset filters
- Access help

Monitor Tab

On the **Monitor** tab, you can monitor Data Integration Services and objects that the Data Integration Services run.

After you configure the monitoring Model repository on the domain, you can view monitoring statistics in the Administrator tool.

You can monitor the following objects on the **Monitor** tab:

- Ad hoc jobs. Includes profiles, enterprise data profiles, mappings, scorecards, reference tables, and previews.

- Applications. Includes deployed mappings, logical data objects, SQL data services, web services, and workflows.
- SQL data service connections.
- Requests. Includes SQL data service and web service requests.

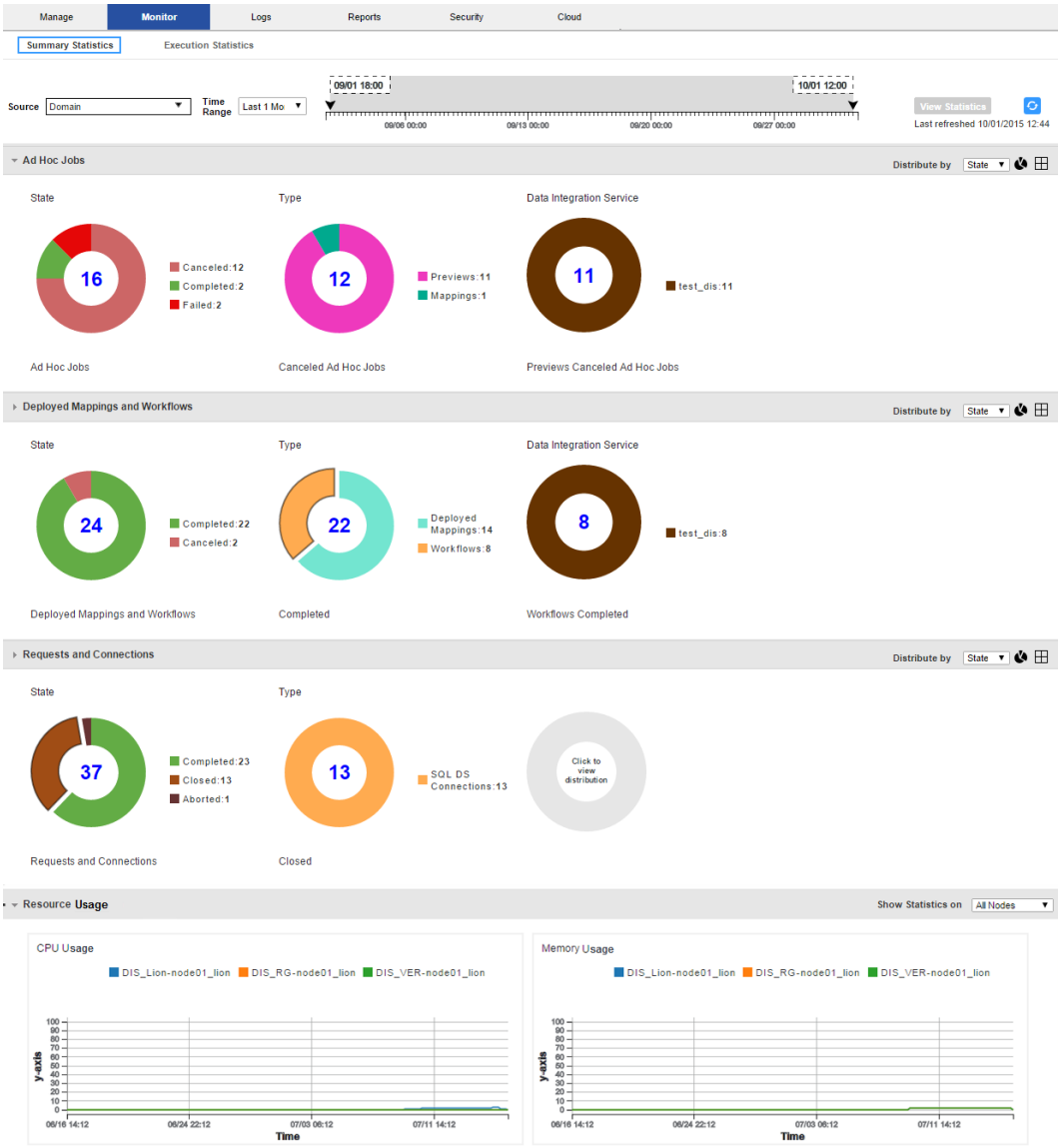
You can select the following views:

- **Summary Statistics** view. View graphical information about object state, distribution, and resource usage across the Data Integration Services.
- **Execution Statistics** view. View properties, statistics, and reports for the objects that the Data Integration Service runs.

Monitor Tab - Summary Statistics View

The **Summary Statistics** view displays historical statistics about Data Integration Services and the jobs that they run.

The following image shows the **Summary Statistics** view:



Use the **Summary Statistics** view to view graphical information about object state, distribution, and resource usage.

The **Summary Statistics** view contains the following components:

Time line

Specify a source and time range for which you want to view statistics.

Ad Hoc Jobs panel

View doughnut or tabular charts for jobs for the selected source and time range. Select sections of the doughnuts to filter by job type, state, and Data Integration Service.

Deployed Mappings and Workflows panel

View doughnut or tabular charts for deployed mappings and workflows for the selected source and time range. Select sections of the doughnuts to filter by type, state, and Data Integration Service.

Requests and Connections panel

View doughnut or tabular charts for Data Integration Service jobs for the selected source and time range. Select sections of the doughnuts to filter by object type, state, and Data Integration Service.

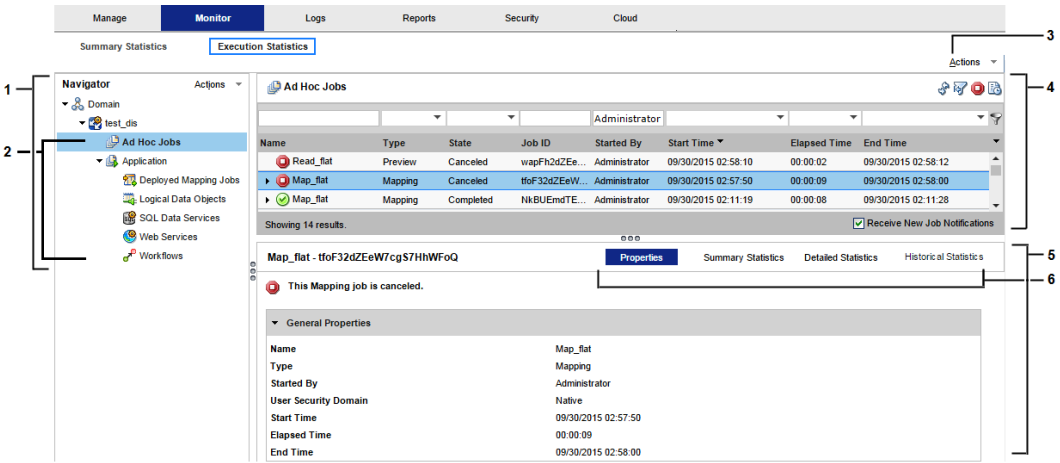
Resource Usage panel

View CPU and memory usage of all of the Data Integration Service processes that are running in the domain or on a node in the domain.

Monitor Tab - Execution Statistics View

On the **Execution Statistics** view on the **Monitor** tab, you can monitor Data Integration Services and objects that the Data Integration Services run.

The following image shows the **Execution Statistics** view on the **Monitor** tab:



1. Navigator
2. Folders
3. Actions menu
4. Contents panel
5. Details panel
6. Views in the details panel

When you select an object in the Navigator, you can view details about the object and monitor the object.

You can select the following types of objects in the Navigator in the **Execution Statistics** view:

Domain

View the states and properties of Data Integration Services in the domain.

Data Integration Service

View general properties about the Data Integration Service, and view statistics about objects that the Data Integration Service runs.

Folder

View a list of objects in the folder. The folder is a logical grouping of objects. When you select a folder, a list of objects appears in the contents panel. The contents panel shows multiple columns that show properties about each object. You can configure the columns that appear in the contents panel.

The following table shows the folders that appear in the Navigator:

Folder	Location
Ad Hoc Jobs	Appears under the Data Integration Service.
Deployed Mapping Jobs	Appears under the corresponding application.
Logical Data Objects	Appears under the corresponding application.
SQL Data Services	Appears under the corresponding application.
Web Services	Appears under the corresponding application.
Workflows	Appears under the corresponding application.

Views in the Execution Statistics View

When you select an integration object in the Navigator or an object link in the contents panel of the **Execution Statistics** view, multiple views of information appear in the contents panel. The views show information about the selected object, such as properties, run-time statistics, and run-time reports.

Depending on the type of object you select in the Navigator, the contents panel might display the following views:

Properties view

Shows general properties and run-time statistics about the selected object. General properties might include the name and description of the object. Statistics vary based on the selected object type.

Statistics view

Shows historical statistics about jobs in an application, or jobs that the Data Integration Service ran. For example, when you select an application, you can view the number of deployed mapping jobs that failed in the last four hours.

Reports view

Shows reports for the selected object. The reports contain key metrics for the object. For example, you can view reports to determine the longest running jobs on a Data Integration Service during a particular time period.

Summary Statistics view

Shows throughput and resource usage statistics for ad hoc mapping jobs, deployed mapping jobs, or mappings in a workflow.

Detailed Statistics view

Shows graphs of the throughput and resource usage for ad hoc mapping jobs, deployed mapping jobs, or mappings in a workflow.

Historical Statistics view

Shows averaged data from multiple runs for a specific job. For example, you can view the minimum, maximum, and average duration of the mapping job. You can view the average amount of CPU that the job consumes when it runs.

Connections view

Shows connections defined for the selected object. You can view statistics about each connection, such as the number of closed, aborted, and total connections.

Requests view

Shows details about requests. There are two types of requests: SQL queries and web service requests. Users can use a third-party client tool to run SQL queries against the virtual tables in an SQL data service. Users can use a web service client to run web service requests against a web service. Each web service request runs a web service operation.

A request is a web services request or an SQL query that a user runs against a virtual table in an SQL data service.

Virtual Tables view

Shows virtual tables defined in an SQL data service. You can also view properties and cache refresh details for each virtual table.

Operations view

Shows the operations defined for the web service.

Statistics in the Execution Statistics View

When you select a Data Integration Service or an application in the Navigator in the **Execution Statistics** view, the **Statistics** section in the **Properties** view shows statistics for jobs that run on the Data Integration Service.

The following table describes the types of jobs and statistics that you can view:

Object Type	Statistics
Ad Hoc Jobs	Displays the following statistics about ad hoc jobs: - Total. Total number of jobs. - Failed. Number of failed jobs. - Aborted. Number of aborted jobs. The Data Integration Service was recycled, or disabled in the abort mode when the job was running. - Completed. Number of completed jobs. - Canceled. Number of canceled jobs.
Applications	Displays the following statistics about application: - Total. Total number of applications. - Running. Number of running applications. - Failed. Number of failed applications. - Stopped. Number of stopped applications. - Disabled. Number of disabled applications.
Deployed Mapping Jobs	Displays the following statistics about deployed mapping jobs: - Total. Total number of deployed mappings. - Failed. Number of failed mapping jobs. - Aborted. Number of aborted mapping jobs. - Completed. Number of completed mapping jobs. - Canceled. Number of canceled mapping jobs.

Object Type	Statistics
Connections	Displays the following statistics about SQL data service connections to virtual databases: - Total. Total number of connections. - Closed. Number of database connections on which SQL data service requests have previously run. - Aborted. Number of connections that were manually aborted, or that were aborted when the Data Integration Service was recycled or disabled in abort mode.
Requests	Displays the following statistics about SQL data service and web service requests: - Total. Total number of requests. - Completed. Number of completed requests. - Aborted. Requests that were aborted when the Data Integration Service was recycled, or disabled in abort mode. - Failed. Number of failed requests.
Workflows	Displays the following statistics about workflows: - Total. Total number of workflow instances. - Completed. Number of completed workflow instances. - Canceled. Number of canceled workflow instances. - Aborted. Number of aborted workflow instances. - Failed. Number of failed workflow instances.

Reports on the Execution Statistics View

You can view monitoring reports in the **Reports** view of the **Execution Statistics** view. The **Reports** view appears when you select the appropriate object in the Navigator. You can view reports to monitor objects deployed to a Data Integration Service, such as ad hoc jobs, web services, SQL data services, and workflows.

The reports that appear in the **Reports view** vary depending on which object you select. To view reports in the **Reports** view, you must configure them in **Monitor tab Actions** > **Report and Statistic Settings**. By default, no reports appear in the **Reports** view.

You can view the following monitoring reports:

Longest Duration Ad Hoc Jobs

Shows ad hoc jobs that ran the longest during the specified time period. The report shows the job name, ID, type, state, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Longest Duration Mapping Jobs

Shows mapping jobs that ran the longest during the specified time period. The report shows the job name, state, ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Longest Duration Profile Jobs

Shows profile jobs that ran the longest during the specified time period. The report shows the job name, state, ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Longest Duration Reference Table Jobs

Shows reference table process jobs that ran the longest during the specified time period. Reference table jobs are jobs where you export or import reference table data. The report shows the job name, state, ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Longest Duration Scorecard Jobs

Shows scorecard jobs that ran the longest during the specified time period. The report shows the job name, state, ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Longest Duration SQL Data Service Connections

Shows SQL data service connections that were open the longest during the specified time period. The report shows the connection ID, SQL data service, connection state, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service, an SQL data service, or an application in the **Monitor** tab.

Longest Duration SQL Data Service Requests

Shows SQL data service requests that ran the longest during the specified time period. The report shows the request ID, SQL data service, request state, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service, an SQL data service, or an application in the **Monitor** tab.

Longest Duration Web Service Requests

Shows web service requests that were open the longest during the specified time period. The report shows the request ID, web service operation, request state, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service, a web service, or an application in the **Monitor** tab.

Longest Duration Workflows

Shows all workflows that were running the longest during the specified time period. The report shows the workflow name, state, instance ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service or an application in the **Monitor** tab.

Longest Duration Workflows Excluding Human Tasks

Shows workflows that do not include a Human task that were running the longest during the specified time period. The report shows the workflow name, state, instance ID, and duration. You can view this report in the **Reports** view when you monitor a Data Integration Service or an application in the **Monitor** tab.

Minimum, Maximum, and Average Duration Report

Shows the minimum, maximum, and average duration for SQL data service and web service requests during a specified time period. You can view this report in the **Reports** view when you monitor a Data Integration Service, an SQL data service, a web service, or an application in the **Monitor** tab.

Most Active IP for SQL Data Service Requests

Shows the total number of SQL data service requests from each IP address during the specified time period. You can view this report in the **Reports** view when you monitor a Data Integration Service, an SQL data service, or an application in the **Monitor** tab.

Most Active SQL Data Service Connections

Shows SQL data service connections that received the most connection requests during the specified time period. The report shows the connection ID, SQL data service, and the total number of connection requests. You can view this report in the **Reports** view when you monitor a Data Integration Service, an application, or an SQL data service in the **Monitor** tab.

Most Active Users for Ad Hoc Jobs

Shows users that ran the most number of ad hoc jobs during the specified time period. The report shows the user name and the total number of jobs that the user ran. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Most Active Web Service Client IP

Shows IP addresses that received the most number of web service requests during the specified time period. You can view this report in the **Reports** view when you monitor a Data Integration Service, an application, a web service, or web service operation in the **Monitor** tab.

Most Frequent Errors for Ad Hoc Jobs

Shows the most frequent errors for ad hoc jobs, regardless of job type, during the specified time period. The report shows the job type, error ID, and error count. You can view this report in the **Reports** view when you monitor a Data Integration Service in the **Monitor** tab.

Most Frequent Errors for SQL Data Service Requests

Shows the most frequent errors for SQL data service requests during the specified time period. The report shows the error ID and error count. You can view this report in the **Reports** view when you monitor a Data Integration Service, an SQL data service, or an application in the **Monitor** tab.

Most Frequent Faults for Web Service Requests

Shows the most frequent faults for web service requests during the specified time period. The report shows the fault ID and fault count. You can view this report in the **Reports** view when you monitor a Data Integration Service, a web service, or an application in the **Monitor** tab.

Summary Statistics View

You can view throughput and resource information for mapping jobs in the **Execution Statistics** view.

When you select an ad hoc mapping job, deployed mapping job, or mapping in a workflow in the contents panel, the details panel displays the **Summary Statistics** view. The **Summary Statistics** view displays run-time statistics about the throughput and resource usage for the job. The run-time period begins when the Data Integration Service starts reading from the first row.
You can view the following throughput statistics for the job:

- Source. The name of the mapping source file.
- Target name. The name of the target file.
- Rows. Number of rows read for source and target.
- Average Rows/Sec. Average number of rows read per second for source and target.
- Bytes. Number of bytes read for source and target.
- Average Bytes/Sec. Average number of bytes read per second for source and target
- First Row Accessed. The date and time when the Data Integration Service started reading the first row in the source file.
- Dropped rows. Number of source rows that the Data Integration Service did not read.
- Rejected Rows. Number of target rows that the Data Integration Service did not write to target.

You can view the following resource usage statistics for the job:

- Executing node. Node where the Data Integration Service that ran the job is running.
- Average CPU usage. The average amount of CPU that the Data Integration Service used to run the job.
- Average Memory usage. The average amount of memory that the Data Integration Service used to run the job.

Detailed Statistics View

You can view graphs of the throughput and resource information for mapping jobs that run longer than one minute.

When you select an ad hoc mapping job, deployed mapping job, or mapping in a workflow in the contents panel of the **Execution Statistics** view, the details panel displays the **Detailed Statistics** view. The **Detailed Statistics** view displays run-time statistics about the throughput and resources usage for the job. The run-time period begins when the Data Integration Service starts reading from the first row.
The **Detailed Statistics** view displays the following graphs:

Throughput graph

Plots the number of rows read and written against the run time of the job.

CPU Usage graph

Plots the percentage of the Data Integration Service CPU that was allocated to run the job against the run time of the job.

Memory Usage graph

Plots the amount of memory, in megabytes, that the Data Integration Service allocated to run the job against the run time of the job.

Logs Tab

The **Logs** tab shows logs.

On the **Logs** tab, you can view the following types of logs:

- Domain log. Domain log events are log events generated from the domain functions that the Service Manager performs.
- Service log. Service log events are log events generated by each application service.
- User Activity log. User Activity log events monitor user activity in the domain.

The **Logs** tab displays the following components for each type of log:

- Filter. Configure filter options for the logs.
- Log viewer. Displays log events based on the filter criteria.
- Reset filter. Reset the filter criteria.
- Copy rows. Copy the log text of the selected rows.
- **Actions** menu. Contains options to save, purge, and manage logs. It also contains filter options.

Reports Tab

The **Reports** tab shows domain reports.

On the **Reports** tab, you can run the following domain reports:

- License Management Report. Run a report to monitor the number of software options purchased for a license and the number of times a license exceeds usage limits. Run a report to monitor the usage of logical CPUs. You run the report for a license.
- Web Services Report. Run a report to analyze the performance of web services running on a Web Services Hub. You run the report for a time interval.

Security Tab

You administer Informatica security on the Security tab of the Administrator tool.

The Security tab has the following components:

- Search section. Search for users, groups, or roles by name.
- Navigator. The Navigator appears in the left pane and displays groups, users, and roles.
- Contents panel. The contents panel displays properties and options based on the object selected in the Navigator and the tab selected in the contents panel.
- Security Actions menu. Contains options to create or delete a group, user, or role. You can manage LDAP and operating system profiles. You can also view users that have privileges for a service.

Using the Search Section

Use the Search section to search for users, groups, and roles by name. Search is not case sensitive.

1. In the Search section, select whether you want to search for users, groups, or roles.
2. Enter the name or partial name to search for.

   You can include an asterisk (*) in a name to use a wildcard character in the search. For example, enter “ad*” to search for all objects starting with “ad”. Enter “*ad” to search for all objects ending with “ad”.
3. Click Go.

   The Search Results section appears and displays a maximum of 100 objects. If your search returns more than 100 objects, narrow your search criteria to refine the search results.
4. Select an object in the Search Results section to display information about the object in the contents panel.

Using the Security Navigator

The Navigator appears in the contents panel of the Security tab. When you select an object in the Navigator, the contents panel displays information about the object.

The Navigator on the Security tab displays one of the following sections based on what you are viewing:

- Groups section. Select a group to view the properties of the group, the users assigned to the group, and the roles and privileges assigned to the group.

- Users section. Select a user to view the properties of the user, the groups the user belongs to, and the roles and privileges assigned to the user.
- Roles section. Select a role to view the properties of the role, the users and groups that have the role assigned to them, and the privileges assigned to the role.
- Operating Profiles section. Select an operating profile to view the properties of the operating system profile, and the permissions assigned to users and groups that use the operating system profile.
- LDAP Configuration section. Select a configuration to view the LDAP server connection details, the LDAP security domain that contains users and groups imported from the LDAP directory service, and the LDAP synchronization schedule.

The Navigator provides different ways to complete a task. You can use any of the following methods to manage groups, users, and roles:

- Click the **Actions** menu. Each section of the Navigator includes an Actions menu to manage groups, users, roles, operating system profiles, or LDAP configurations.
- Right-click an object. Right-click an object in the Navigator to display the options available in the Actions menu.
- Use keyboard shortcuts. Use keyboard shortcuts to move to different sections of the Navigator.

Groups

A group is a collection of users and groups that can have the same privileges, roles, and permissions.

The Groups section of the Navigator organizes groups into security domain folders. A security domain is a collection of user accounts and groups in an Informatica domain. Native authentication uses the Native security domain which contains the users and groups created and managed in the Administrator tool. LDAP authentication uses LDAP security domains which contain users and groups imported from the LDAP directory service.

When you select a security domain folder in the Groups section of the Navigator, the contents panel displays all groups belonging to the security domain.

When you select a group in the Navigator, the contents panel displays the following tabs:

- Overview. Displays general properties of the group and users assigned to the group.
- Privileges. Displays the privileges and roles assigned to the group for the domain and for application services in the domain.
- Permissions. Displays the level of access that users within the group have perform tasks on domain objects, including nodes, grids and application services . Also displays the level of access that users within the group have to perform tasks on connection objects and operating system profiles.

Users

A user with an account in the Informatica domain can log in to the following application clients:

- Informatica Administrator
- Data Analyzer
- Informatica Developer
- Informatica Analyst
- Jaspersoft

The Users section of the Navigator organizes users into security domain folders. A security domain is a collection of user accounts and groups in an Informatica domain. Native authentication uses the Native

security domain which contains the users and groups created and managed in the Administrator tool. LDAP authentication uses LDAP security domains which contain users and groups imported from the LDAP directory service.

When you select a security domain folder in the Users section of the Navigator, the contents panel displays all users belonging to the security domain. Right-click a user and select Navigate to Item to display the user details in the contents panel.

When you select a user in the Navigator, the contents panel displays the following tabs:

- Overview. Displays general properties of the user and all groups to which the user belongs.
- Privileges. Displays the privileges and roles assigned to the user for the domain and for application services in the domain.

Roles

A role is a collection of privileges that you assign to a user or group. Privileges determine the actions that users can perform. You assign a role to users and groups for the domain and for application services in the domain.

The Roles section of the Navigator organizes roles into the following folders:

- System-defined Roles. Contains roles that you cannot edit or delete. The Administrator role is a system-defined role.
- Custom Roles. Contains roles that you can create, edit, and delete. The Administrator tool includes some custom roles that you can edit and assign to users and groups.

When you select a folder in the Roles section of the Navigator, the contents panel displays all roles belonging to the folder.

When you select a role in the Navigator, the contents panel displays the following tabs:

- Overview. Displays general properties of the role and the users and groups that have the role assigned for the domain and application services.
- Privileges. Displays the privileges assigned to the role for the domain and application services.

Service States

You can identify the state of an application service by the icon that displays in the Administrator tool.

The following table describes the icons that indicate service states:

State	Icon
Available	
Disabled	
Unavailable	

Process States

You can identify the state of a Data Integration Service process by the icon that displays in the Administrator tool.

The state icons also indicate the type of node on which the process runs. If the primary node has high availability, a yellow diamond is superimposed on the process state icon. If the process runs on a grid, a grid icon is superimposed on the process state icon.

The following table describes the icons that indicate process states:

State	Icon
Aborted	
Aborted (with high availability)	
Aborted (Grid)	
Disabled	
Disabled (with high availability)	
Disabled (Grid)	
Failed	
Failed (with high availability)	
Failed (Grid)	
Running	
Running (with high availability)	
Running (Grid)	
Standing by or Delayed	
Standing by or Delayed (with high availability)	
Standing by or Delayed (Grid)	

State	Icon
Starting	
Starting (with high availability)	
Starting (Grid)	
Stopped	
Stopped (with high availability)	
Stopped (Grid)	
Stopping	
Stopping (with high availability)	
Stopping (Grid)	

Job States

You can identify the state of a job by the icon that displays in the Administrator tool.

The following table describes the icons associated with each job state:

State	Icon
Aborted	
Canceled	
Completed	
Failed	
Queued or Pending	
Running	

State	Icon
Starting	
Stopped	
Stopping	
Terminated	
Unknown	

Informatica Administrator Accessibility Overview

You can use a screen reader and keyboard shortcuts to navigate and work with the Administrator tool interface.

To turn the JAWS Virtual PC cursor on and off, use the keyboard shortcut **Insert+Z**.

Note: To use the JAWS screen reader with the Administrator tool, you must use Internet Explorer 11.

Keyboard Shortcuts

You can use keyboard shortcuts to navigate and work with the Administrator tool interface.

You can add, edit, and change values in the Administrator tool. Keyboard focus in the Administrator tool is indicated by a blue border around the interface label. A dotted line appears around a selected object indicating that the object is in focus. Tooltips appear when the label item receives keyboard focus or on mouse-over. The navigation order of objects in the editor is from top to bottom and left to right.

You can perform the following tasks with keyboard shortcuts:

Navigate among elements and select an element

Press **Tab**.

Select the previous object

Press **Shift+Tab**.

Navigate among perspective tabs

Press the **Left** or **Right** arrow key.

Select or clear a check box or radio button

Press the **Space** bar.

Upload files using the File Upload button

Press the **Space** bar.

Navigate through records in a dialog box

Press the **Up** or **Down** arrow key.

Select and open a drop-down menu item with sub-menus

Press the **Space** arrow key. To go back to the main menu, press **Esc**.

Edit the value of grid content such as the Access and Revoke fields in the Assign Permission and Edit Direct Permissions dialog box

Press the **Space** bar.

Note: You must enter appropriate values for all the form elements marked with an asterisk (*).

Move focus from Update Frequency drop-down menu to Time Range check box in the Statistics and Reports list grid in the Report and Statistic Settings dialog box of the Monitor tab or Monitoring tool

Press **Esc**.

You cannot access the split bars in the Administrator tool and increase or decrease the size of the panels using the keyboard. You cannot select multiple items with the **Ctrl** key in the Audit Reports tab under Security.

CHAPTER 4

Using the Domain View

This chapter includes the following topics:

- About the Domain View, 59
- Dependency Graph, 60
- Command History, 62
- History View, 62

About the Domain View

The Domain view displays an overview of the status of the domain and the objects it contains. You can use the Domain view to review current and historical information about the domain.

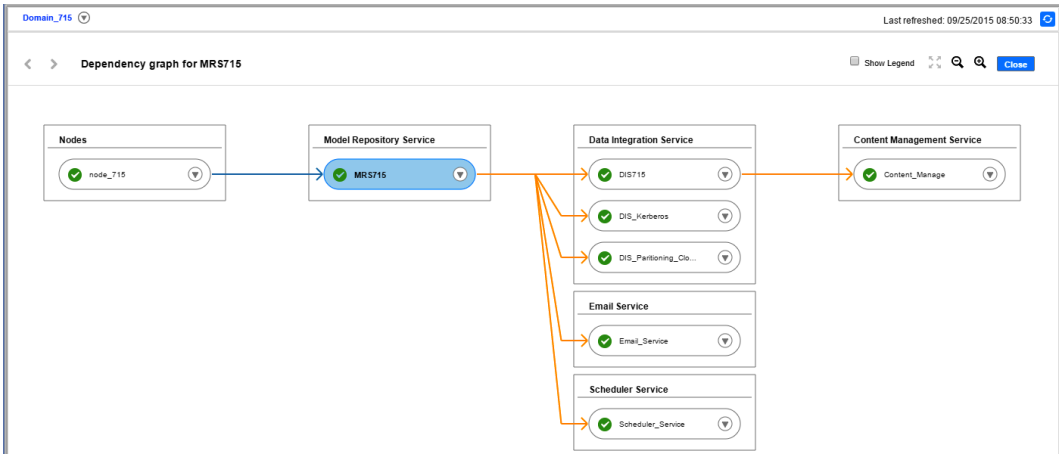
Use the Domain view to perform the following tasks:

- View current status, resource usage, and details for the domain and objects in the domain.
- View dependencies among objects in the domain.
- Perform domain actions such as shutting down the domain, enabling and disabling services, and shutting down nodes.
- View recent service commands that users issued from the Administrator tool.
- View historical information about status, resource usage, and events in the domain.

Dependency Graph

The **Dependency** graph displays dependencies among services, nodes, and grids in the Informatica domain.

The following image shows the **Dependency** graph for a Model Repository Service:



You can use the **Dependency** graph to perform the following tasks:

- View dependencies among nodes, services, and grids.
- Shut down a node.
- Enable, disable, or recycle a service.
- Disable or recycle services that depend on other services.

When you view dependencies for an object, the **Dependency** graph displays the upstream and downstream dependencies. Upstream dependencies are objects on which the selected object depends. Downstream dependencies are objects that depend on the selected object.

When you enable, disable, or recycle services from the **Dependency** graph, the actions appear in the **Command History** panel.

Viewing Dependencies for Application Services, Nodes, and Grids

You can view dependencies among application services, nodes, and grids in the Informatica domain.

1. In the Administrator tool, click the **Manage** tab.
2. In the contents panel, click the **Actions** menu for a domain object, and then select **View Dependencies**.

   The **Dependency** graph opens and displays the object and its direct dependencies.
   The **Dependency** graph displays domain objects connected by blue and orange lines, as follows:

   - Blue lines indicate service-to-node and service-to-grid dependencies.
   - Dashed blue lines indicate backup node-to-service dependencies.
   - Orange lines indicate service-to-service dependencies, such as a Data Integration Service-to-Content Management Service dependency, or a Model Repository Service-to-Data Integration Service dependency.

The following table describes the information that appears in the **Dependency** graph based on the object:

Domain Object	Upstream Dependencies	Downstream Dependencies
Node	N/A	Services that run on the node.
Node running in a grid	N/A	The node has the following downstream dependencies: - Grid in which the node runs. - Service process that runs on the grid. - Service processes that run on the node, but not in the grid.
Service	Node on which the service process runs.	Services that depend on the service.
Service running on a grid	The service has the following upstream dependencies: - Nodes on which the service process runs. - The grid on which the service processes run.	Services that depend on the service.
Service running in HA mode	Primary and backup nodes on which the service processes can run.	Services that depend on the service.
Grid	Nodes assigned to the grid.	Services that are running on the grid.

3. In the **Dependency** graph, you can optionally complete the following tasks:
   - Select **Show Legend** to view information about the icons and lines used in the graph.
   - Click and drag to view different parts of the graph.
   - Zoom in or zoom out of the graph.
   - To exit the **Dependency** graph, click **Close**.

Recycling or Disabling Downstream Services

You can recycle or disable downstream services in the **Dependency** graph.

Downstream services are services that depend on other services. For example, a Data Integration Service depends on a Model Repository Service. You recycle or disable downstream services using the Actions menu for the service on which they depend. When you disable downstream services, the service processes abort.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Actions** menu for a domain object, and then select **View Dependencies**.

   The **Dependency** graph opens and displays the object and its direct dependencies.
3. Click **Actions** > **Recycle Downstream Dependents** or **Actions** > **Disable Downstream Dependents**.

   The Recycle Downstream Dependents or Disable Downstream Dependents window appears.
4. Optionally, choose whether the action is **Planned** or **Unplanned**.
5. Optionally, enter comments about the action.
6. Select the services that you would like to recycle or disable.
7. Click **Recycle Services** or **Disable Services**.

Command History

The **Command History** panel on the **Domain** view displays recent service lifecycle commands that users issued from the Administrator tool. Service lifecycle commands include enable, disable, and recycle.

To view the command history, click **Domain Actions** > **View Command History**.

You can view the following information about the commands in the **Command History** panel:

- Service Name. Name of the service for which the command was issued.
- Service Type.
- Command.
- Status. Can be Failed, Success, or Queued.
- Status Updated
- Comments. Comments that users entered when they recycled or disabled the service.
- Message. Log messages associated with the command.

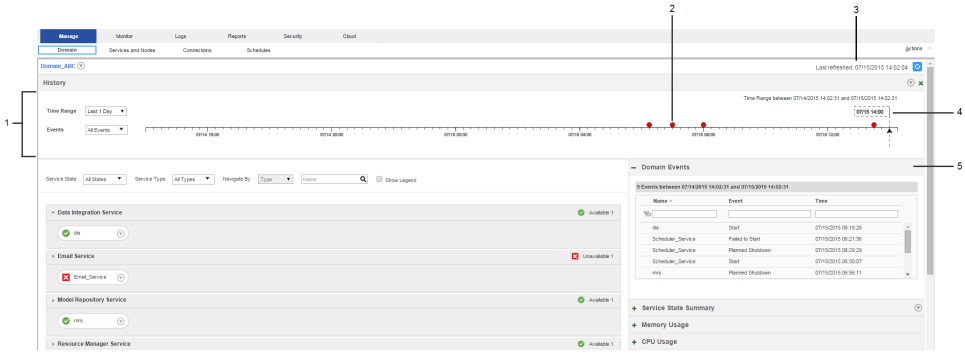
Optionally, you can show or hide columns in the command history. To change the columns, right-click the column header, and then select or clear columns.

Note: The command history is erased when you shut down or restart the master gateway node.

History View

The **History** view displays historical data for the domain, a service, or a node. You can view historical data for the contents panel, Service State Summary, resource usage indicators, and Details panel. You can also view information about events in the domain.

The following image shows the **History** view:



1. Time line
2. Event circle
3. Last Refreshed
4. Time line slider
5. Events panel

The **History** view has the following components:

Time line

Use the time line to choose which time range and events you would like to view. When events occur around a point in time, the point is marked with a red event circle. When you select a circle, the Events panel displays the events that occurred around that point in time. By default, the time line displays crashes and unplanned shutdowns that occurred in the last one day.

Events panel

Displays events that occurred during a specified time range. When you open the **History** view, the panel lists crashes and unplanned shutdowns that occurred in the last one day.

Contents panel

Displays current and historical domain contents and states. When you open the **History** view, the contents panel shows domain objects and states at the last refresh. When you drag the time line slider to a point in time, the contents panel shows domain objects and states at that point in time.

Service State Summary

Displays current and historical service state summaries. When you open the **History** view from the domain Actions menu, the Service State Summary appears and displays the number and states of services at the last refresh. When you drag the time line slider to a point in time, the Service State Summary displays the number of services that were available or unavailable at that time.

Details panel

Displays current and historical state for a service or node. When you open the **History** view, the Details panel displays the state of the service or node at the last refresh. When you drag the time line slider to a point in time, the Details panel displays the object state at that point in time.

Resource usage indicators

Display current and historical resource usage information. When you open the **History** view, the usage indicators display the usage statistics as they appeared at the last refresh. When you drag the time line slider to a point in time, the indicators display statistics for that point in time.

The data that you view in the History view is stored in the monitoring Model repository. Before you can view historical data, you must configure the monitoring Model repository in the **Manage** > **Services and Nodes** > **Monitoring Configuration** tab. You can view the per-minute data for up to two weeks in the past. After two weeks, the statistics that you view are hourly averages. You can view hourly averaged data for up to one year in the past.

Note: The monitoring Model Repository Service that you configure to store the historical data cannot capture data about events that occur while it is unavailable or disabled. For example, Start events for the monitoring Model Repository Service and Crash events for the node that the monitoring Model Repository Service process runs on do not appear in the History view.

Viewing History

You can view historical statistics for the domain, a service, or a node.

The amount of history that you can view depends on the monitoring Model repository options that you configure in the Monitoring Configuration tab. You must configure a monitoring Model repository in the Monitoring Configuration tab before you can view historical statistics.

1. To access the **History** view, click the **Actions** menu for the domain, a service, or a node, and then select **View History**.

   The **History** view opens. The auto-refresh pauses at the Last Refreshed time stamp. The time line and **Events** panel display and display crashes and unplanned shutdowns that occurred in the last one day.

2. To change the time range, select a time range in the **Time Range** list.

When you change the time range, the time line refreshes and displays the selected time range.

3. To choose a custom time range, select **Custom** in the **Time Range** list.

   You can select a custom time range from one hour to one month.

4. To change the events that display, select events in the **Events** list.

   You can view information about the following events:

   - Crash
   - Failed to Start
   - Unplanned Shutdown
   - Planned Shutdown
   - Start

   When you select different events, the time line refreshes and reflects the change.

5. To view historical statistics about the domain, drag the time line slider to a point in time.

   The **History** view refreshes and displays the domain status at that point in time.

6. To exit the **History** view, click **Close**.

Note: Current services display as Unavailable when the time range begins before they were created.

Viewing Events

You can view events for the domain, a node, or a service.

1. To view events, click the **Actions** menu for the domain, a node, or a service, and then choose **View History**.

   The **History** view opens and displays the Events panel. By default, the Events panel lists crashes and unplanned shutdowns that have occurred in the last one day.

1. You can view the following information about events:

   - Object name
   - Object type
   - Event type
   - Time the event occurred
   - Name of the associated node
   - Comments about recycled or disabled services
   - Comments about shut down nodes

2. Optionally, you can complete the following actions in the Events panel:

Option	Description
Search events	Type the search criteria in the search field, and then press enter. To clear the search, click the Reset Filters icon.
Sort a column	To sort a column in ascending order, click the column header. To sort the column in descending order, click the column header again.

Option	Description
Add or remove columns	To change the columns that appear in the Events panel, right-click a column header, and then select or deselect a column.
Reset filter	When you click a circle on the time line, you can clear the selection by clicking Reset filters.

CHAPTER 5

Domain Management

This chapter includes the following topics:

- Domain Management Overview, 66
- Alert Management, 67
- Folder Management, 69
- Domain Security Management, 70
- User Security Management, 71
- Application Service Management, 71
- Gateway Configuration, 75
- Domain Configuration Management, 76
- Shutting Down a Domain, 79
- Domain Properties, 80

Domain Management Overview

An Informatica domain is a collection of nodes and services that define the Informatica environment. To manage the domain, you manage the nodes and services within the domain.

Use the Administrator tool to complete the following tasks:

- Manage alerts. Configure, enable, and disable domain and service alerts for users.
- Create folders. Create folders to organize domain objects and manage security by setting permission on folders.
- Manage domain security. Configure secure communication between domain components.
- Manage user security. Assign privileges and permissions to users and groups.
- Manage application services. Enable, disable, recycle, and remove application services. Enable and disable service processes.
- Manage nodes. Configure node properties, such as the backup directory and resources, and shut down nodes.
- Configure gateway nodes. Configure nodes to serve as a gateway.
- Shut down the domain. Shut down the domain to complete administrative tasks on the domain.
- Manage domain configuration. Back up the domain configuration on a regular basis. You might need to restore the domain configuration from a backup to migrate the configuration to another database user

account. You might also need to reset the database information for the domain configuration if it changes.

- Complete domain tasks. You can monitor the statuses of all application services and nodes, view dependencies among application services and nodes, and shut down the domain.
- Configure domain properties. For example, you can change the database properties, SMTP properties for alerts, and domain resiliency properties.

To manage nodes and services through a single interface, all nodes and services must be in the same domain. You cannot access multiple Informatica domains in the same Administrator tool window. You can share metadata between domains when you register or unregister a local repository in the local Informatica domain with a global repository in another Informatica domain.

Alert Management

Alerts provide users with domain and service alerts. Domain alerts provide notification about node failure and master gateway election. Service alerts provide notification about service process failover.

To use the alerts, complete the following tasks:

- Configure the SMTP settings for the outgoing email server.
- Subscribe to alerts.

After you configure the SMTP settings, users can subscribe to domain and service alerts.

Configuring SMTP Settings

You configure the SMTP settings for the outgoing mail server to enable alerts.

Configure SMTP settings on the domain **Properties** view.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. In the Navigator, select the domain.
4. In the contents panel, click the **Properties** view.
5. In the SMTP Configuration section, click **Edit**.
6. Edit the SMTP settings.

Property	Description
Host Name	The SMTP outbound mail server host name. For example, enter the Microsoft Exchange Server for Microsoft Outlook.
Port	Port used by the outgoing mail server. Valid values are from 1 to 65535. Default is 25.
User name	The user name for authentication upon sending if required by the outbound mail server.

Property	Description
Password	The user password for authentication upon sending if required by the outbound mail server.
Sender Email Address	The email address that the Service Manager uses in the From field when sending notification emails. If you leave this field blank, the Service Manager uses `Administrator@<host name>` as the sender.

7. Click **OK**.

Subscribing to Alerts

After you complete the SMTP configuration, you can subscribe to alerts.

1. Verify that the domain administrator has entered a valid email address for your user account on the **Security** page.

   If the email address or the SMTP configuration is not valid, the Service Manager cannot deliver the alert notification.
2. In the Administrator tool header area, click **Manage** > **Preferences**.

   The **Preferences** page appears.
3. In the User Preferences section, click **Edit**.

   The **Edit Preferences** dialog box appears.
4. Select **Subscribe for Alerts**.
5. Click **OK**.
6. Click **OK**.

The Service Manager sends alert notification emails based on your domain privileges and permissions.

The following table lists the alert types and events for notification emails:

Alert Type	Event
Domain	Node Failure Master Gateway Election
Service	Service Process Failover

Viewing Alerts

When you subscribe to alerts, you can receive domain and service notification emails for certain events. When a domain or service event occurs that triggers a notification, you can track the alert status in the following ways:

- The Service Manager sends an alert notification email to all subscribers with the appropriate privilege and permission on the domain or service.
- The Log Manager logs alert notification delivery success or failure in the domain or service log.

For example, the Service Manager sends the following notification email to all alert subscribers with the appropriate privilege and permission on the service that failed:

```
From: Administrator@<database host>
To: Jon Smith
```

```
Subject: Alert message of type [Service] for object [HR_811].
The service process on node [node01] for service [HR_811] terminated unexpectedly.
```

In addition, the Log Manager writes the following message to the service log:

```
ALERT_10009 Alert message [service process failover] of type [service] for object
[HR_811] was successfully sent.
```

You can review the domain or service logs for undeliverable alert notification emails. In the domain log, filter by Alerts as the category. In the service logs, search on the message code ALERT. When the Service Manager cannot send an alert notification email, the following message appears in the related domain or service log:

```
ALERT_10004: Unable to send alert of type [alert type] for object [object name], alert
message [alert message], with error [error].
```

Folder Management

Use folders in the domain to organize objects and to manage security.

Folders can contain nodes, services, grids, licenses, and other folders. You might want to use folders to group services by type. For example, you can create a folder called IntegrationServices and move all Integration Services to the folder. Or, you might want to create folders to group all services for a functional area, such as Sales or Finance.

When you assign a user permission on the folder, the user inherits permission on all objects in the folder.

You can perform the following tasks with folders:

- View services and nodes. View all services in the folder and the nodes where they run. Click a node or service name to access the properties for that node or service.
- Create folders. Create folders to group objects in the domain.
- Move objects to folders. When you move an object to a folder, folder users inherit permission on the object in the folder. When you move a folder to another folder, the other folder becomes a parent of the moved folder.
- Remove folders. When you remove a folder, you can delete the objects in the folder or move them to the parent folder.

Note: The System_Services folder is created for you when you create the domain, and contains all of the system services. A system service is an application service that can have a single instance in the domain. You cannot delete, move, or edit the properties or contents of the System_Services folder.

Creating a Folder

You can create a folder in the domain or in another folder.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain or folder in which you want to create a folder.
3. On the Navigator Actions menu, click New > Folder.

4. Edit the following properties:

Node Property	Description
Name	Name of the folder. The name is not case sensitive and must be unique within the domain. It cannot exceed 80 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the folder. The description cannot exceed 765 characters.
Path	Location in the Navigator.

5. Click OK.

Moving Objects to a Folder

When you move an object to a folder, folder users inherit permission on the object. When you move a folder to another folder, the moved folder becomes a child object of the folder where it resides.

Note: The domain serves as a folder when you move objects in and out of folders.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select an object.
3. On the Navigator Actions menu, select Move to Folder.
4. In the Select Folder dialog box, select a folder, and click OK.

Removing a Folder

When you remove a folder, you can delete the objects in the folder or move them to the parent folder.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a folder.
3. On the Navigator Actions menu, select Delete.
4. Confirm that you want to delete the folder.

   You can delete the contents only if you have the appropriate privileges and permissions on all objects in the folder.
5. Choose to wait until all processes complete or to abort all processes.
6. Click OK.

Domain Security Management

You can configure Informatica domain components to use the Secure Sockets Layer (SSL) protocol or the Transport Layer Security (TLS) protocol to encrypt connections with other components. When you enable SSL or TLS for domain components, you ensure secure communication.

You can configure secure communication in the following ways:

Between services within the domain

You can configure secure communication between services within the domain.

Between the domain and external components

You can configure secure communication between Informatica domain components and web browsers or web service clients.

Each method of configuring secure communication is independent of the other methods. When you configure secure communication for one set of components, you do not need to configure secure communication for any other set.

Note: If you change a secure domain to a non-secure domain or from a non-secure domain to a secure domain, you must delete the domain configuration in the Developer tool and configure the domain again in the client.

User Security Management

You manage user security within the domain with privileges and permissions.

Privileges determine the actions that users can complete on domain objects. Permissions define the level of access a user has to a domain object. Domain objects include the domain, folders, nodes, grids, licenses, database connections, operating system profiles, and application services.

Even if a user has the domain privilege to complete certain actions, the user might also require permission to complete the action on a particular object. For example, a user has the Manage Services domain privilege which grants the user the ability to edit application services. However, the user also must have permission on the application service. A user with the Manage Services domain privilege and permission on the Development Repository Service but not on the Production Repository Service can edit the Development Repository Service but not the Production Repository Service.

To log in to the Administrator tool, a user must have the Access Informatica Administrator domain privilege. If a user has the Access Informatica Administrator privilege and permission on an object, but does not have the domain privilege that grants the ability to modify the object type, then the user can view the object. For example, if a user has permission on a node, but does not have the Manage Nodes and Grids privilege, the user can view the node properties but cannot configure, shut down, or remove the node.

If a user does not have permission on a selected object in the Navigator, the contents panel displays a message indicating that permission on the object is denied.

Application Service Management

You can perform the following common administration tasks for application services:

- Enable and disable services and service processes.
- Configure the domain to restart service processes.
- Remove an application service.
- Troubleshoot problems with an application service.

Note: You can perform all of the common administration tasks for system services, except for removing the system service.

Enabling and Disabling Services and Service Processes

You can enable and disable application services and service processes in the Administrator tool. When a service is enabled, there must be at least one service process enabled and running for the service to be available. By default, all service processes are enabled.

The behavior of a service when it starts service processes depends on how it is configured:

- If the service is configured for high availability, then the service starts the service process on the primary node. The service processes on the backup nodes are in Standing By state.
- If the service is configured to run on a grid, then the service starts service processes on all nodes that have the service role.

A service does not start a disabled service process in any situation.

The state of a service depends on the state of its processes. A service can have the following states:

- Available. You have enabled the service and at least one service process is running. The service is available to process requests.
- Unavailable. You have enabled the service and none of its processes are running. This can be because the service processes are disabled or failed to start. The service is not available to process requests.
- Disabled. You have disabled the service.

You can disable a service to perform a management task. You might want to disable the service process on a node if you need to shut down the node for maintenance. When you disable a service, all associated service processes stop, but they remain enabled.

The following table describes the different states of a service and its processes:

Service Process Configuration	Service Process State	Description
Enabled	Running	The service process is running on the node.
Enabled	Standing By	The service process is enabled but is not running because another service process is running as the primary service process. It is on standby to run in case of service failover.
Disabled	Disabled	The service is enabled but the service process is not running on the node.
Enabled	Stopped	The service is unavailable.
Enabled	Failed	The service and service process are enabled, but the service process could not start.

Enable and Upgrade the Application Services

To enable an application service, select the service in Informatica Administrator and click **Actions** > **Enable Service**.

Enable the following Informatica application services:

1. Model Repository Service
2. Data Integration Service
3. Content Management Service

After you enable the Informatica application services, run the service upgrade wizard to upgrade the services.

Upgrade the Model Repository Service, Data Integration Service, and the Content Management Service Using the Service Upgrade Wizard

Use the service upgrade wizard to upgrade application services and the database contents of the databases associated with the services. The service upgrade wizard displays upgraded services in a list along with services and associated databases that require an upgrade. You can also save the current or previous upgrade report.

1. In the Informatica Administrator header area click **Manage** > **Upgrade**.
2. Select the application services and associated databases to upgrade.
3. Optionally, specify if you want to **Automatically recycle services after upgrade**.

   If you choose to automatically recycle application services after the upgrade, the upgrade wizard restarts the services after they have been upgraded.
4. Click **Next**.
5. If dependency errors exist, the **Dependency Errors** dialog box appears. Review the dependency errors and click **OK**. Then, resolve dependency errors and click **Next**.
6. Enter the repository login information.
7. Click **Next**.

   The service upgrade wizard upgrades each application service and associated database and displays the status and processing details.
8. When the upgrade completes, the **Summary** section displays the list of application services and their upgrade status. Click each service to view the upgrade details in the **Service Details** section.
9. Optionally, click **Save Report** to save the upgrade details to a file.

   If you choose not to save the report, you can click **Save Previous Report** the next time you launch the service upgrade wizard.
10. Click **Close**.
11. If you did not choose to automatically recycle application services after the upgrade, restart the upgraded services.

You can view the upgrade report and save the upgrade report. The second time you run the service upgrade wizard, the Save Previous Report option appears in the service upgrade wizard. If you did not save the upgrade report after upgrading services, you can select this option to view or save the previous upgrade report.

Disable the Application Services

You must disable the following Informatica application services before you back up the catalog:

1. Informatica Cluster Service
2. Content Management Service
3. Data Integration Service
4. Model Repository Service

To disable an application service, select the service in Informatica Administrator and click **Actions** > **Disable Service**.

Viewing Service Processes

You can view the state of a service process on the Processes view of a service. You can view the state of all service processes on the Overview view of the domain.

To view the state of a service process:

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a service.
3. In the contents panel, select the Processes view.

   The Processes view displays the state of the processes.

Configuring Restart for Service Processes

If an application service process becomes unavailable while a node is running, the domain tries to restart the process on the same node based on the restart options configured in the domain properties.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the Properties view, configure the following restart properties:

Domain Property	Description
Maximum Restart Attempts	Number of times within a specified period that the domain attempts to restart an application service process when it fails. The value must be greater than or equal to 1. Default is 3.
Within Restart Period (sec)	Maximum period of time that the domain spends attempting to restart an application service process when it fails. If a service fails to start after the specified number of attempts within this period of time, the service does not restart. Default is 900.

Removing Application Services

You can remove an application service using the Administrator tool. Before removing an application service, you must disable it.

Note: You cannot remove a system service.

Disable the service before you delete the service to ensure that the service is not running any processes. If you do not disable the service, you may have to choose to wait until all processes complete or abort all processes when you delete the service.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the application service.
3. In the **Manage** tab Actions menu, select **Delete**.
4. In the warning message that appears, click **Yes** to stop other services that depend on the application service.
5. If the **Disable Service** dialog box appears, choose to wait until all processes complete or abort all processes, and then click **OK**.

Troubleshooting Application Services

I think that a service is using incorrect environment variable values. How can I find out which environment variable values are used by a service.

Set the error severity level for the node to debug. When the service starts on the node, the Domain log will display the environment variables that the service is using.

Gateway Configuration

A domain requires at least one node configured as a gateway node. You can configure multiple gateway nodes as backups.

One gateway node in the domain serves as the master gateway node for the domain. The Service Manager on the master gateway node accepts service requests and manages the domain and services in the domain. If the domain has one gateway node and it becomes unavailable, the domain cannot accept service requests. If the domain has multiple gateway nodes and the master gateway node becomes unavailable, the Service Managers on the other gateway nodes elect a new master gateway node. The new master gateway node accepts service requests. Only one gateway node can be the master gateway node at any particular time.

You can make the following changes to the gateway configuration for the domain:

Convert a worker node to serve as a gateway node.

You can convert a worker node to serve as a gateway node if the worker node is running and has the service role enabled. When you convert a worker node to a gateway node, you must specify the log directory for the node. If you have multiple gateway nodes, configure all gateway nodes to write log files to the same directory on a shared disk.

After you convert a worker node to a gateway node, the Service Manager on the master gateway node writes the domain configuration database connection to the nodemeta.xml file of the new gateway node.

Convert a gateway node to serve as a worker node.

You can convert a gateway node to serve as a worker node if another node in the domain is configured as a gateway node.

If you convert a master gateway node to serve as a worker node, you must restart the node to make the Service Managers elect a new master gateway node. If you do not restart the node, the node continues as the master gateway node until you restart the node or the node becomes unavailable.

Configuring the Gateway and Worker Nodes

You can convert an existing worker node to a gateway node. Or, you can convert an existing gateway node to a worker node.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the contents panel, select the **Properties** view.
4. In the **Properties** view, click **Edit** in the **Gateway Configuration Properties** section.
5. To convert a worker node to a gateway node, complete the following steps:
   a. Select the check box next to the node.
   b. If the domain uses a secure domain configuration database, specify the truststore file and password for the database.
   c. Configure the directory path for the log files for each node that you convert to a gateway node.

      If you have multiple gateway nodes, configure all gateway nodes to write log files to the same directory on a shared disk.

      Note: You must use the infacmd isp SwitchToWorkerNode command to convert a a worker node to a gateway node in a domain configured to use SAML authentication. See the *Informatica Command Reference* for instructions on using the infacmd isp SwitchToWorkerNode command.
6. To convert a gateway node to a worker node, clear the check box next to the node.

7. Click **OK**.

Domain Configuration Management

The Service Manager on the master gateway node manages the domain configuration. The domain configuration is a set of metadata tables stored in a relational database that is accessible by all gateway nodes in the domain. Each time you make a change to the domain, the Service Manager writes the change to the domain configuration. For example, when you add a node to the domain, the Service Manager adds the node information to the domain configuration. The gateway nodes use a JDBC connection to access the domain configuration database.

You can perform the following domain configuration management tasks:

- Back up the domain configuration. Back up the domain configuration on a regular basis. You may need to restore the domain configuration from a backup if the domain configuration in the database becomes corrupt.
- Restore the domain configuration. You may need to restore the domain configuration if you migrate the domain configuration to another database user account. Or, you may need to restore the backup domain configuration to a database user account.
- Migrate the domain configuration. You may need to migrate the domain configuration to another database user account.
- Configure the connection to the domain configuration database. Each gateway node must have access to the domain configuration database. You configure the database connection when you create a domain. If you change the database connection information or migrate the domain configuration to a new database, you must update the database connection information for each gateway node.
- Configure custom properties. Configure domain properties that are unique to your environment or that apply in special cases. Use custom properties only if Informatica Global Customer Support instructs you to do so.

Note: The domain configuration database and the Model repository cannot use the same database user schema.

Backing Up the Domain Configuration

Back up the domain configuration on a regular basis. You may need to restore the domain configuration from a backup file if the domain configuration in the database becomes corrupt.

Run the *infasetup* BackupDomain command to back up the domain configuration to a binary file.

Note: If the infasetup BackupDomain command fails with a Java memory error, increase the system memory available for infasetup. To increase system memory, set the -Xmx value in the INFA_JAVA_COMD_OPTS environment variable.

When you run this command, infasetup backs up the domain configuration database tables. To restore the domain to another database, you must back up the ISP_RUN_LOG table contents manually to get the previous workflow and session logs.

Additionally, use the database backup utility to manually back up additional repository tables that the infasetup command does not back up.

Restoring the Domain Configuration

You can restore domain configuration from a backup file. You may need to restore the domain configuration if the domain configuration in the database becomes inconsistent or if you want to migrate the domain configuration to another database.

Informatica restores the domain configuration from the current version. If you have a backup file from an earlier product version, you must use the earlier version to restore the domain configuration.

You can restore the domain configuration to the same or a different database user account. If you restore the domain configuration to a database user account with existing domain configuration, you must configure the command to overwrite the existing domain configuration. If you do not configure the command to overwrite the existing domain configuration, the command fails.

Each node in a domain has a host name and port number. When you restore the domain configuration, you can disassociate the host names and port numbers for all nodes in the domain. You might do this if you want to run the nodes on different machines. After you restore the domain configuration, you can assign new host names and port numbers to the nodes. Run the *infasetup* DefineGatewayNode or DefineWorkerNode command to assign a new host name and port number to a node.

If you restore the domain configuration to another database, you must reset the database connections for all gateway nodes.

Important: You lose all data in the summary tables when you restore the domain configuration.

Complete the following tasks to restore the domain:

1. Disable the application services. Disable the application services in complete mode to ensure that you do not abort any running service process. You must disable the application services to ensure that no service process is running when you shut down the domain.
2. Shut down the domain. You must shut down the domain to ensure that no change to the domain occurs while you are restoring the domain.
3. Run the *infasetup* RestoreDomain command to restore the domain configuration to a database. The RestoreDomain command restores the domain configuration in the backup file to the specified database user account.
4. Assign new host names and port numbers to the nodes in the domain if you disassociated the previous host names and port numbers when you restored the domain configuration. Run the *infasetup* DefineGatewayNode or DefineWorkerNode command to assign a new host name and port number to a node.
5. Reset the database connections for all gateway nodes if you restored the domain configuration to another database. All gateway nodes must have a valid connection to the domain configuration database.

Migrating the Domain Configuration

You can migrate the domain configuration to another database user account. You may need to migrate the domain configuration if you no longer support the existing database user account. For example, if your company requires all departments to migrate to a new database type, you must migrate the domain configuration.

1. Shut down all application services in the domain.
2. Shut down the domain.
3. Back up the domain configuration.
4. Create the database user account where you want to restore the domain configuration.

5. Restore the domain configuration backup to the database user account.
6. Update the database connection for each gateway node.
7. Start all nodes in the domain.
8. Enable all application services in the domain.

Important: Summary tables are lost when you restore the domain configuration.

Step 1. Disable All Application Services

You must disable all application services to disable all service processes. If you do not disable an application service and a user starts a service process while you are backing up and restoring the domain, the service process changes may be lost and data may become corrupt.

Disable application services in complete mode to ensure that you do not abort running service processes.

Disable the application services in the following order:

1. Analyst Service
2. Catalog Service
3. Cluster Service
4. Data Integration Service
5. Model Repository Service

Step 2. Shut Down the Domain

You must shut down the domain to ensure that users do not modify the domain while you are migrating the domain configuration. For example, if the domain is running when you are backing up the domain configuration, users can create new services and objects. Also, if you do not shut down the domain and you restore the domain configuration to a different database, the domain becomes inoperative. The connections between the gateway nodes and the domain configuration database become invalid. The gateway nodes shut down because they cannot connect to the domain configuration database. A domain is inoperative if it has no running gateway node.

Step 3. Back Up the Domain Configuration

Run the *infasetup* BackupDomain command to back up the domain configuration to a binary file.

Step 4. Create a Database User Account

Create a database user account if you want to restore the domain configuration to a new database user account.

Step 5. Restore the Domain Configuration

Run the *infasetup* RestoreDomain command to restore the domain configuration to a database. The RestoreDomain command restores the domain configuration in the backup file to the specified database user account.

Step 6. Update the Database Connection

If you restore the domain configuration to a different database user account, you must update the database connection information for each gateway node in the domain. Gateway nodes must have a connection to the domain configuration database to retrieve and update domain configuration.

Step 7. Start All Nodes in the Domain

Start all nodes in the domain. You must start the nodes to enable services to run.

1. Shut down the gateway node that you want to update.
2. Run the *infasetup* UpdateGatewayNode command to update the gateway node.
3. Start the gateway node.
4. Repeat this process for each gateway node.

Step 8. Enable All Application Services

Enable all application services that you previously shut down. Application services must be enabled to run service processes.

Updating the Domain Configuration Database Connection

All gateway nodes must have a connection to the domain configuration database to retrieve and update domain configuration. When you create a gateway node or configure a node to serve as a gateway, you specify the database connection, including the database user name and password. If you migrate the domain configuration to a different database or change the database user name or password, you must update the database connection for each gateway node. For example, as part of a security policy, your company may require you to change the password for the domain configuration database every three months.

To update the node with the new database connection information, complete the following steps:

1. Shut down the gateway node.
2. Run the *infasetup* UpdateGatewayNode command.

If you change the user or password, you must update the node.

To update the node after you change the user or password, complete the following steps:

1. Shut down the gateway node.
2. Run the *infasetup* UpdateGatewayNode command.

If you change the host name or port number, you must redefine the node.

To redefine the node after you change the host name or port number, complete the following steps:

1. Shut down the gateway node.
2. In the Administrator tool, remove the node association.
3. Run the *infasetup* DefineGatewayNode command.

Shutting Down a Domain

To run administrative tasks on a domain, you might need to shut down the domain. For example, to back up and restore a domain configuration, you must first shut down the domain.

When you shut down a domain, the Service Manager on the master gateway node stops all application services and Informatica services in the domain. Any service processes running on nodes in the domain are aborted. To avoid possible loss of data or metadata and allow running processes to complete, you can shut down each node from the Administrator tool or from the operating system.

Before you shut down a domain, verify that all processes, including workflows, have completed and no users are logged in to repositories in the domain.

1. Click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. In the Navigator, select the domain.
4. Click **Manage tab Actions** > **Shut Down Domain** .

   The **Shut Down Domain** dialog box lists the processes that are running in the domain.
5. Click **Shut down**.

   The **Shutdown Down Domain** dialog box displays a warning message.
6. Click **Shut down**.

   The Service Manager on the master gateway node shuts down the application services and Informatica services on each node in the domain.
7. To restart the domain, restart Informatica services on the gateway and worker nodes in the domain.

Domain Properties

On the **Manage** tab, you can configure domain properties including database properties, gateway configuration, and service levels.

To view and edit properties, click the **Manage** tab. In the Navigator, select a domain. Then click the **Properties** view in the contents panel. The contents panel shows the properties for the domain.

You can configure the properties to change the domain. You cannot change the database properties in the Admin Console. You need to change these properties using the Command UpdateGatewayNode. You can change SMTP properties for alerts, and the domain resiliency properties.

You can configure the following domain properties:

- General properties. Edit general properties, such as service resilience and dispatch mode.
- Database properties. View the database properties, such as database name and database host.
- Gateway configuration properties. Configure a node to serve as gateway and specify the location to write log events.
- Service level management. Create and configure service levels.
- SMTP configuration. Edit the SMTP settings for the outgoing mail server to enable alerts.
- Custom properties. Edit custom properties that are unique to the Informatica environment or that apply in special cases. When you create a domain, it has no custom properties. Use custom properties only at the request of Informatica Global Customer Support.

General Properties

In the General Properties area, you can configure general properties for the domain.

To edit general properties, click **Edit**.

The following table describes the properties that you can edit in the General Properties area:

Property	Description
Name	Read-only. The name of the domain.
Resilience Timeout	<need info>
Limit on Resilience Timeout	<need info>
Restart Period	The maximum amount of time in seconds that the domain spends trying to restart an application service process. Valid values are from 0 to 1000000.
Maximum Restart Attempts within Restart Period	The number of times that the domain tries to restart an application service process. Valid values are from 0 to 1000. If you set the value as 0, the domain does not try to restart the service process.
Dispatch Mode	<need info>
Enable Secure Communication	Configures services to use the TLS protocol to transfer data securely within the domain. When you enable secure communication for the domain, services use secure connections to communicate with other Informatica application services and clients. Verify that all domain nodes are available before you enable secure communication for the domain. If a node is not available, the secure communication changes cannot be applied to the Service Manager of the node. To apply changes, restart the domain. Set this property to True or False.
Service Resilience Timeout	The maximum number of seconds that application clients and application services can try to connect to the Data Integration Service or to the Model Repository Service. The default is 180 seconds.

Database Properties

In the Database Properties area, you can view the database properties for the domain, such as database name and database host. You cannot edit these properties in the Admin Console. You need to update these properties using the command UpdateGatewayNode.

The following table describes the Database properties :

Property	Description
Database Type	The type of database that stores the domain configuration metadata.
Database Host	The name of the machine hosting the database.
Database Port	The port number used by the database.
Database Name	The name of the database.
Database User	The user account for the database containing the domain configuration information.
Database TLS enabled	Indicates whether the database for the domain configuration repository is a secure database. True if the domain configuration repository database is secure. You can use a secure domain configuration repository if secure communication is enabled for the Informatica domain.

Note: The service manager uses the DataDirect drivers included with the Informatica installation. Informatica does not support the use of any other database driver.

Gateway Configuration Properties

In the Gateway Configuration Properties area, you can configure a node to serve as gateway for a domain and specify the directory where the Service Manager on this node writes the log event files.

If you edit gateway configuration properties, previous logs do not appear. Also, the changed properties apply to restart and failover scenarios only.

To edit gateway configuration properties, click **Edit**.

To sort gateway configuration properties, click the header of the column by which you want to sort.

The following table describes the properties that you can edit in the Gateway Configuration Properties area:

Property	Description
Node Name	Read-only. The name of the node.
Status	The status of the node.
Gateway	To configure the node as a gateway node, select this option. If the domain uses a secure domain configuration database, you must specify the truststore file and password for the database. To configure the node as a worker node, clear this option.
Log Directory Path	The directory path for the log event files. If the Log Manager cannot write to the directory path, it writes log events to the node.log file on the master gateway node.

Secure Domain Configuration Repository

If you configure a node as a gateway node and the domain uses a secure domain configuration database, you must specify the truststore file and password for the secure database.

If you configure multiple gateway nodes for the domain, set the database truststore file and password for all gateway nodes.

The following table describes the database truststore properties:

Property	Description
Database Truststore Password	Password for the truststore file.
Database Truststore Location	Path and file name of the truststore file for the secure database.

Note: To use a secure domain configuration repository database, the secure communication option must be enabled for the domain.

Service Level Management

In the Service Level Management area, you can view, add, and edit service levels.

Service levels set priorities among tasks that are waiting to be dispatched. When multiple tasks are in the dispatch queue, the Load Balancer uses service levels to determine the order in which to dispatch tasks from the queue.

Because service levels are domain properties, you can use the same service levels for all repositories in a domain. You create and edit service levels in the domain properties or by using infacmd.

You can edit but you cannot delete the Default service level, which has a dispatch priority of 5 and a maximum dispatch wait time of 1800 seconds.

To add a service level, click **Add**.

To edit a service level, click the link for the service level.

To delete a service level, select the service level and click the Delete button.

The following table describes the properties that you can edit in the Service Level Management area:

Property	Description
Name	The name of the service level. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with the @ character. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : / ? . < > | ! ( )] [After you add a service level, you cannot change its name.
Dispatch Priority	A number that sets the dispatch priority for the service level. The Load Balancer dispatches high priority tasks before low priority tasks. Dispatch priority 1 is the highest priority. Valid values are from 1 to 10. Default is 5.
Maximum Dispatch Wait Time (seconds)	The amount of time in seconds that the Load Balancer waits before it changes the dispatch priority for a task to the highest priority. Setting this property ensures that no task waits forever in the dispatch queue. Valid values are from 1 to 86400. Default is 1800.

SMTP Configuration

Use the SMTP Configuration properties to configure SMTP settings for the domain. The outgoing mail server uses the SMTP settings to send alerts and scorecard notifications.

The following table describes the properties that you can edit in the SMTP Configuration area:

Property	Description
Host Name	The SMTP outbound mail server host name. For example, enter the Microsoft Exchange Server for Microsoft Outlook.
Port	Port used by the outgoing mail server. Valid values are from 1 to 65535. Default is 25.
User name	The user name for authentication upon sending if required by the outbound mail server.

Property	Description
Password	The user password for authentication upon sending if required by the outbound mail server.
Sender Email Address	The email address that the Service Manager uses in the From field when sending notification emails. If you leave this field blank, the Service Manager uses `Administrator@<host name>` as the sender.

Custom Properties for the Domain

Configure custom properties that are unique to specific environments.

You might need to apply custom properties in special cases. When you define a custom property, enter the property name and an initial value. Define custom properties only at the request of Informatica Global Customer Support.

CHAPTER 6

Nodes

This chapter includes the following topics:

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Nodes Overview

A node is the logical representation of a machine in the domain. When you configure a domain with multiple nodes, you can scale service processing across multiple machines. The Service Manager runs on all nodes in the domain to support the domain and application services. If the Service Manager is not running, the node is not available.

An installation on multiple machines consists of a master gateway node, which hosts the domain, and additional gateway nodes and worker nodes that run Informatica application services. The node type determines whether the node can serve as a gateway or a worker node and determines the domain functions that the node performs. You define the node type when you install Informatica services and join the node to the domain. You can use the Administrator tool to change the node type after installation.

By default, each node in the domain can run application services and computational processes. The node role determines whether a node can run application services, computational processes, or both. If the node has the service role, you can view the application service processes running on the node. Before you remove or shut down a node, verify that all running processes are stopped. You might need to shut down the node if you need to perform maintenance on the machine or to ensure that domain configuration changes take effect.

Use the Manage tab of the Administrator tool to manage nodes, including configuring node properties, updating a node role, and removing nodes from a domain. The properties that you can configure depend on the node role.

If your license includes grid, you can configure the Data Integration Service to run on a grid. A grid is an alias assigned to a group of nodes. When you run jobs on a grid of nodes, you improve scalability and performance by distributing jobs to processes running on multiple nodes in the grid.

Node Types

The node type determines whether the node can serve as a gateway or worker node and determines the domain functions that the node performs.

You define the node type when you install Informatica services and join the node to the domain. You can use the Administrator tool to change the node type after installation. You change the node type in the gateway configuration properties for the domain.

Gateway Nodes

A gateway node is any node that you configure to serve as a gateway for the domain. A gateway node can run application services and perform computations, and it can serve as a master gateway node. One gateway node acts as the master gateway at any given time. The master gateway node is the entry point to the domain.

The Service Manager on the master gateway node performs all domain functions on the master gateway node. The Service Managers running on other gateway nodes perform limited domain functions on those nodes.

You can configure more than one node to serve as a gateway. If the master gateway node becomes unavailable, the Service Managers on other gateway nodes elect another master gateway node. If you configure only one node to serve as the gateway and the node becomes unavailable, the domain cannot accept service requests.

Worker Nodes

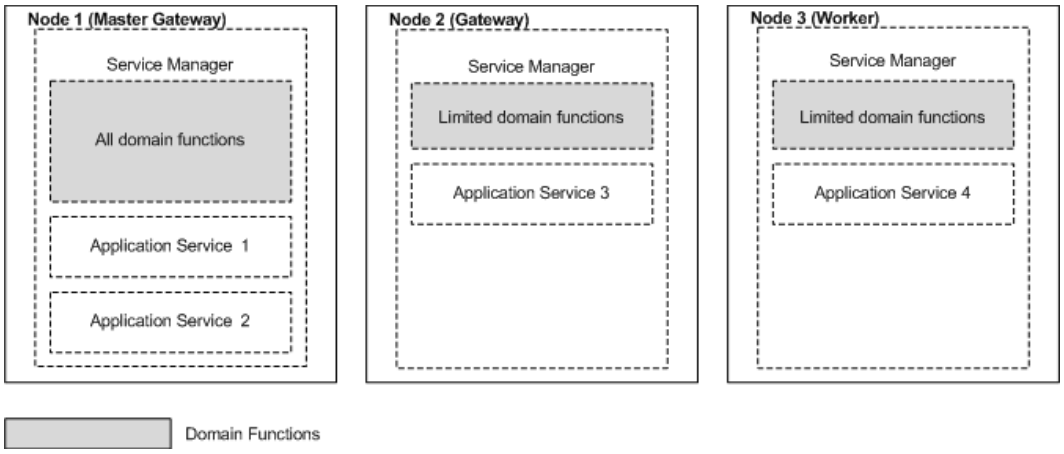
A worker node is any node that you do not configure to serve as a gateway for the domain. A worker node can run application services and perform computations, but it cannot serve as a gateway. The Service Manager performs limited domain functions on a worker node.

Example Domain with Multiple Nodes

This example domain contains three nodes. Each node has both the service and compute roles enabled.

All nodes run the Service Manager. Node 1 is the master gateway node and runs two application services. Node 2 is a back-up gateway node and runs one application service. Node 3 is a worker node and runs one application service. If Node 1 becomes unavailable, Node 2 is elected as the new master gateway node. The Service Manager on Node 2 then performs all domain functions. When Node 1 restarts, it becomes a back-up gateway node and the Service Manager performs limited domain functions.

The following image shows a domain with two gateway nodes and one worker node:



Node Roles

The node role defines the purpose of the node. A node with the service role can run application services. A node with the compute role can perform computations requested by remote application services. A node with both roles can run application services and locally perform computations for those services.

By default, each gateway and worker node has both the service and compute roles enabled. Each node must have at least one role enabled.

You can configure a Data Integration Service grid so that some nodes are dedicated to running application service processes while other nodes are dedicated to performing computations. When you enable only the compute role on a node in a Data Integration Service grid, the node does not have to run the service process. The machine uses all available processing power to run mappings. You can add additional nodes with only the compute role to the grid to increase scalability of Data Integration Service mappings.

For more information about setting up a Data Integration Service grid, see the *Informatica Application Service Guide*.

Service Role

A node with the service role can run application services.

When you enable the service role on a node, the Service Manager supports application services configured to run on that node.

A node requires the service role in the following situations:

- The node is a gateway node.
- The node is configured as a primary or back-up node for an application service.
- The node is assigned to a Data Integration Service grid and a service process is running on the node.

Compute Role

A node with the compute role can perform computations requested by remote application services.

When a node has the compute role, the Service Manager manages the containers on the node. A container is an allocation of memory and CPU resources. An application service uses the container to remotely perform computations on the node. For example, a Data Integration Service grid includes Node 1 with the service role and Node 2 with the compute role. The Data Integration Service process that runs on Node 1 runs a mapping within a container on Node 2.

A node requires the compute role when the Data Integration Service runs jobs on the node. When the Data Integration Service runs on a single node, the node must have both the service and compute roles. When the Data Integration Service runs on a grid, at least one of the nodes in the grid must have the compute role.

A node does not require the compute role when the Data Integration Service does not run jobs on the node. In this case, you can disable the compute role on the node. However, because the container management function of the Service Manager is a lightweight process, enabling or disabling the compute role does not have a performance impact.

When you disable the compute role on a node, you must specify whether to stop, complete, or abort computations that might be running on the node.

Updating the Node Role

By default, each node has both the service and compute roles. If a node is assigned to a Data Integration Service grid that is configured to run jobs in separate remote processes, you might want to update the node role.

Enable only the service role to dedicate the node to running the Data Integration Service process. Enable only the compute role to dedicate the node to running Data Integration Service mappings.

Note: Before you can disable the service role on a node, you must shut down all application service processes running on the node and remove the node as a primary or back-up node for any application service. You cannot disable the service role on a gateway node.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a node assigned to a Data Integration Service grid.
3. In the Properties view, click **Edit** for the general properties.

   The **Edit General Properties** dialog box appears.
4. Select or clear the service and compute roles to update the node role.
5. Click **OK**.
6. If you disabled the compute role, the **Disable Compute Role** dialog box appears. Perform the following steps:
   a. Select one of the following modes to disable the compute role:
      - Complete. Allows jobs to run to completion before disabling the role.
      - Stop. Stops all jobs and then disables the role.
      - Abort. Tries to stop all jobs before aborting them and disabling the role.
   b. Click **OK**.
7. If you updated the role on a node assigned to a Data Integration Service or a Data Integration Service grid, recycle the Data Integration Service.

Viewing Processes on a Node with the Service Role

You can view the status of all application service processes configured to run on a node with the service role. Before you shut down or remove a node, you can view the status of each application service process to determine which service processes you need to disable.

When a node does not have the service role, no application service processes run on the node.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a node with the service role.
3. In the contents panel, select the **Processes** view.

   The view displays the status of each application service process configured to run on the node.

Define and Add Nodes

To create a node, you define the node as a gateway or worker node and then add the node to the domain.

Use either of the following programs to define a node:

Informatica installer

Run the installer on each machine you want to define as a node.

infasetup command line program

Run the infasetup DefineGatewayNode or infasetup DefineWorkerNode command on each machine that you want to define as a node. You might use infasetup to define a node if you decide to move a node from one domain to another domain.

When the Informatica installer or infasetup defines a node, the program creates `nodemeta.xml`. This file is the node configuration file for the node. A gateway node uses information in `nodemeta.xml` to connect to the domain configuration database. A worker node uses the information in `nodemeta.xml` to connect to the domain. The file is stored in the following directory on each node:

```
<Informatica installation directory>/isp/config
```

When you define a node using the Informatica installer, the installer adds the node to the domain with both the service and compute roles enabled. When you log in to the Administrator tool, the node appears in the Navigator.

When you define a node with infasetup, you must manually add the node to the domain. You can add a node to the domain in the Administrator tool or with the infacmd isp AddDomainNode command. When you add the node, you specify the roles to enable on the node.

You can use the Administrator tool to add a node to the domain before you define the node. In this case, the Administrator tool displays a message saying that you need to run the Informatica installer to associate the node with a physical host name and port number. The name that you enter for the node must be the same name that you use when you define the node.

Adding Nodes to the Domain

You can use the Administrator tool to add a node to the domain.

Use the Administrator tool to add a node to the domain in the following situations:

- After you run the infasetup DefineGatewayNode or infasetup DefineWorkerNode command.

- When you decide to add the node before running the Informatica installer or infasetup command line program to define the node.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the folder where you want to add the node. If you do not want the node to appear in a folder, select the domain.
3. On the Navigator Actions menu, click **New** > **Node**.

   The **Create Node** dialog box appears.
4. Enter the node name.

   The name must be the same node name that you use when you define the node.
5. If you want to change the folder for the node, click **Browse** and choose a new folder or the domain.
6. Optionally update the node role.

   By default, each node has both the service and compute roles. If a node is assigned to a Data Integration Service grid, you might want to update the node role to dedicate the node to running the Data Integration Service process or to running mappings.
7. Click **OK**.

   If you add a node to the domain before you define the node using the Informatica installer or infasetup, the Administrator tool displays a message saying that you need to run the installer to associate the node with a physical host name and port number.

Configuring Node Properties

You configure node properties on the Properties view for the node. You can configure properties such as the node roles, error severity level, and minimum and maximum port numbers.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a node.
3. Click the **Properties** view.

   The Properties view displays the node properties in separate sections.
4. In the **Properties** view, click **Edit** for the section that contains the property you want to set.
5. Edit the following properties:

Node Property	Description
Name	Name of the node. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the node. The description cannot exceed 765 characters.
Host Name	Host name of the machine represented by the node.

Node Property	Description
Port	Port number used by the node.
Gateway Node	Indicates whether the node can serve as a gateway. If this property is disabled, then the node is a worker node.
Service Role	Indicates whether the node has the service role. If enabled, application services can run on the node. If disabled, application services cannot run on the node. Disable the property only if the node is assigned to a Data Integration Service grid and you want to dedicate the node to running mappings. Default is enabled.
Compute Role	Indicates whether the node has the compute role. If enabled, the node can perform computations. If disabled, the node cannot perform computations. A node requires the compute role when the Data Integration Service runs jobs on the node. If the Data Integration Service does not run jobs on the node, you can disable the compute role. However, enabling or disabling the compute role does not have a performance impact. Default is enabled.
Backup Directory	Directory to store repository backup files. The directory must be accessible by the node.
Error Severity Level	Level of error logging for the node. These messages are written to the Log Manager application service and Service Manager log files. Set one of the following message levels: - **ERROR**. Writes ERROR code messages to the log. - **WARNING**. Writes WARNING and ERROR code messages to the log. - **INFO**. Writes INFO, WARNING, and ERROR code messages to the log. - **TRACING**. Writes TRACE, INFO, WARNING, and ERROR code messages to the log. - **DEBUG**. Writes DEBUG, TRACE, INFO, WARNING, and ERROR code messages to the log. Default is **WARNING**.
Minimum Port Number	Minimum port number used by service processes on the node. To apply changes, restart Informatica services. The default value is the value entered when the node was defined.
Maximum Port Number	Maximum port number used by service processes on the node. To apply changes, restart Informatica services. The default value is the value entered when the node was defined.
CPU Profile Benchmark	Ranks the node's CPU performance against a baseline system. For example, if the CPU is running 1.5 times as fast as the baseline machine, the value of this property is 1.5. You can calculate the benchmark by clicking **Actions** > **Recalculate CPU Profile Benchmark**. The calculation takes approximately five minutes and uses 100% of one CPU on the machine. Or, you can update the value manually. Default is 1.0. Minimum is 0.001. Maximum is 1,000,000. Used in adaptive dispatch mode. Ignored in round-robin and metric-based dispatch modes.
Maximum CPU Run Queue Length	Maximum number of runnable threads waiting for CPU resources on the node. Set this threshold to a low number to preserve computing resources for other applications. Set this threshold to a high value, such as 200, to cause the Load Balancer to ignore it. Default is 10. Minimum is 0. Maximum is 1,000,000,000. Used in metric-based and adaptive dispatch modes. Ignored in round-robin dispatch mode.

Node Property	Description
Maximum Memory %	Maximum percentage of virtual memory allocated on the node relative to the total physical memory size. Set this threshold to a value greater than 100% to allow the allocation of virtual memory to exceed the physical memory size when dispatching tasks. Set this threshold to a high value, such as 1,000, if you want the Load Balancer to ignore it. Default is 150. Minimum is 0. Maximum is 1,000,000,000. Used in metric-based and adaptive dispatch modes. Ignored in round-robin dispatch mode.
Log Collection Directory	The directory that stores the logs for the application service when you run the log aggregator. The directory must be accessible from all the nodes in the domain. If the log collection directory is not accessible by other nodes, the aggregated logs do not appear in the aggregated logs listgrid. The users who run node processes must have read-write permissions on the directory. Configure the log collection directory for the master gateway node in the domain.
Core Dump Directory	The directory that stores the core dump files for the domain processes when you run the log aggregator. Configure the core dump directory for all the nodes in the domain.

6. Click **OK**.

Shutting Down and Restarting the Node

Some administrative tasks might require you to shut down a node. For example, you might need to perform maintenance or benchmarking on a machine. You might also need to shut down and restart a node for some configuration changes to take effect. For example, if you change the shared directory for the Log Manager or domain, you must shut down the node and restart it to update the configuration files.

You can shut down a node from the Administrator tool or from the operating system. When you shut down a node, you stop Informatica services and abort all application service processes and computations running on the node.

To restart a node, start Informatica services on the node.

Warning: To avoid loss of data or metadata when you shut down a node, disable all running application service processes in complete mode.

Shutting Down a Node from the Administrator Tool

When you shut down a node from the Administrator tool, you can view all application service processes running on the node.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a node.
3. In the Navigator **Actions** menu, select **Shutdown Node**.

   If the node has the service role, the Administrator tool displays the list of application service processes running on that node.
4. Optionally, choose whether the shutdown is planned or unplanned.

5. Optionally, enter comments about the shutdown.
6. Click **OK** to stop all service processes and shut down the node, or click **Cancel** to cancel the operation.

Starting or Stopping a Node on Windows

On Windows, use the Control Panel to start and stop the Informatica service.

1. Open the Windows Control Panel.
2. Select **Administrative Tools**.
3. Right-click **Services** and select **Run as Administrator**.
4. Right-click the Informatica service.
5. If the service is running, click **Stop**.

   If the service is stopped, click **Start**.

Starting or Stopping a Node on UNIX

On UNIX, run infaservice.sh to start and stop the Informatica daemon. By default, infaservice.sh is installed in the following directory:

```
<InformaticaInstallationDir>/tomcat/bin
```

1. Go to the directory where infaservice.sh is located.
2. At the command prompt, enter the following command to start the daemon:

   ```
   infaservice.sh startup
   ```

   Enter the following command to stop the daemon:

   ```
   infaservice.sh shutdown
   ```

   Note: If you use a softlink to specify the location of infaservice.sh, set the INFA_HOME environment variable to the location of the Informatica installation directory.

Removing the Node Association

You can remove the host name and port number associated with a node. When you remove the node association, the node remains in the domain, but it is not associated with a host machine.

To associate a different host machine with the node, you must run the installation program or infasetup DefineGatewayNode or infasetup DefineWorkerNode command on the new host machine, and then restart the node on the new host machine.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Navigator, select a node.
3. In the **Services and Nodes** view **Actions** menu, select **Remove Node Association**.

Removing a Node

When you remove a node from a domain, it is no longer visible in the Navigator. If the node is running when you remove it, the node shuts down and aborts all application service processes.

Note: To avoid loss of data or metadata when you remove a node, disable all running application service processes in complete mode.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select a node.
3. In the Navigator **Actions** menu, select **Delete**.
4. In the warning message that appears, click **OK**.

CHAPTER 7

Connections

This chapter includes the following topics:

- Connection Management Overview, 95
- Connection Management, 95
- Pass-through Security, 98
- Pooling Properties in Connection Objects, 100

Connection Management Overview

A connection is a repository object that defines a connection in the domain configuration repository.

The Data Integration Service uses database connections to process jobs for the Developer tool and the Analyst tool. Jobs include mappings, data profiles, scorecards, and SQL data services.

You can create and manage connections in the Administrator tool, the Developer tool, and the Analyst tool.

The tasks that you can perform in each tool depend on the tool that you use. For example, you can create an SAP NetWeaver connection in the Developer tool and manage it in the Administrator tool, but you cannot create or manage it in the Analyst tool.

Connection Management

After you create a connection, you can view the connection, configure connection properties, and delete the connection.

After you create a connection, you can perform the following actions on the connection:

Configure connection pooling.

Configure connection pooling to optimize processing for the Data Integration Service. Connection pooling is a framework to cache connections.

View connection properties.

View the connection properties through the **Connections** view on the **Manage** tab.

Edit the connection.

You can change the connection name and the description. You can also edit connection details such as the user name, password, and connection strings. When you update a database connection that has connection pooling disabled, all updates take effect immediately.

The Data Integration Service identifies connections by the connection ID instead of the connection name. When you rename a connection, the Developer tool and the Analyst tool update the jobs that use the connection.

Deployed applications and parameter files identify a connection by name, not by connection ID. Therefore, when you rename a connection, you must redeploy all applications that use the connection. You must also update all parameter files that use the connection parameter.

Delete the connection.

When you delete a connection, objects that use the connection are no longer valid. If you accidentally delete a connection, you can re-create it by creating another connection with the same connection ID as the deleted connection.

Refresh the connections list.

You can refresh the connections list to see the latest list of connections for the domain. Refresh the connections list after a user adds, deletes, or renames a connection in the Developer tool or the Analyst tool.

Creating a Connection

In the Administrator tool, you can create relational database, social media, and file systems connections.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.
3. In the Navigator, select the domain.
4. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
5. In the **New Connection** dialog box, select the connection type, and then click **OK**.

   The **New Connection** wizard appears.
6. Enter the connection properties.

   The connection properties that you enter depend on the connection type. Click **Next** to go to the next page of the **New Connection** wizard.
7. When you finish entering connection properties, you can click **Test Connection** to test the connection.
8. Click **Finish**.

Refreshing the Connections List

Refresh the connections list to see the latest list of connections in the domain.

The Administrator tool displays the latest list of connections when you start the Administrator tool. You might want to refresh the connections list when a user adds, deletes, or renames a connection in the Developer tool or the Analyst tool.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.

The Navigator shows all connections in the domain.

3. In the Navigator, select the domain.
4. Click **Actions** > **Refresh**.

Viewing a Connection

View connections in the Administrator tool.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.

   The Navigator shows all connections in the domain.
3. In the Navigator, select the domain.

   The contents panel shows all connections for the domain.
4. To filter the connections that appear in the contents panel, enter filter criteria and click the Filter button.

   The contents panel shows the connections that meet the filter criteria.
5. To remove the filter criteria, click the Reset Filters button.

   The contents panel shows all connections in the domain.
6. To sort the connections, click in the header for the column by which you want to sort the connections.

   By default, connections are sorted by name.
7. To add or remove columns from the contents panel, right-click a column header.

   If you have Read permission on the connection, you can view the data in the **Created By** column. Otherwise, this column is empty.
8. To view the connection details, select a connection in the Navigator.

   The contents panel shows the connection details.

Configuring Pooling for a Connection

Configure pooling for a connection in the Administrator tool.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.
3. In the Domain Navigator, select a connection.

   The contents panel shows the connection properties.
4. In the contents panel, click the **Pooling** view.
5. In the **Pooling Properties** area, click **Edit**.

   The **Edit Pooling Properties** dialog box appears.
6. Edit the pooling properties and click **OK**.

Editing and Testing a Connection

In the Administrator tool, you can edit connections that you created in the Administrator tool, the Analyst tool, the Developer tool, or by running the infacmd isp CreateConnection command. You can test relational database connections.

1. In the Administrator tool, click the **Manage** tab.

2. Click the **Connections** view.

   The Navigator shows all connections in the domain.
3. In the Navigator, select a connection.

   The contents panel shows properties for the connection.
4. In the contents panel, select the **Properties** view or the **Pooling** view.
5. To edit properties in a section, click **Edit**.

   Edit the properties and click **OK**.

   Note: If you change a connection name, you must redeploy all applications that use the connection. You must also update all parameter files that use the connection parameter.
6. To test a database connection, select the connection in the Navigator.

   Click **Actions** > **Test Connection** on the **Manage** tab.

   Note: The **Test Connection** button tests the connection string of the metadata access properties and not the data access properties.

   A message box displays the result of the test.

Deleting a Connection

You can delete a database connection in the Administrator tool.

When you delete a connection in the Administrator tool, you also delete it from the Developer tool and the Analyst tool.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.

   The Navigator shows all connections in the domain.
3. In the Navigator, select a connection.
4. In the Navigator, click **Actions** > **Delete**.

Pass-through Security

Pass-through security is the capability to connect to an SQL data service or an external source with the client user credentials instead of the credentials from a connection object.

Users might have access to different sets of data based on the job in the organization. Client systems restrict access to databases by the user name and the password. When you create an SQL data service, you might combine data from different systems to create one view of the data. However, when you define the connection to the SQL data service, the connection has one user name and password.

If you configure pass-through security, you can restrict users from some of the data in an SQL data service based on their user name. When a user connects to the SQL data service, the Data Integration Service ignores the user name and the password in the connection object. The user connects with the client user name or the LDAP user name.

A web service operation mapping might need to use a connection object to access data. If you configure pass-through security and the web service uses WS-Security, the web service operation mapping connects to a source using the user name and password provided in the web service SOAP request.

Configure pass-through security for a connection in the connection properties of the Administrator tool or with infacmd dis UpdateServiceOptions. You can set pass-through security for connections to deployed applications. You cannot set pass-through security in the Developer tool. Only SQL data services and web services recognize the pass-through security configuration.

Example

An organization combines employee data from multiple databases to present a single view of employee data in an SQL data service. The SQL data service contains data from the Employee and Compensation databases. The Employee database contains name, address, and department information. The Compensation database contains salary and stock option information.

A user might have access to the Employee database but not the Compensation database. When the user runs a query against the SQL data service, the Data Integration Service replaces the credentials in each database connection with the user name and the user password. The query fails if the user includes salary information from the Compensation database.

Pass-Through Security with Data Object Caching

To use data object caching with pass-through security, you must enable caching in the pass-through security properties for the Data Integration Service.

When you deploy an SQL data service or a web service, you can choose to cache the logical data objects in a database. You must specify the database in which to store the data object cache. The Data Integration Service validates the user credentials for access to the cache database. If a user can connect to the cache database, the user has access to all tables in the cache. The Data Integration Service does not validate user credentials against the source databases when caching is enabled.

For example, you configure caching for the EmployeeSQLDS SQL data service and enable pass-through security for connections. The Data Integration Service caches tables from the Compensation and the Employee databases. A user might not have access to the Compensation database. However, if the user has access to the cache database, the user can select compensation data in an SQL query.

When you configure pass-through security, the default is to disallow data object caching for data objects that depend on pass-through connections. When you enable data object caching with pass-through security, verify that you do not allow unauthorized users access to some of the data in the cache. When you enable caching for pass-through security connections, you enable data object caching for all pass-through security connections.

Adding Pass-Through Security

Enable pass-through security for a connection in the connection properties. Enable data object caching for pass-through security connections in the pass-through security properties of the Data Integration Service.

1. Select a connection.
2. Click the **Properties** view.
3. Edit the connection properties.

   The **Edit Connection Properties** dialog box appears.
4. To choose pass-through security for the connection, select the **Pass-through Security Enabled** option.
5. Optionally, select the Data Integration Service for which you want to enable object caching for pass-through security.
6. Click the **Properties** view.
7. Edit the pass-through security options.

The **Edit Pass-through Security Properties** dialog box appears.

8. Select **Allow Caching** to allow data object caching for the SQL data service or web service. This applies to all connections.
9. Click **OK**.

You must recycle the Data Integration Service to enable caching for the connections.

Pooling Properties in Connection Objects

You can edit connection pooling properties in the **Pooling** view for a database connection.

The number of connection pool libraries depends on the number of running Data Integration Service processes or DTM processes. Each Data Integration Service process or DTM process maintains its own connection pool library. The values of the pooling properties are for each connection pool library.

For example, if you set maximum connections to 15, then each connection pool library can have a maximum of 15 idle connections in the pool. If the Data Integration Service runs jobs in separate local processes and three DTM processes are running, then you can have a maximum of 45 idle connection instances.

To decrease the total number of idle connection instances, set the minimum number of connections to 0 and decrease the maximum idle time for each database connection.

The following list describes database connection pooling properties that you can edit in the **Pooling** view for a database connection:

Enable Connection Pooling

Enables connection pooling. When you enable connection pooling, each connection pool retains idle connection instances in memory. To delete the pools of idle connections, you must restart the Data Integration Service.

If connection pooling is disabled, the DTM process or the Data Integration Service process stops all pooling activity. The DTM process or the Data Integration Service process creates a connection instance each time it processes a job. It drops the instance when it finishes processing the job.

Default is enabled for DB2 for i5/OS, DB2 for z/OS, IBM DB2, Microsoft SQL Server, Oracle, and ODBC connections. Default is disabled for Adabas, IMS, Sequential, and VSAM connections.

Minimum # of Connections

The minimum number of idle connection instances that a pool maintains for a database connection after the maximum idle time is met. Set this value to be equal to or less than the maximum number of idle connection instances. Default is 0.

Maximum # of Connections

The maximum number of idle connection instances that a pool maintains for a database connection before the maximum idle time is met. Set this value to be more than the minimum number of idle connection instances. Default is 15.

Maximum Idle Time

The number of seconds that a connection instance that exceeds the minimum number of connection instances can remain idle before the connection pool drops it. The connection pool ignores the idle time when the connection instance does not exceed the minimum number of idle connection instances. Default is 120.

CHAPTER 8

Schedules

This chapter includes the following topics:

- Schedules Overview, 101
- Create and Edit Schedules, 101
- Pausing and Resuming a Schedule, 104
- Removing Jobs from a Schedule, 105
- Deleting a Schedule, 105

Schedules Overview

You can create a schedule to indicate when to run deployed mappings and workflows. You can also schedule the date and time to run profiles and scorecards.

When you create a schedule, you configure when the schedule runs. You can add jobs to the schedule when you create it or you can save the schedule and add jobs to it later.

You might define a schedule to automate routine tasks or manage resource usage in the domain. For example, you can schedule large mappings to run at different times to prevent the mappings from overloading the nodes. You can also pause the schedule to prevent the jobs from running during holidays or during domain maintenance.

Create and Edit Schedules

Before you schedule a job, create a schedule that defines when the job runs. Jobs can run one time or at an interval. You can schedule jobs when you create the schedule or you can add jobs to the schedule later.

You can select a time zone when you configure the date and time to run the schedule. By default, when you enter a start time for a schedule, the time zone is the time zone of your client machine. If the Data Integration Service runs on a machine in a different time zone from the client, you need to ensure that the job runs at a specific time for that time zone. You can choose the time zone when you define the schedule.

You can schedule deployed mappings and workflows including mappings and workflows for profiles and scorecards. Optionally, you can configure a parameter file or parameter set for the job. Configure a parameter file or set when you want to run the deployed mapping or workflow with different settings. If the Scheduler Service runs on multiple nodes, you must store the parameter file in a directory that all of the nodes can access. Configure the directory in the Storage Properties for the Scheduler Service.

Optionally, you can configure the **Run As User** property for the job. You might configure the property if you need to schedule an object that you do not have permission on. When you configure the **Run As User** property, you run the job as a user in the domain who has permission on the object.

After you create a schedule, you can edit the schedule. When you edit a schedule, you can change the recurrence options, schedule jobs, or change the job parameters.

Creating a Schedule

Create a schedule to run deployed mappings, deployed workflows, profiles, or scorecards at a specified time.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Schedules** view.
3. Click **Actions** > **Create Schedule**.

   The **Properties** dialog box appears.
4. Enter a name and an optional description for the schedule.
5. In the **Recurrence** section, choose Run Once to run the schedule one time, or choose Recurring Schedule to run the schedule on an interval.
6. If you are creating a recurring schedule, specify whether to run the job on a Daily, Weekly, or Monthly interval.

   The following table describes the recurrence options that you can configure:

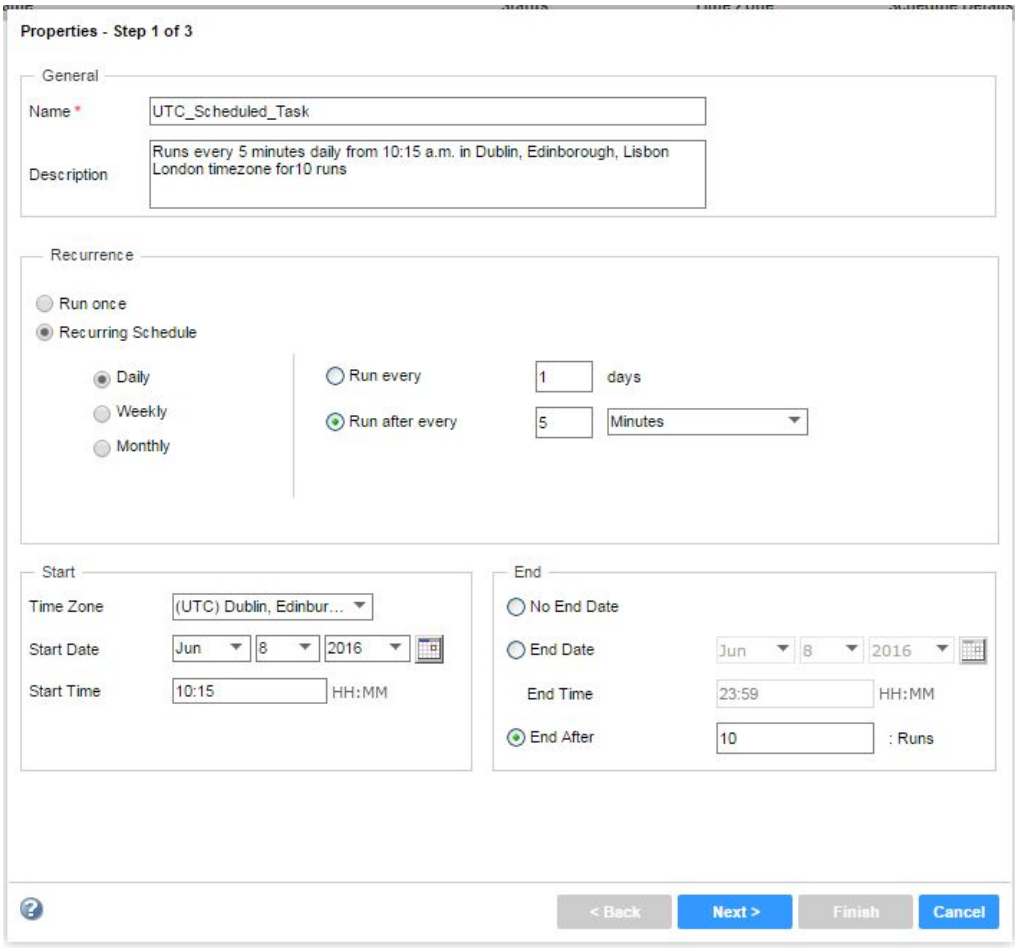
Option	Description
Daily - Run every	Run the job every n days
Daily - Run after every	Run the job every n minutes or hours
Weekly	Run the job on a day or days of the week
Monthly - Run every n day	Run the job on a day of the month
Monthly - Run every	Run the job on a day even n week of the month
Monthly - Run every last day of the month	Run the job on the last day of the month

7. In the **Start** section, configure a start date and time for the schedule.
8. Select the time zone for the schedule from the list.

   The default time zone is the time zone of the client machine.
9. If you are creating a recurring schedule, you can optionally configure an end date and time.

   You can end the recurrence on a specific date or after n runs.

The following image shows the Scheduler Wizard:



10. Click **Next**.

    The **Applications** dialog box appears.

11. Expand a Data Integration Service and select jobs to run on the schedule.
12. Optionally, upload a parameter file to define parameters for the mapping or workflow.
    a. Select a mapping or a workflow.

       The **Application Properties** section appears.

    b. Click **Upload**.

       The **Upload Parameter File** dialog box appears.

    c. Click **Choose Files**.
    d. Browse for a file, and then click **Open**.
    e. Click **OK**.
13. Optionally, select a parameter set.
14. Optionally, configure the **Run As User** property to run the job as a different user.
    a. Click **Change**.

       The **Run As** dialog box appears.

    b. Expand the user list.

c. Select a user.

d. Enter the password for the user.

e. Click **OK**.

15. Click **Next**.

    The **Review** window appears and lists the schedule properties.

16. Click **Finish** to create the schedule.

    The schedule appears in the list of schedules in the Navigator.

Editing a Schedule

You can edit a schedule in the Administrator tool. You can edit a schedule to change recurrence options, schedule jobs, or run jobs with different parameters.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Schedules** view.

   The Domain Navigator lists the schedules in the domain.
3. In the Domain Navigator, select a schedule.

   The contents panel shows the schedule properties.
4. To edit the schedule, click **Actions** > **Edit Schedule**.

Pausing and Resuming a Schedule

You can pause and resume a schedule from the Administrator tool. For example, you might want to pause a schedule when you perform maintenance on the Scheduler Service.

When you pause a schedule, the jobs that run on the schedule stop running until you resume the schedule.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Schedules** view.

   The **All Schedules** view lists the schedules in the domain.
3. Select a schedule. Use the **Ctrl** and **Shift** keys to select multiple schedules.
4. To pause the schedule, click **Pause**.

   The **Pause Confirmation** message appears.
5. Click **OK.**

   The **Schedule Status Change** message appears.
6. Click **OK**.

   The schedule status changes to Paused.
7. To resume the schedule, select the schedule, and then click **Resume**.

   The **Resume Confirmation** message appears.
8. Click **OK**.

   The **Schedule Status Change** message appears.

9. Click **OK**.

   The schedule status changes to Scheduled.

Removing Jobs from a Schedule

You can remove a deployed mapping, profile, scorecard, or workflow job from a schedule.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Schedules** view.
3. Click the **Scheduled Jobs** view.

   A list of scheduled jobs appears in the contents panel.
4. Select the job that you would like to remove from a schedule.
5. Click **Actions** > **Remove Schedule Association**.

   The **Remove Job Confirmation** message displays.
6. Click **OK**.

   The job is removed from the schedule and the **Scheduled Jobs** view.

Deleting a Schedule

You can delete a schedule in the Administrator tool.

When you delete a schedule, any jobs that are running finish running, and all future runs are canceled. If you do not want the jobs to stop running, add them to a different schedule before you delete the schedule.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Schedules** view.

   The **All Schedules** view lists all of the schedules in the domain.
3. In the **All Schedules** view, select a schedule. Use the **Ctrl** and **Shift** keys to select multiple schedules.
4. Click **Actions** > **Delete Schedule**.

   The **Delete Confirmation** dialog box appears.
5. Click **OK** to delete the schedule.

CHAPTER 9

Domain Object Export and Import

This chapter includes the following topics:

- Domain Object Export and Import Overview, 106
- Export Process, 106
- View Domain Objects, 107
- Import Process, 114

Domain Object Export and Import Overview

You can use the command line to migrate objects between two different domains of the same version.

You might migrate domain objects from a development environment to a test or production environment.

To export and import domain objects, use the following infacmd isp commands:

ExportDomainObjects

Exports native users, native groups, roles, and connections to an XML file.

ImportDomainObjects

Imports native users, native groups, roles, and connections into an Informatica domain.

You can use an infacmd control file to filter the objects during the export or import.

You can also use the infacmd xrf generateReadableViewXML command to generate a readable XML file from an export file. You can review the readable XML file to determine if you need to filter the objects that you import.

Export Process

You can use the command line to export domain objects from a domain.

Perform the following tasks to export domain objects:

1. Determine the domain objects that you want to export.
2. If you do not want to export all domain objects, create an export control file to filter the objects that are exported.
3. Run the infacmd isp exportDomainObjects command to export the domain objects.

The command exports the domain objects to an export file. You can use this file to import the objects into another domain.

Rules and Guidelines for Exporting Domain Objects

Review the following rules and guidelines before you export domain objects:

- When you export a user, by default, you do not export the user password. If you do not export the password, the administrator must reset the password for the user after the user is imported into the domain. However, when you run the infacmd isp exportDomainObjects command, you can choose to export an encrypted version of the password.
- When you export a user, you do not export the associated groups of the user. If applicable, assign the user to the group after you import the user and group.
- When you export a group, you export all sub-groups and users in the group.
- You cannot export the Administrator user, the Administrator role, the Everyone group, or LDAP users or groups. To replicate LDAP users and groups in an Informatica domain, import the LDAP users and groups directly from the LDAP directory service.
- To export native users and groups from domains of different versions, use the infacmd isp exportUsersAndGroups command.
- When you export a connection, by default, you do not export the connection password. If you do not export the password, the administrator must reset the password for the connection after the connection is imported into the domain. However, when you run the infacmd isp exportDomainObjects command, you can choose to export an encrypted version of the password.

View Domain Objects

You can view domain object names and properties in the export XML file.

Run infacmd xrf generateReadableViewXML command, to create a readable XML from the export file.

The following section provides a sample readable XML file:

```
<global:View xmlns:global="http://global" xmlns:connection="http://connection"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="
  http://connection connection.xsd http://global globalSchemaDomain.xsd http://global
  globalSchema.xsd">
  <NativeUser isAdmin="false" name="admin" securityDomain="Native" viewId="0">
    <UserInfo email="" fullName="admin" phone="" viewId="1"/>
  </NativeUser>
  <User isAdmin="false" name="User1" securityDomain="Native" viewId="15">
    <UserInfo email="" fullName="NewUSer" phone="" viewId="16"/>
  </User>
  <Group name="TestGroup1" securityDomain="Native" viewId="182">
    <UserRef name="User1" securityDomain="Native" viewId="183"/>
    <UserRef name="User6" securityDomain="Native" viewId="188"/>
  </Group>
  <Role customRole="false" name="Administrator" viewId="242">
    <Description viewId="243">Provides all privilege and permission access to an
Informatica service.</Description>
    <ServicePrivilegeDefinition name="PwxListenerService" viewId="244">
      <Privilege category="" isEnabled="true" name="close" viewId="245"/>
      <Privilege category="" isEnabled="true" name="closeforce" viewId="246"/>
      <Privilege category="" isEnabled="false" name="Management Commands" viewId="249"/>
      <Privilege category="" isEnabled="false" name="Informational Commands"
viewId="250"/>
    </ServicePrivilegeDefinition>
  </Role>
```

```
  <Connection connectionString="inqa85sql25@qa90"
connectionType="SQLServerNativeConnection"
    domainName="" environmentSQL="" name="conn4" ownerName=""
    schemaName="" transactionSQL="" userName="dummy" viewId="7512">
    <ConnectionPool maxIdleTime="120" minConnections="0" usePool="true" viewId="7514"/>
  </Connection>
 </global:View>
```

Viewable Domain Object Names

You can view the following domain object names and properties in the readable XML file:

User

The following table lists the properties and the type:

Property	Type
name	string
securityDomain	string
admin	boolean
UserInfo	List<UserInfo>

UserInfo

The following table lists the properties and the type:

Property	Type
description	string
email	string
fullName	string
phone	string

Role

The following table lists the properties and the type:

Property	Type
name	string
description	string
customRole	boolean
servicePrivilege	List<ServicePrivilegeDef>

ServicePrivilegeDef

The following table lists the properties and the type:

Property	Type
name	string
privileges	List<Privilege>

Privilege

The following table lists the properties and the type:

Property	Type
name	string
enable	boolean
category	string

Group

The following table lists the properties and the type:

Property	Type
name	string
securityDomain	string
description	string
UserRefs	List<UserRef>

GroupRef

The following table lists the properties and the type:

Property	Type
name	string
securityDomain	string

UserRef

- name
- securityDomain

ConnectInfo

The following table lists the properties and the type:

Property	Type
id	string
name	string
connectionType	string
ConnectionPoolAttributes	List<ConnectionPoolAttributes>

ConnectionPoolAttributes

The following table lists the properties and the type:

Property	Type
maxIdleTime	int
minConnections	int
poolSize	int
usePool	boolean

Supported Connection Types

- DB2iNativeConnection
- DB2NativeConnection
- DB2zNativeConnection
- JDBCConnection
- ODBCNativeConnection
- OracleNativeConnection
- PWXMetaConnection
- SAPConnection
- SDKConnection
- SQLServerNativeConnection
- SybaseNativeConnection
- TeradataNativeConnection
- URLLocation
- WebServiceConnection
- NRDBMetaConnection
- NRDBNativeConnection

- RelationalBaseSDKConnection

DB2iNativeConnection Properties

- connectionType
- connectionString
- username
- environmentSQL
- libraryList
- location
- databaseFileOverrides

DB2NativeConnection Properties

- connectionType
- connectionString
- username
- environmentSQL
- tableSpace
- transactionSQL

DB2zNativeConnection Properties

- connectionType
- connectionString
- username
- environmentSQL
- location

JDBCConnection Properties

- connectionType
- connectionString
- username
- dataStoreType

ODBCNativeConnection Properties

- connectionType
- connectionString
- username
- environmentSQL
- transactionSQL
- odbcProvider

OracleNativeConnection Properties

- connectionType
- connectionString
- username

- environmentSQL
- transactionSQL

PWXMetaConnection Properties

- connectionType
- databaseName
- userName
- dataStoreType
- dbType
- hostName
- location
- port

SAPConnection Properties

- connectionType
- userName
- description
- dataStoreType

SDKConnection Properties

- connectionType
- sdkConnectionType
- dataSourceType

SQLServerNativeConnection Properties

- connectionType
- connectionString
- username
- environmentSQL
- transactionSQL
- domainName
- ownerName
- schemaName

TeradataNativeConnection Properties

- connectionType
- username
- environmentSQL
- transactionSQL
- dataSourceName
- databaseName

TeradataNativeConnection Properties

- connectionType

- username
- environmentSQL
- transactionSQL
- connectionString

URLLocation Properties

- connectionType
- locatorURL

WebServiceConnection Properties

- connectionType
- url
- userName
- wsseType
- httpAuthenticationType

NRDBNativeConnection Properties

- connectionType
- userName
- location

NRDBMetaConnection Properties

- connectionType
- username
- location
- dataStoreType
- hostName
- port
- databaseType
- databaseName
- extensions

RelationalBaseSDKConnection Properties

- connectionType
- databaseName
- connectionString
- domainName
- environmentSQL
- hostName
- owner
- ispSvcName
- metadataDataStorageType
- metadataConnectionString

- metadataConnectionUserName

Import Process

You can use the command line to import domain objects from an export file into a domain.

Perform the following tasks to import domain objects:

1. Run the infacmd xrf generateReadableViewXML command to generate a readable XML file from an export file. Review the domain objects in the readable XML file and determine the objects that you want to import.
2. If you do not want to import all domain objects in the export file, create an import control file to filter the objects that are imported.
3. Run the infacmd isp importDomainObjects command to import the domain objects into the specified domain.
4. After you import the objects, you may still have to create other domain objects such as application services and folders.

Rules and Guidelines for Importing Domain Objects

Review the following rules and guidelines before you import domain objects:

- When you import a group, you import all sub-groups and users in the group.
- To import native users and groups from domains of different versions, use the infacmd isp importUsersAndGroups command.
- After you import a user or group, you cannot rename the user or group.
- You import roles independently of users and groups. Assign roles to users and groups after you import the roles, users, and groups.
- You cannot import the Administrator group, the Administrator user, the Administrator role, the Everyone group, or LDAP users or groups.

Conflict Resolution

A conflict occurs when you try to import an object with a name that exists for an object in the target domain. Configure the conflict resolution to determine how to handle conflicts during the import.

You can define a conflict resolution strategy through the command line or control file when you import the objects. The control file takes precedence if you define conflict resolution in the command line and control file. The import fails if there is a conflict and you did not define a conflict resolution strategy.

You can configure one of the following conflict resolution strategies:

Reuse

Reuses the object in the target domain.

Rename

Renames the source object. You can provide a name in the control file, or else the name is generated. A generated name has a number appended to the end of the name.

Replace

Replaces the target object with the source object.

Merge

Merges the source and target objects into one group. For example, if you merge groups with the same name, users and sub-groups from both groups are merged into the group in the target domain.

You cannot define the merge conflict resolution strategy through the command line. Use a control file to define the merge conflict resolution strategy. You must include the group object type section with merge as the conflict resolution policy with reuse, replace, or rename for all conflicting users in the control file.

For example, specify the merge conflict resolution strategy for the following groups:

- Group A with users a1, a2, b1, b2 in the source domain.
- Group A with users a1, a2, a3 b1, b2 in the target domain

You get the following results in the group after merge in the target domain:

- a1, a2, b1, b2 if you choose reuse or replace
- a1, a2, a3, b1, b2 if you choose rename.

CHAPTER 10

License Management

This chapter includes the following topics:

- License Management Overview, 116
- Types of License Keys, 118
- Creating a License Object, 119
- Assigning a License to a Service, 120
- Unassigning a License from a Service, 120
- Updating a License, 121
- Removing a License, 121
- Viewing License Details, 122

License Management Overview

The Service Manager on the master gateway node manages Informatica licenses.

A license enables you to perform the following tasks:

- Run application services, such as the Analyst Service and Data Integration Service.
- Use add-on options, such as grid and high availability.
- Access particular types of connections, such as Oracle, Teradata, Microsoft SQL Server, and IBM MQ Series.
- Use Metadata Exchange options, such as Metadata Exchange for Cognos and Metadata Exchange for Rational Rose.

When you install Informatica, the installation program creates a license object in the domain based on the license key that you used during installation.

You assign a license object to each application service to enable the service.

You can create additional license objects in the domain. Based on your project requirements, you may need multiple license objects. For example, you may have two license objects, where each license object allows you to run services on a different operating system. You might also use multiple license objects to manage multiple projects in the same domain. One project may require access to particular database types, while the other project does not.

License Validation

The Service Manager validates application service processes when they start. The Service Manager validates the following information for each service process:

- Product version. Verifies that you are running the appropriate version of the Informatica services.
- Platform. Verifies that the Informatica services are running on a licensed operating system.
- Expiration date. Verifies that the license is not expired. If the license expires, no application service assigned to the license can start. You must assign a valid license to the Informatica services to start them.
- Connectivity. Verifies connections that the Informatica services have permission to use.
- Metadata Exchange options. Determines the Metadata Exchange options that are available for use. For example, the Service Manager verifies that you have access to the Metadata Exchange for Business Objects Designer.

Licensing Log Events

The Service Manager generates log events and writes them to the Log Manager. It generates log events for the following actions:

- You create or delete a license.
- You apply an incremental license key to a license.
- You assign an application service to a license.
- You unassign a license from an application service.
- The license expires.
- The Service Manager encounters an error, such as a validation error.

The log events include the user name and the time associated with the event.

You must have permission on the domain to view the logs for Licensing events.

The Licensing events appear in the domain logs.

License Management Tasks

You can perform the following tasks to manage the licenses:

- Create the license in the Administrator tool. You use a license key to create a license in the Administrator tool.
- Assign a license to each application service. Assign a license to each application service to enable the service.
- Unassign a license from an application service. Unassign a license from an application service if you want to discontinue the service or migrate the service from a development environment to a production environment. After you unassign a license from a service, you cannot enable the service until you assign another valid license to it.
- Update the license.
- Remove the license. Remove a license if it is obsolete.
- Configure user permissions on a license.
- View license details. You may need to review the licenses to determine details, such as expiration date and the maximum number of licensed CPUs. You may want to review these details to ensure you are in compliance with the license. Use the Administrator tool to determine the details for each license.

- Monitor license usage and licensed options. You can monitor the usage of logical CPUs. You can monitor the number of software options purchased for a license and the number of times a license exceeds usage limits in the License Management Report.

You can perform all of these tasks in the Administrator tool or by using *infacmd isp* commands.

Types of License Keys

Informatica provides license keys in license files. The license key is encrypted. When you create the license from the license key file, the Service Manager decrypts the license key and enables the purchased options.

You create a license from a license key file. You apply license keys to the license to enable additional options. Informatica uses the following types of license keys:

- Original keys. Informatica generates an original key based on your contract. Informatica may provide multiple original keys depending on your contract.
- Incremental keys. Informatica generates incremental keys based on updates to an existing license, such as an extended license period or an additional option.

Note: Informatica licenses typically change with each version. Use a license key file valid for the current version to ensure that your installation includes all functionality.

Original Keys

Original keys identify the contract, product, and licensed features. Licensed features include the Informatica edition, deployment type, number of authorized CPUs, and authorized Informatica options and connectivity. You use the original keys to install Informatica and create licenses for services. You must have a license key to install Informatica. The installation program creates a license object for the domain in the Administrator tool. You can use other original keys to create more licenses in the same domain. You use a different original license key for each license object.

Incremental Keys

You use incremental license keys to update an existing license. You add an incremental key to an existing license to add or remove options. For example, if an existing license does not allow high availability, you can add an incremental key with the high availability option to the existing license.

The Service Manager updates the license expiration date if the expiration date of an incremental key is later than the expiration date of an original key. The Service Manager uses the latest expiration date. A license object can have different expiration dates for options in the license. For example, the IBM DB2 relational connectivity option may expire on 12/01/2006, and the session on grid option may expire on 04/01/06.

The Service Manager validates the incremental key against the original key used to create the license. An error appears if the keys are not compatible.

Creating a License Object

You can create a license object in a domain and assign the license to application services. You can create the license in the Administrator tool using a license key file. The license key file contains an encrypted original key. You use the original key to create the license.

You can also use the *infacmd isp* AddLicense command to add a license to the domain.

Use the following guidelines to create a license:

- Use a valid license key file. The license key file must contain an original license key. The license key file must not be expired.
- You cannot use the same license key file for multiple licenses. Each license must have a unique original key.
- Enter a unique name for each license. You create a name for the license when you create the license. The name must be unique among all objects in the domain.
- Put the license key file in a location that is accessible by the Administrator tool computer. When you create the license object, you must specify the location of the license key file.

After you create the license, you can change the description. To change the description of a license, select the license in Navigator of the Administrator tool, and then click Edit.

1. In the Administrator tool, click **Actions** > **New** > **License**.

   The **Create License** window appears.

2. Enter the following options:

Option	Description
Name	Name of the license. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the license. The description cannot exceed 765 characters.
Path	Path of the domain in which you create the license. Read-only field. Optionally, click **Browse** and select a domain in the **Select Folder** window. Optionally, click **Create Folder** to create a folder for the domain.
License File	File containing the original key. Click **Browse** to locate the file.

   If you try to create a license using an incremental key, a message appears that states you cannot apply an incremental key before you add an original key.

   You must use an original key to create a license.

3. Click **Create**.

Assigning a License to a Service

Assign a license to an application service before you can enable the service. When you assign a license to a service, the Service Manager updates the license metadata. You can also use the *infacmd isp* AssignLicense command to assign a license to a service.

1. Select the license in the **Domain Navigator** of the Administrator tool.
2. Click the **Assigned Services** tab.
3. In the **License** tab, click **Actions** > **Edit Assigned Services.**

   The **Assign or Unassign this license to the services** window appears.
4. Select the services under **Unassigned Services**, and click Add.

   Use Ctrl-click to select multiple services. Use Shift-click to select a range of services. Optionally, click **Add all** to assign all services.
5. Click **OK**.

Rules and Guidelines for Assigning a License to a Service

Use the following rules and guidelines when you assign licenses:

- You can assign licenses to disabled services.
- If you want to assign a license to a service that has a license assigned to it, you must first unassign the existing license from the service.
- To start a service with backup nodes, you must assign it to a license with high availability.
- To restart a service automatically, you must assign the service to a license with high availability.

Unassigning a License from a Service

You might need to unassign a license from a service if the service becomes obsolete or if you want to discontinue a service. You might want to discontinue a service if you are using more CPUs than you are licensed to use.

You can use the Administrator tool or the *infacmd isp* UnassignLicense command to unassign a license from a service.

You must disable a service before you can unassign a license from it. After you unassign the license from the service, you cannot enable the service. You must assign a valid license to the service to reenable it.

You must disable the service before you can unassign the license. If you try to unassign a license from an enabled service, a message appears that states you cannot remove the service because it is running.

1. Select the license in the **Domain Navigator** of the Administrator tool.
2. Click the **Assigned Services** tab.
3. In the **License** tab, click **Actions** > **Edit Assigned Services.**

   The **Assign or Unassign this license to the services** window apears.
4. Select the service under **Assigned Service**s, and then click **Remove**. Optionally, click **Remove all** to unassign all assigned services.
5. Click **OK**.

Updating a License

You can update the current license in the Informatica domain with an incremental license key..

When you add an incremental key to a license, the Service Manager adds or removes licensed options and updates the license expiration date.

You can also use the infacmd isp `UpdateLicense` command to add an incremental key to a license.

After you update the license you must restart Informatica services for the changes to take effect.

Use the following guidelines to update a license:

- Verify that the license key file is accessible by the Administrator tool computer. When you update the license object, you must specify the location of the license key file.
- The incremental key must be compatible with the original key. An error appears if the keys are not compatible.

The Service Manager validates the license key against the original key based on the following information:

- Serial number
- Deployment type
- Distributor
- Informatica edition
- Informatica version

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. Select a **license** in the Domain Navigator.
3. Click **Manage Actions** > **Add Incremental Key**.

   The **Update License** window appears.
4. Click **Choose Files** to browse for a license key file.
5. Click **OK**.
6. In the **License Details** section of the **Properties** view, click **Edit** to edit the description of the license.
7. Click **OK**.

Removing a License

You can remove a license from a domain using the Administrator tool or the *infacmd isp* RemoveLicense command.

Before you remove a license, disable all services assigned to the license. If you do not disable the services, all running service processes abort when you remove the license. When you remove a license, the Service Manager unassigns the license from each assigned service and removes the license from the domain. To re-enable a service, assign another license to it.

If you remove a license, you can still view License Usage logs in the Log Viewer for this license, but you cannot run the License Report on this license.

To remove a license from the domain:

1. Select the license in the **Domain Navigator** of the Administrator tool.

2. Click **Actions** > **Delete.**

Viewing License Details

You can view license details using the Administrator tool or the *infacmd isp* ShowLicense command.

The license details are based on all license keys applied to the license. The Service Manager updates the existing license details when you add a new incremental key to the license.

You might review license details to determine options that are available for use. You may also review the license details and license usage logs when monitoring licenses.

For example, you can determine the number of CPUs your company is licensed to use for each operating system.

To view license details, select the license in the **Domain Navigator**.

The Administrator tool displays the license properties in the following sections:

- License Details. View license details on the **Properties** tab. Shows license attributes, such as the license object name, description, and expiration date.
- Supported Platforms. View supported platforms on the **Properties** tab. Shows the operating systems and how many CPUs are supported for each operating system.
- Repositories. View the licensed repositories on the **Properties** tab. Shows the maximum number of licensed repositories.
- Assigned Services. View application services that are assigned to the license on the **Assigned Services** tab.
- Connections. View the licensed connections on the **Options** tab. Shows all licensed connections. The license enables you to use connections, such as DB2 and Oracle database connections.

You can also run the License Management Report to monitor licenses.

License Details

You can use the license details to view high-level information about the license. Use this license information when you audit the licensing usage.

The general properties for the license appear in the **License Details** section of the **Properties** tab.

The following table describes the general properties for a license:

Property	Description
Name	Name of the license.
Description	Description of the license.
Location	Path to the license in the Navigator.
License Version	Version of license.
Distributed By	Distributor of the product.

Property	Description
Issued On	Date when the license was issued to the customer.
Expires On	Date when the license expires.
Validity Period	Period for which the license is valid.
Serial Number	Serial number of the license. The serial number identifies the customer or project. The original and incremental keys for a license have the same serial number.
Deployment Level	Level of deployment. Values are "Development" and "Production."

You can also use the license event logs to view audit summary reports. You must have permission on the domain to view the logs for license events.

Supported Platforms

You assign a license to each service. The service can run on any operating system supported by the license. One product license can support multiple operating system platforms.

The supported platforms for the license appear in the Supported Platforms section of the **Properties** tab.

The following table describes the supported platform properties for a license:

Property	Description
Description	Name of the supported operating system.
Logical CPUs	Number of CPUs you can run on the operating system.
Issued On	Date on which the license was issued.
Expires	Date on which the license expires.

Repositories

The maximum number of active repositories for the license appear in the Repositories section of the Properties tab.

The following table describes the repository properties for a license:

Property	Description
Description	Name of the repository.
Instances	Number of repository instances running on the operating system.
Issued On	Date on which the license was issued for this option.
Expires	Date on which the license expires for this option.

Connections

The license enables you to use connections such as DB2 and Oracle database connections.

The connections for the license appear in the Connections section of the **Options** tab.

Metadata Exchange Options

The license enables you to use Metadata Exchange options such as Metadata Exchange for Business Objects Designer and Metadata Exchange for Microstrategy.

The Metadata Exchange options for the license appear in the Metadata Exchange Options section of the **Options** tab.

CHAPTER 11

Monitoring

This chapter includes the following topics:

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Monitoring Overview

In Informatica Administrator, you can monitor statistics for Data Integration Service jobs on the **Monitor** tab.

After you configure the monitoring Model repository at the domain level, you can view monitoring statistics in the Administrator tool. The **Monitor** tab displays current and historical information about multiple Data Integration Service and integration objects. Use the **Summary Statistics** view to view graphical summaries of object state and distribution across multiple Data Integration Service. You can also view graphs of the memory and CPU that the Data Integration Services used to run the objects. Use the **Execution Statistics** view to monitor properties, run-time statistics, and run-time reports for integration objects.

You can monitor the following objects:

- Ad hoc jobs
- Applications
- Logical data objects
- SQL data services
- Web services
- Workflows

Monitoring is a domain function that the Service Manager performs. The Service Manager stores the monitoring configuration in the monitoring Model repository. The Service Manager also persists, updates, retrieves, and publishes run-time statistics for integration objects in the monitoring Model repository.

You can also access monitoring from the following tools:

The Informatica Monitoring tool

The Monitoring tool is a direct link to the **Monitor** tab of the Administrator tool. The Monitoring tool is useful if you do not need access to other Administrator tool features. You must have at least one monitoring privilege to access the Monitoring tool. You can access the Monitoring tool from the following URL:

```
http://<Administrator tool host><Administrator tool port>/monitoring/
```

The Analyst tool

You can monitor objects on the **Job Status** tab in the Analyst tool. The **Job Status** tab shows the status of Analyst tool jobs, such as profile jobs, scorecard jobs, and jobs that load mapping specification results to the target.

The Developer tool

You can open the Monitoring tool from the Developer tool. When you monitor from the Developer tool, you can view jobs that users run from the Developer tool. The Monitoring tool shows the status of Developer tool jobs, such as mapping jobs.

Configure Monitoring

You can configure the domain to display statistics and reports about objects in the domain. Statistics and reports appear in the **History** view on the **Manage** tab and in the **Summary Statistics** and **Execution Statistics** views on the **Monitor** tab. To view statistics and reports, you must configure monitoring for the domain.

When you configure monitoring in the domain, the Data Integration Services store statistics and reports in a Model repository. The statistics include historical information about objects that the Data Integration Services run. The reports show key metrics about integration objects.

If you do not configure monitoring, then some views on the **Manage** and **Monitor** tabs do not have content. The workflow graph is also empty, and notifications disappear when you refresh the page.

To set up monitoring statistics and reports, perform the following tasks:

1. Configure monitoring settings. Configure a Model repository to store run-time statistics for objects that the Data Integration Services run.
2. Configure reports and statistics views. Choose which statistics appear in the **Statistics** and **Reports** views.

Configuring Monitoring Settings

Configure monitoring settings for the domain when you want to view historical information about the domain. When you configure monitoring settings, you specify a Model repository to store run-time statistics about objects that the Data Integration Services run.

Create the Model repository contents before you configure the monitoring settings. If you create the contents after you configure the monitoring settings, then you must recycle the Model Repository Service after the contents are created.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. Click the **Monitoring Configuration** view.

   The current monitoring configuration displays.
3. Click **Edit** to change the monitoring configuration.
4. Edit the following options:

Option	Description
Model Repository Service	Name of the Model repository that stores the historical information. The Model repository must not be integrated with a version control system.
Username	User name to access the Model Repository Service.
Password	Password of the user name to access the Model Repository Service.
Modify Password	Modify the Model Repository Service password.
Security Domain	Name of the security domain that the Model repository user belongs to.
Preserve Summary Historical Data	Number of days that the Model repository saves averaged data. If purging is disabled, then the Model repository saves the data indefinitely. Default is 180. Minimum is 0. Maximum is 366.
Preserve Detailed Historical Data	Number of days that the Model repository saves per-minute data. If purging is disabled, then the Model repository saves the data indefinitely. Default is 14. Minimum is 1. Maximum is 14.
Purge Statistics Every	Interval, in days, at which the Model Repository Service purges data that is older than the values configured in the **Preserve** options. Default is 1 day.
Days At	Time of day when the Model Repository Service purges statistics. Default is 1:00 a.m.
Maximum Number of Sortable Records	Maximum number of records that can be sorted in the **Monitor** tab. If the number of records on the **Monitor** tab is greater than this value, then you can only sort by **Start Time** and **End Time**. Default is 3,000.
Maximum Delay for Update Notifications	Maximum time, in seconds, that the Data Integration Service buffers statistics before it stores them in the Model repository and displays them in the **Monitor** tab. If the Data Integration Service shuts down unexpectedly before it stores the statistics in the Model repository, then the statistics are lost. Default is 10.
Show Milliseconds	Include milliseconds for date and time fields in the **Monitor** tab.

5. Click **OK**.
6. Click **Save**.

To apply the settings, you must restart all of the Data Integration Services.

Step 2. Configure Reports and Statistics Views

By default, the **Statistics** and **Reports** views on the **Execution Statistics** view are empty. To view statistics and reports, you must configure the Report and Statistic settings in the domain. These settings apply to all Data Integration Services in the domain.

Before you configure statistics and reports, you must specify a monitoring Model Repository Service in the Monitoring Configuration tab and enable the monitoring Model Repository Service.

1. In the Administrator tool, click the **Monitor** > **Execution Statistics** tab.
2. Click **Actions** > **Report and Statistic Settings**.
3. In the **Report and Statistic Settings** dialog box, click the **Statistics** tab.
4. Configure the time ranges that you want to use for statistics, and select the frequency at which the statistics assigned to each time range should be updated.
5. In the **Default Time Range** list, select a default time range to appear for all statistics.
6. Click the **Reports** tab.
7. Enable the time ranges that you want to use for reports, and select the frequency at which the reports assigned to each time range should be updated.
8. In the **Default Time Range** list, select a default time range to appear for all reports.
9. Click **Select Reports**.
10. In the **Select Reports** dialog box, add the reports that you want to run to the **Selected Reports** box.
11. Organize the reports in the order in which you want to view them on the **Monitor** tab.
12. Click **OK** to close the **Select Reports** dialog box.
13. Click **OK** in the **Report and Statistic Settings** dialog box.

Optimizing Monitoring Performance

When you configure monitoring in the domain, you configure the monitoring Model repository to store monitoring statistics. You also configure how often the monitoring Model Repository Service purges statistics. You can configure the repository database and purge settings to minimize resource consumption and maximize monitoring performance.

To optimize monitoring performance, enable the **Purge Statistics** option in the Monitoring Configuration. When you enable statistics purging, you configure how often the monitoring Model Repository Service purges data. Use the **Preserve** options in the Monitoring Configuration to configure how long Detailed and Summary data are saved in the monitoring Model repository before they are purged.

For optimum monitoring performance, consider the following guidelines when you configure the domain for monitoring:

- Create the monitoring Model repository to store monitoring data. Specify the monitoring Model Repository Service when you configure monitoring in the domain.
- Configure the monitoring Model Repository Service on the machine where you configure the domain.
- Verify that the monitoring Model repository database is tuned as required for the following properties:
  - Maximum Heap Size property
  - Java Stack Size property
  - Memory settings

- Hibernate Connection Pool Size property

- Purge monitoring data during off-hours to limit the impact on other database operations.
- Purge monitoring data daily.

Summary Statistics

The **Summary Statistics** view shows information about the Data Integration Services and the objects that a Data Integration Service runs.

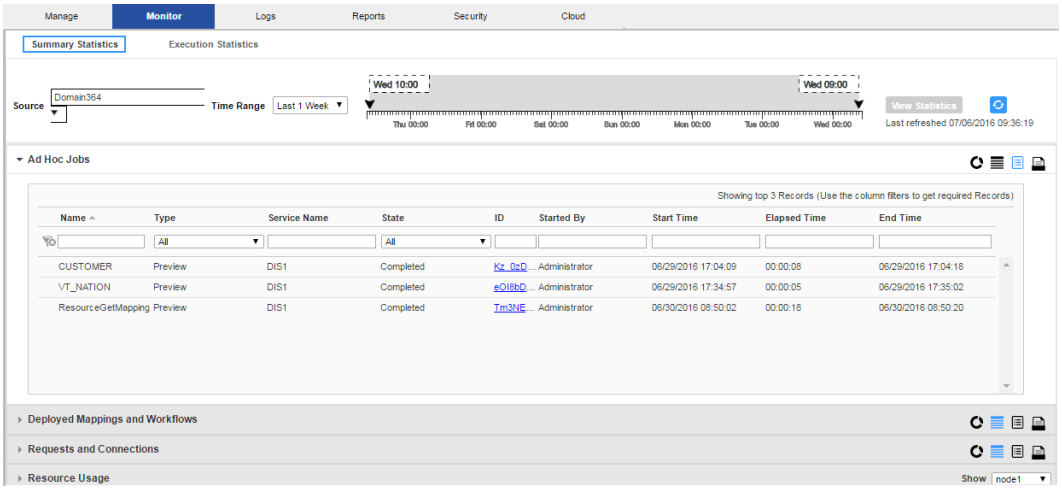
Use the **Summary Statistics** view to view summaries of object distribution, object state, and Data Integration Service resource usage for a specified time range. You can view statistics for the domain, or by Data Integration Service or application.

You can view statistics about the following objects:

- Ad hoc jobs. Jobs that users ran from the Developer tool or the Analyst tool.
- Deployed mappings and workflows. Mappings and workflows that are deployed in an application.
- Requests and connections. Deployed SQL data services and web services.
- Resource Usage. CPU and memory usage of the Data Integration Service processes that are running in the domain or on a node in the domain.

For example, you can view all of the jobs in the domain that failed over the last eight hours. You can review resource usage for that time range to see if the jobs failed due to resource issues on the node.

The following image shows the **Summary Statistics** view with a list of ad hoc jobs in the **Details** panel:



When you select a source and time range you can choose one of the following options to view the data:

- Graphical distribution. Displays doughnut and line charts that summarize object distribution and resource usage for a time range. The doughnut charts show distributions of objects by state, type, and the Data Integration Service that ran them. The line charts compare resource usage for the Data Integration Service that ran the jobs to resource usage for all of the processes that ran on the node during that time range.
- Tabular distribution. View the total completed, running, canceled, aborted, and failed jobs.
- Details. View a list of the jobs, requests, or connections that comprise the summary statistics. You can click the Job ID to view the specific job in the Execution Statistics.

- Export data. Export the detail data for a specific object type to a .csv file.

The **Summary Statistics** view displays statistics using data that is stored in the monitoring Model repository. You must configure a monitoring Model Repository Service in the Monitoring Configuration tab before you can view **Summary Statistics**.

Viewing Summary Statistics

You can view summary and detail information about ad hoc jobs, deployed mappings and workflows, requests and connections, and resource usage for Data Integration Services in the domain.

1. Click the **Monitor** tab.
2. Click the **Summary Statistics** view.

   The time line appears.
3. In the **Source** field, choose the source for which you want to view statistics.
4. In the **Time Range** field, choose the time range for which you want to view statistics.
5. Optionally, choose **Custom** to specify a custom time range.
6. Click **View Statistics**.

   The object and resource usage panels appear.
7. Expand an object type to view statistics for that object type.
8. Select whether to view a graphical distribution, a tabular distribution, or a detail list of the data. Or, choose to export a .csv file.
9. Optionally, you can perform the following actions in the **Resource Usage** panel:

Option	Description
Show	Show all nodes or one node.
Click the magnifying glass	Enlarge the chart.
Drag	Zoom in on a section of the chart.
Reset Zoom	View the chart at the default size.

Monitor Data Integration Services

You can monitor Data Integration Services on the **Execution Statistics** view on the **Monitor** tab.

When you select a Data Integration Service in the Navigator, the contents panel shows the following views:

- **Properties** view
- **Reports** view

Properties View for a Data Integration Service

The **Properties** view shows the general properties and run-time statistics for objects that the selected Data Integration Service ran.

When you select a Data Integration Service in the Navigator, you can view the general properties and run-time statistics.

General Properties for a Data Integration Service

You can view general properties, such as the service name, object type, and description. The Persist Statistics Enabled property indicates whether the Data Integration Service stores persisted statistics in the monitoring Model repository. This option is true when you configure the global settings for the domain.

You can also view information about objects that the Data Integration Service runs. To view information about an object, select the object in the Navigator or contents panel. Depending on the object type, details about the object appear in the contents panel or details panel.

Statistics for a Data Integration Service

You can view run-time statistics about objects that the Data Integration Service runs. Select the object type and time period to display the statistics. You can view statistics about jobs, applications, connections, requests, and workflows. For example, you can view the number of failed, canceled, and completed profiling jobs in the last four hours.

Reports View for a Data Integration Service

The **Reports** view shows reports about objects that the selected Data Integration Service runs.

When you select a Data Integration Service in Navigator of the **Monitor** tab, the **Reports** view shows reports about jobs. For example, you can view the Most Active Users for Jobs report to determine users that ran the most jobs during a specific time period. Click a link in the report to show more details about the objects included in the link. For example, you can click the number of failed deployed mappings to see details about each deployed mapping that failed.

Monitor Ad Hoc Jobs

You can monitor ad hoc jobs on the **Monitor** tab. Ad hoc jobs are jobs that users run from the Developer tool or the Analyst tool.

An ad hoc job is a preview, scorecard, profile, mapping, audit, or reference table process that a user runs from the Developer tool or the Analyst tool. When a user runs a job, a Data Integration Service runs the job process and the job appears in the Monitor tab.

You can run up to five jobs at a time from the Developer tool. All remaining jobs are queued, and do not appear in the Monitor tab until they run.

By default, you can monitor jobs that you run. If you have the appropriate monitoring privilege, you can also view jobs that other users run.

When you select **Ad Hoc Jobs** in the Navigator of the **Execution Statistics** view, a list of jobs appears in the contents panel. The contents panel groups related jobs based on the job type. You can expand a job type to view the related jobs under it. For example, when you run a profile job, the Data Integration Service converts the job into a mapping. The mapping appears under the profile job in the contents panel.

When you select a job in the contents panel, you can view logs for the job, view context for the job, or cancel the job. You can also view properties for the job in the details panel. Depending on the type of job, the details panel might show general properties, mapping properties, or statistics.

When you select Ad Hoc jobs in the Navigator of the Execution Statistics view, a list of jobs appears in the contents panel. The contents panel groups related jobs based on the job type. You can expand a job type to view the related jobs under it.

The following list describes the types of properties and statistics that can appear in the details panel:

Properties

Shows the general properties about the selected job, such as the name, job type, user who ran the job, and start time of the job. If the job ran on a grid, the details panel shows the node that the job ran on.

Mapping Properties

You can view mapping properties when you select a profile or scorecard job in the contents panel. These jobs have an associated mapping. You can view mapping properties such as the request ID, the mapping name, and the log file name. You can also view throughput and resource usage statistics for the associated mappings.

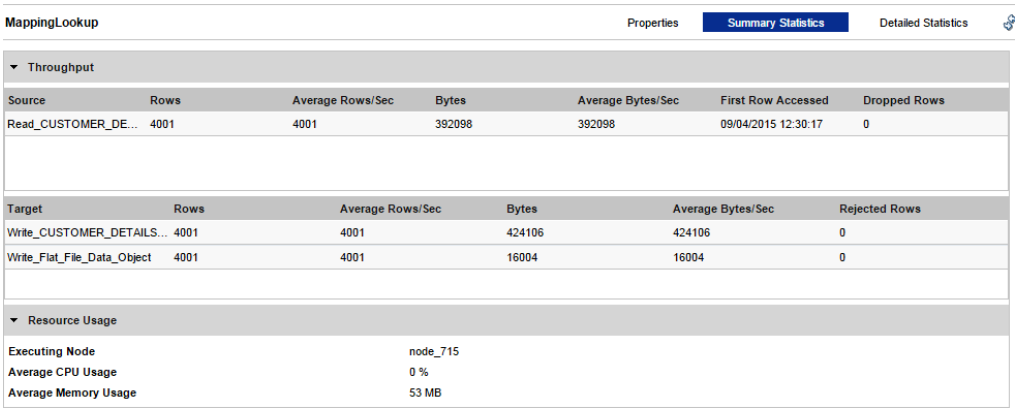
Blaze Execution Plan

You can view a Blaze execution plan when you run a mapping with the Blaze engine in the Hadoop environment. The Blaze execution plan displays the Blaze engine script that the Data Integration Service generates based on the mapping logic, the unique identifier for the script, and the tasks that the script depends on.

Summary Statistics

You can view summary statistics you select an ad hoc mapping job, deployed mapping job, or mapping object in a workflow in the contents panel. The **Summary Statistics** view displays throughput and resource usage statistics for the job run.

The following image shows the **Summary Statistics** view for a mapping job:



You can also view the summary statistics for the source or target on the **Summary Statistics** tab, such as processed rows and bytes.

Note: Only the **number of rows processed** appears in the Summary Statistics for the Hive source and target. The remaining property values appears as 0 or N/A for the Hive source or target.

Detailed Statistics

You can view detailed statistics for an ad hoc mapping job, deployed mapping job, or mapping object in a workflow in the contents panel. Detailed statistics appear for jobs that run longer than one minute in separate local processes. The **Detailed Statistics** view displays graphs of the throughput and resource usage statistics for the job run.

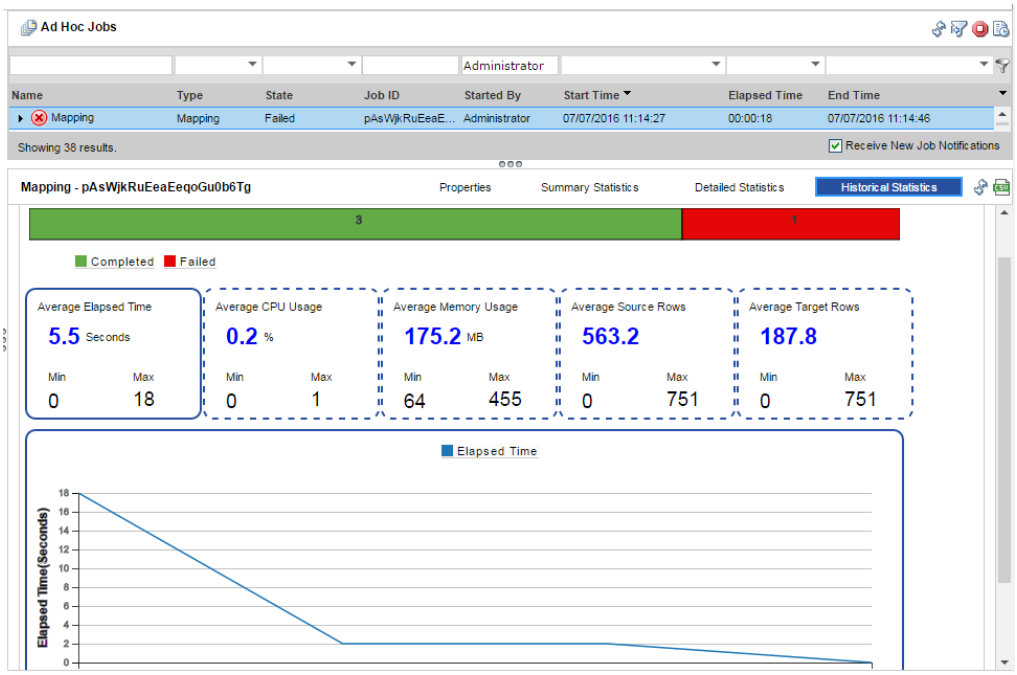
The following image shows the **Detailed Statistics** view for a mapping job in a workflow:



Historical Statistics

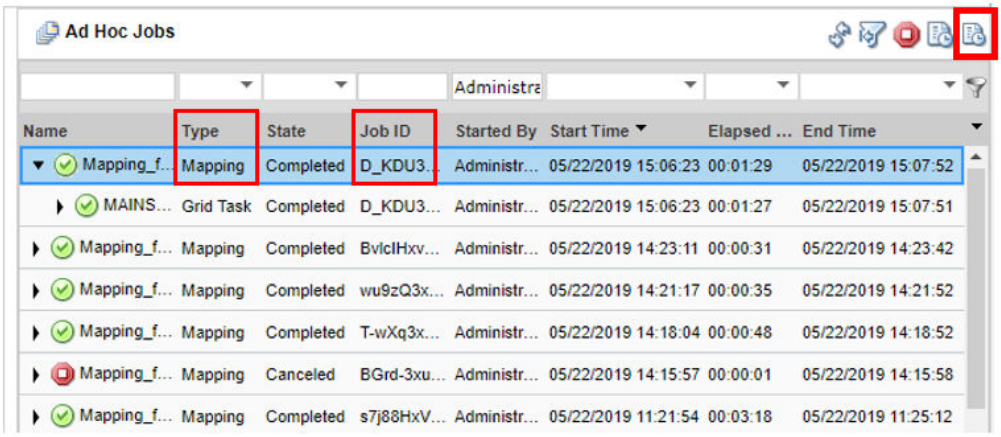
You can view historical statistics when you select an ad hoc mapping job, deployed mapping job, or mapping object in a workflow in the **Contents** panel. The **Historical Statistics** view shows averaged data from the last 500 runs for a specific job. For example, you can view the minimum, maximum, and average duration of the mapping job. You can view the average amount of CPU that the job consumes when it runs. You can choose to view averages for mapping jobs with different states, such as completed, canceled, or aborted. A bar graph shows the number of jobs in each state. Click a link below the bar graph to view statistics for a specific state.

The following image shows the **Historical Statistics** view for a mapping job that completed three times and failed one time:



Aggregated Cluster Logs

You can get aggregated cluster logs for deployed Hadoop mappings, Databricks mappings, auto-deploy cluster jobs, local files, and Spark data preview jobs based on the job ID. You can get a .zip or tar.gz file of the aggregated cluster logs for a job and write the compressed aggregated log file to a target directory. You can also use the infacmd ms `fetchAggregatedClusterLogs` command or use the REST API to collect the aggregated cluster logs for a job.



For more information about the infacmd ms `fetchAggregatedClusterLogs` command, see the *Informatica 10.4.0 Command Reference*.

Viewing Logs for an Ad Hoc Job

You can download the logs for a job to view the job details.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service and select **Ad Hoc Jobs**.
4. In the contents panel, select a job.
5. Click **Actions** > **View Logs for Selected Object**.

   A dialog box appears with the option to open or save the log file.

Canceling an Ad Hoc Job

You can cancel a running job. You may want to cancel a job that hangs or that is taking an excessive amount of time to complete.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service and select **Ad Hoc Jobs**.
4. In the contents panel, select a job.
5. Click **Actions** > **Cancel Selected Object**.

Viewing Summary Statistics for an Ad Hoc Job

You can view throughput and resource usage statistics for ad hoc mapping jobs.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service and select **Ad Hoc Jobs**.

   A list of jobs appears in the contents panel.
4. In the contents panel, select a job.

   The details panel displays the Properties for the job.
5. Click the **Summary Statistics** view in the details panel.

   The **Summary Statistics** view displays throughput and resource usage statistics for the source and target.

Optionally, you can sort the statistics in ascending or descending order. Click a column header to sort the column in ascending order. Click the column header again to sort the column in descending order.

Viewing Detailed Statistics for an Ad Hoc Job

You can view graphs of the throughput and resource usage for ad hoc mapping jobs that run in separate local processes. Detailed statistics appear for jobs that run longer than one minute.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service and select **Ad Hoc Jobs**.

   A list of jobs appears in the contents panel.

4. In the contents panel, select a job.

   The details panel displays the Properties for the job.

5. Click the **Detailed Statistics** view in the details panel.

   The **Detailed Statistics** view displays the throughput graph and resource usage graphs.

Optionally, you can complete the following tasks in the **Detailed Statistics** view:

Task	Description
Enlarge a graph	Move the cursor over a graph, and then click the magnifying glass icon.
Enlarge a section of an enlarged graph	Drag the cursor to select an area to enlarge.
Switch between rows and bytes in the throughput graph	Click the Bytes option or the Rows option.
Choose which statistics are plotted on the throughput graph	In the throughput field, select the sources and targets that you want to view.

Monitor Applications

You can monitor applications on the **Monitor** tab.

When you select an application in the Navigator of the **Execution Statistics** view, the contents panel shows the following views:

- **Properties** view
- **Reports** view

You can expand an application in the Navigator to monitor the application components.

Properties View for an Application

The **Properties** view shows general properties and run-time statistics about each application and the objects in an application. Applications can include deployed mapping jobs, logical data objects, SQL data services, web services, and workflows.

When you select an application in the Navigator of the **Execution Statistics** view, you can view general properties and run-time statistics.

General Properties

You can view general properties, such as the name and description of the application. You can also view additional information about the objects in an application. To view information about an object, select the folder in the Navigator and the object in the contents panel. Details about the object appear in the details panel.

Statistics

You can view run-time statistics about an application and about the jobs, connections, requests, and workflows associated with the application. For example, you can view the number of enabled and

disabled applications, number of aborted connections, and number of completed, failed, and canceled jobs and workflows.

Reports View for an Application

The **Reports** view shows monitoring reports about the selected application.

When you select an application in the Navigator of the **Execution Statistics** view, the **Reports** view shows reports about the application components.

Monitor Deployed Mapping Jobs

You can monitor deployed mapping jobs on the **Execution Statistics** view on the **Monitor** tab, or in the Monitoring tool.

You can view information about deployed mapping jobs in an application.

You can monitor a deployed mapping in the following locations:

- Monitoring tool. In the Developer tool, click the **Menu** button in the Progress view and select **Monitor Jobs**. Select the Data Integration Service that runs the mapping and click **OK**. The Monitoring tool opens. Expand an application in the Navigator and select the **Deployed Mapping Jobs** folder. A list of deployed mapping jobs appears in the contents panel.
  You can view mapping execution statistics in the Monitoring tool. The REST Operations Hub generates statistics based on the API configuration.
- Administrator tool. Expand an application in the Navigator and select the **Deployed Mapping Jobs** folder. A list of deployed mapping jobs appears in the contents panel.

The contents panel shows properties about each deployed mapping job, such as Job ID, name of the mapping, state of the job, and start time of the job. If you run the job on a grid, the contents panel also shows the node that the Data Integration Service that runs the process is running on.

Select a deployed mapping job in the contents panel to view logs for the job, reissue the job, cancel the job, or view statistics about the job. You can view statistics about throughput and resource usage for the job run.

Viewing Logs for a Deployed Mapping Job

You can download the logs for a deployed mapping job to view the job details.

Note: The log contents for a deployed mapping job depend on how the Data Integration Service is configured. For more information about logs when a Data Integration Service grid is configured to run jobs in separate remote processes, see the *Informatica Application Service Guide*.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service.
4. Expand an application and select **Deployed Mapping Jobs**.

   A list of mapping jobs appear in the contents panel.
5. Select a mapping job.
6. Click **Actions** > **View Logs for Selected Object**.

   A dialog box appears with the option to open or save the log file.

Reissuing a Deployed Mapping Job

You can reissue a deployed mapping job when the mapping jobs fails. When you reissue a deployed mapping job, the Data Integration Service runs the job again.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **Deployed Mapping Jobs**.

   The contents panel displays a list of deployed mapping jobs.
4. Select a deployed mapping job.
5. Click **Actions** > **Reissue Selected Object**.

Canceling a Deployed Mapping Job

You can cancel a deployed mapping job. You might want to cancel a deployed mapping job that hangs or that is taking an excessive amount of time to complete.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service.
4. Expand an application and select **Deployed Mapping Jobs**.

   The contents panel displays a list of deployed mapping jobs.
5. Select a deployed mapping job.
6. Click **Actions** > **Cancel Selected Job**.

Viewing Summary Statistics for a Deployed Mapping Job

You can view throughput and resource usage statistics for deployed mapping jobs.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service.
4. Expand an application and select **Deployed Mapping Jobs**.

   A list of mapping jobs appears in the contents panel.
5. Select a mapping job.

   The details panel displays the Properties for the mapping job.
6. Click the **Summary Statistics** view.

   The **Summary Statistics** view displays throughput and resource usage statistics for the source and target.

Optionally, you can sort the statistics in ascending or descending order. Click a column header to sort the column in ascending order. Click the column header again to sort the column in descending order.

Viewing Detailed Statistics for a Deployed Mapping Job

You can view graphs of the throughput and resource usage for deployed mapping jobs that run in separate local processes. Detailed statistics appear for jobs that run longer than one minute.

1. In the Administrator tool, click the **Monitor** tab.

2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service.
4. Expand an application and select **Deployed Mapping Jobs**.

   A list of mapping jobs appears in the contents panel.
5. Select a mapping job.

   The details panel displays the Properties for the mapping job.
6. Click the **Detailed Statistics** view.

   The **Detailed Statistics** view displays the throughput graph and resource usage graphs.

Optionally, you can complete the following tasks in the **Detailed Statistics** view:

Task	Description
Enlarge a graph	Move the cursor over a graph, and then click the magnifying glass icon.
Enlarge a section of an enlarged graph	Drag the cursor to select an area to enlarge.
Switch between rows and bytes in the throughput graph	Click the Bytes option or the Rows option.
Choose which statistics are plotted on the throughput graph	In the throughput field, select the sources and targets that you want to view.

Monitor Logical Data Objects

You can monitor logical data objects on the **Execution Statistics** view on the **Monitor** tab or the Monitoring tool.

You can view information about logical data objects included in an application. To monitor a logical data object, expand a Data Integration Service in the Navigator. Expand an application, and then select the **Logical Data Objects** folder. A list of logical data objects appears in the contents panel. The contents panel shows properties about each logical data object.

Select a logical data object in the contents panel to download the logs for a data object.

When you select a logical data object in the contents panel, the details panel shows the following views:

- **Properties** view
- **Cache Refresh Runs** view

Properties View for a Logical Data Object

The **Properties** view shows general properties and run-time statistics about the selected object.

You can view properties such as the data object name, logical data object model, folder path, cache state, and last cache refresh information.

Cache Refresh Runs View for a Logical Data Object

The **Cache Refresh Runs** view shows cache refresh details about the selected logical data object.

The **Cache Refresh Runs** view shows cache refresh details such as the cache run ID, request count, and row count.

Viewing Logs for Data Object Cache Refresh Runs

You can download the logs for data object cache refresh runs to view the cache refresh run details.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **Logical Data Objects**.

   The contents panel displays a list of logical data objects.
4. Select a logical data object.

   Details about the selected data object appear in the details panel.
5. Select the **Cache Refresh Runs** view.
6. Click **View Logs for Selected Object**.

Monitor SQL Data Services

You can monitor SQL data services on the **Execution Statistics** view on the **Monitor** tab. An SQL data service is a virtual database that you can query. It contains a schema and other objects that represent underlying physical data.

You can view information about the SQL data services included in an application. To monitor an SQL data service, expand an application in the Navigator and select the **SQL Data Services** folder. A list of SQL data services appears in the contents panel. The contents panel shows properties about each SQL data service, such as the name, description, and state.

When you select an SQL data service in the contents panel, the contents panel shows the following views:

- **Properties** view
- **Connections** view
- **Requests** view
- **Virtual Tables** view
- **Reports** view

Properties View for an SQL Data Service

The **Properties** view shows general properties and run-time statistics for an SQL data service.

When you select an SQL data service in the contents panel of the **Properties** view, you can view the general properties and run-time statistics.

General Properties for an SQL Data Service

You can view general properties, such as the SQL data service name and the description.

Statistics for an SQL Data Service

You can view run-time statistics about connections and requests for the SQL data service. Sample statistics include the number of connections to the SQL data service, the number of requests, and the number of aborted connections.

Connections View for an SQL Data Service

The **Connections** view displays properties about connections from third-party clients. The view shows properties such as the connection ID, state of the connection, connect time, elapsed time, and disconnect time.

When you select a connection in the contents panel, you can abort the connection or access the **Properties** view and **Requests** view in the details panel.

Properties View

The **Properties** view in the details panel shows the user who is using the connection, the state of the connection, and the connect time.

Requests View

The **Requests** view in the details panel shows information about the requests for the SQL connection. Each connection can have more than one request. The view shows request properties such as request ID, user name, state of the request, start time, elapsed time, and end time.

Aborting a Connection

You can abort a connection to prevent it from sending more requests to the SQL data service.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **SQL Data Services**.

   The contents panel lists the SQL data services in the application.
4. Select an SQL data service.

   The contents panel displays multiple views for the SQL data service.
5. Click the **Connections** view.

   The contents panel lists connections to the SQL data service.
6. Select a connection.
7. Click **Actions** > **Abort Selected Connection**.

Requests View for an SQL Data Service

The **Requests** view displays properties for requests for each SQL connection.

The **Requests** view shows properties about the requests for the SQL connection. Each connection can have more than one request. The view shows request properties such as request ID, connection ID, user name, state of the request, start time, elapsed time, and end time.

Select a request in the contents panel to view additional information about the request in the details panel.

Aborting an SQL Data Service Connection Request

You can abort an SQL Data Service connection request. You might want to abort a connection request that hangs or that is taking an excessive amount of time to complete.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **SQL Data Services**.

   The contents panel displays a list of SQL data services.
4. Select an SQL data service.
5. Click the **Requests** view.

   A list of connection requests for the SQL data service appears.
6. Select a request row.
7. Click **Actions** > **Abort Selected Request**.

Viewing Logs for an SQL Data Service Request

You can download the logs for an SQL data service request to view the request details.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **SQL Data Services**.

   The contents panel displays a list of SQL data services.
4. Select an SQL data service.
5. Click the **Requests** view.

   A list of requests for the SQL data service appears.
6. Select a request row.
7. Click **Actions** > **View Logs for Selected Object**.

Virtual Tables View for an SQL Data Service

The **Virtual Tables** view displays properties about the virtual tables in the SQL data service.

The view shows properties about the virtual tables, such as the name and description. When you select a virtual table in the contents panel, you can view the **Properties** view and **Cache Refresh Runs** view in the details panel.

Properties View

The **Properties** view displays general information and run-time statistics about the selected virtual table. General properties include the virtual table name and the schema name. Monitoring statistics include the number of request, the number of rows cached, and the last cache refresh time.

Cache Refresh Runs View

The **Cache Refresh Runs** view displays cache information for the selected virtual table. The view includes the cache run ID, the request count, row count, and the cache hit rate. The cache hit rate is the total number of requests on the cache divided by the total number of requests for the data object.

Viewing Logs for an SQL Data Service Table Cache

You can download the logs for an SQL data service table cache to view the table cache details.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **SQL Data Services**.

   The contents panel displays a list of SQL data services.
4. Select an SQL data service.
5. Click the **Virtual Tables** view.

   A list of virtual tables for the SQL data service appears.
6. Select a table row.

   Details about the selected table appear in the details panel.
7. Select the **Cache Refresh Runs** view.
8. Click **View Logs for Selected Object**.

Reports View for an SQL Data Service

The **Reports** view shows monitoring reports about the selected SQL data service.

When you monitor an SQL data service, the **Reports** view shows reports about the SQL data service. For example, you can view the Most Active SQL Connections report to determine the SQL connections that received the most connection requests during a specific time period.

Monitor Web Services

You can monitor web services on the **Execution Statistics** view on the **Monitor** tab. Web services are business functions that operate over the Web. They describe a collection of operations that are network accessible through standardized XML messaging.

You can view information about web services included in an application. To monitor a web service, expand an application in the Navigator and select the **Web Services** folder. A list of web services appears in the contents panel. The contents panel shows properties about each web service, such as the name, description, and state of each web service.

When you select the link for a web service in the contents panel, the contents panel shows the following views:

- **Properties** view
- **Reports** view
- **Operations** view
- **Requests** view

Properties View for a Web Service

The **Properties** view shows general properties and run-time statistics for a web service.

When you select a web service in the contents panel of the **Properties** view, you can view the general properties and monitoring statistics.

General Properties for a Web Service

You can view general properties about the web service, such as the name and type of object.

Statistics for a Web Service

You can view run-time statistics about web service requests during a specific time period. The **Statistics** section shows the number of completed, failed, and total web service requests.

Reports View for a Web Service

The **Reports** view shows monitoring reports about the selected web service.

When you monitor a web service, the **Reports** view shows reports about the web service. For example, you can view the Most Active WebService Client IP report to determine the IP addresses that received the most number of web service requests during a specific time period.

Operations View for a REST or SOAP Web Service

The **Operations** view shows the name and description of each operation or resource included in the web service. The view also displays properties, requests, and reports about each operation.

When you select a web service operation in the contents panel, the details panel shows the **Properties** view, **Requests** view, and **Reports** view.

Properties View

The **Properties** view shows general properties and statistics about the selected web service operation or resource. General properties include the operation or resource name and type of object. The view also shows statistics about the web service operation during a particular time period. Statistics include the number of completed, failed, and total web service requests.

Requests View

The **Requests** view shows properties about each web service operation, such as request ID, user name, state, start time, elapsed time, and end time. You can filter the list of requests. You can also view logs for the selected web service request.

Reports View for a SOAP Web Service

The **Reports** view shows reports about SOAP web service operations.

Requests View for a Web Service

The **Requests** view shows properties about each web service request, such as request ID, user name, state, start time, elapsed time, and end time. You can filter the list of requests.

When you select a web service request in the contents panel, you can view logs about the request in the details panel. The details panel shows general properties and statistics about the selected web service request. Statistics include the number of completed, failed, and total web service requests.

You can also abort a web service request from the **Requests** view. To abort a web service request, select the workflow request and click **Actions** > **Abort Selected Request** in the contents panel.

Monitor Workflows

You can monitor workflows on the **Execution Statistics** view on the **Monitor** tab.

You can view information about workflow instances that run from a workflow in a deployed application. To monitor a workflow, expand an application in the Navigator and select the **Workflows** folder. A list of workflow instances appears in the contents panel. The contents panel shows properties about each workflow instance, such as the name, state, start time, and recovery properties of each workflow instance. If the workflow instance ran on a grid, the contents panel shows the node that ran each mapping in the workflow instance.

Select a workflow instance in the contents panel to perform the following tasks:

- View logs for the workflow instance.
- View the context of the workflow instance to view other workflow instances that started around the same time as the selected workflow instance.
- Cancel or abort the workflow instance.
- Recover the interrupted workflow instance.

Expand a workflow instance to view properties about the workflow objects.

Workflow Graph

You can view the details of a workflow that you run in the Monitoring tool in a graphical form.

After you run a workflow, you can see the graphical view of the workflow in the Monitoring tool. In the workflow graph, you can see the sequential run of the mapping tasks in the workflow. The workflow graph enables you to view the failure points in a workflow at a glance.

In the workflow graph, you can view the following details of a workflow:

- Mapping tasks in the workflow
- Task details
- Recovery details

You can perform the following tasks from the workflow graph:

- Abort a running workflow
- Cancel a running workflow
- Recover a failed workflow
- View the workflow logs

Viewing a Workflow Graph

You can view a workflow graph that shows the sequential run of the mapping tasks in the workflow.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand an application.
3. Select the **Workflows** folder.

   A list of workflows appears in the contents panel.
4. Select the workflow that you want to view.
5. Click **Actions** > **View Workflow Graph**.

   The workflow graph appears in a new window.

View Workflow Objects

When you expand a workflow instance in the contents panel, you can view properties about workflow objects, such as the name, state, start time, and elapsed time for the object.

Workflow objects include events, tasks, and gateways. When you monitor workflows, you can monitor the tasks that run in a workflow instance. The **Monitor** tab does not display information about events or gateways in the workflow instance.

If an expression in a conditional sequence flow evaluates to false, the Data Integration Service does not run the next object or any of the subsequent objects in that branch. The **Monitor** tab does not list objects that do not run in the workflow instance. When a workflow instance includes objects that do not run, the instance can still successfully complete.

You can expand a task in the contents panel to view information about the work item run by the task. For example, if the workflow contains a mapping task, you can view throughput and resource usage statistics for the mapping run.

Viewing Summary Statistics for Workflow Objects

You can view throughput and resource usage statistics for mapping objects in workflows that run in separate local processes.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select the **Workflows** folder.

   A list of workflows appears in the contents panel.
4. Expand a workflow that contains a mapping object.
5. Expand the mapping task and select the mapping.
6. In the details panel, click the **Summary Statistics** view.

   The **Summary Statistics** view displays throughput and resource usage statistics for the source and target.

Optionally, you can sort the statistics in ascending or descending order. Click a column header to sort the column in ascending order. Click the column header again to sort the column in descending order.

Viewing Detailed Statistics for Workflow Objects

You can view graphs of the throughput and resource usage for mapping objects in workflows that run in separate local processes. Detailed statistics appear for jobs that run longer than one minute.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select the **Workflows** folder.

   A list of workflows appears in the contents panel.
4. Expand a workflow that contains a mapping object.
5. Expand the mapping task and select the mapping.
6. Click the **Detailed Statistics** view in the details panel.

   The **Detailed Statistics** view displays the throughput graph and resource usage graphs.

Optionally, you can complete the following tasks in the **Detailed Statistics** view:

Task	Description
Enlarge a graph	Move the cursor over a graph, and then click the magnifying glass icon.
Enlarge a section of an enlarged graph	Drag the cursor to select an area to enlarge.
Switch between rows and bytes in the throughput graph	Click the Bytes option or the Rows option.
Choose which statistics are plotted on the throughput graph	In the throughput field, select the sources and targets that you want to view.

Workflow States

When you monitor a workflow instance, you can view the state of the workflow instance. If a workflow instance recovers after a task is interrupted, the Monitor adds an entry for the task instance that runs in the recovered workflow.

A workflow instance can have one of the following states:

Aborted

A workflow instance aborts when you choose to abort the workflow instance from the Monitoring tool or using the infacmd wfs abortWorkflow command. You can also choose to abort a running workflow instance when you stop the application that contains the workflow or when you disable the workflow in the application.

Note: A workflow instance also aborts if the active sequence flow in the workflow reaches a Terminate event.

Canceled

You choose to cancel the workflow instance from the **Monitor** tab or by using the infacmd wfs cancelWorkflow command.

The workflow can also enter a canceled state if the Data Integration Service shuts down unexpectedly. If the workflow is not configured for automatic recovery, the service process changes the workflow instance state to Canceled when the service process restarts. Before the Data Integration Service restarts, the workflow state and the task state appear as Running, although the workflow and the task are no longer running. If the workflow is configured for automatic recovery, the service process recovers the workflow instance and reruns the interrupted task when the service process restarts. The service process sets the workflow instance state to Running.

Completed

The Data Integration Service successfully completes the workflow instance. A completed workflow instance might indicate that all tasks, gateways, and sequence flow evaluations either successfully completed or were in a branch that did not run.

A workflow can also enter a Completed state if a Command, Mapping, Notification, or Human task encounters a recoverable error or nonrecoverable error. When the task encounters the error, the Data Integration Service fails the task. The Data Integration Service runs subsequent workflow objects if expressions in the conditional sequence flows evaluate to true or if the sequence flows do not include conditions. If the workflow instance finishes running without another interruption, the Data Integration Service updates the workflow state to Completed.

When the task fails, the Data Integration Service continues to run additional objects in the workflow instance if expressions in the conditional sequence flows evaluate to true or if the sequence flows do not include conditions. If the workflow instance finishes running without another interruption, the Data Integration Service updates the workflow state to completed. A completed workflow instance can contain both failed and completed tasks.

Failed

A workflow instance fails when a workflow error occurs. Workflow errors can occur when the Data Integration Service reads the parameter file at the start of the workflow run, copies workflow parameter and variable values to task input, or evaluates expressions in conditional sequence flows. In addition, a workflow error occurs if an Assignment task or a gateway fails.

When a workflow error occurs, the Data Integration Service stops processing additional objects and fails the workflow instance immediately. Workflow errors are nonrecoverable.

Running

The Data Integration Service is running the workflow instance.

Workflow Object States

Workflows include tasks and gateways. When you monitor a workflow instance, you can view the state of the tasks that run in the workflow instance.

Tasks can have one of the following states:

Aborted

A task aborts in the following situations:

- The task encounters an nonrecoverable error.
- You abort the workflow instance.

  When you abort the workflow instance, the Data Integration Service first aborts the task and then aborts the workflow instance.

  If you choose to abort the workflow instance while an Assignment task is running, the Data Integration Service completes running the task. The Data Integration Service then aborts the workflow instance and does not start running other objects.

Completed

The Data Integration Service successfully completes the task.

Failed

A task fails in the following situations:

- Any task in a workflow not enabled for recovery encounters any type of error.
- An Assignment task in a workflow enabled for recovery encounters any type of error.
- A Command, Mapping, Notification, or Human task with a restart recovery strategy in a workflow enabled for recovery encounters a non recoverable error.
- A Mapping task with a skip recovery strategy in a workflow enabled for recovery encounters any type of error.

Note: A workflow can complete if a task fails. The Data Integration Service runs subsequent workflow objects if expressions in the conditional sequence flows evaluate to true or if the sequence flows do not include conditions. If the workflow instance finishes running without another interruption, the Data Integration Service updates the workflow state to Completed.

Running

The Data Integration Service is running the task.

Mapping Task Work Item States

When you expand a Mapping task, you can view the state of the mapping run. When you expand a restarted Mapping task, you can view the mapping jobs run for each recovery attempt of the workflow instance. If a workflow instance recovers after a Mapping task is interrupted, the Monitor adds an entry for the task instance that runs in the recovered workflow.

You can also view the state of the mapping run from the workflow graph of the workflow that contains the mapping task.

Mappings run by a Mapping task can have one of the following states:

Aborted

The Mapping task aborts while the mapping is running because you choose to abort the workflow instance.

Completed

The Data Integration Service successfully completes the mapping.

Failed

The mapping encounters an error. The mapping and the Mapping task appear as Failed in the Monitor. The states do not depend on the Mapping task recovery strategy.

Running

The Data Integration Service is running the mapping.

Canceling or Aborting a Workflow

You can cancel or abort a workflow instance at any time. You might want to cancel or abort a workflow instance that stops responding or that is taking an excessive amount of time to complete.

When you cancel a workflow instance, the Data Integration Service finishes processing any running task and then stops processing the workflow instance. The service does not start running any subsequent workflow objects.

When you abort a workflow instance, the Data Integration Service attempts to kill the process on any running task. If an Assignment task or a gateway is running, the Data Integration Service completes the task or gateway. After the task aborts or completes, the service aborts the workflow instance. The service does not start running any subsequent workflow objects.

You can also cancel or abort a workflow from the workflow graph.

1. Click the **Execution Statistics** view.
2. In the Navigator, expand a Data Integration Service.
3. Expand an application and select **Workflows**.

   A list of workflow instances appears in the contents panel.
4. Select a workflow instance.
5. Click **Actions** > **Cancel Selected Workflow** or **Actions** > **Abort Selected Workflow**.

Workflow Recovery

Workflow recovery is the completion of a workflow instance from the point of interruption.

When a workflow is enabled for recovery, you can recover a workflow instance if a task encounters a recoverable error, if you cancel the workflow instance, or if the Data Integration Service process shuts down unexpectedly.

View the workflow log to identify the cause of the interruption. After fixing any recoverable errors, you can recover the interrupted workflow instance if it is enabled for recovery.

You cannot change a workflow definition between the interrupted run and the recovery run. If a workflow instance has a recoverable state and you change the workflow metadata in the Developer tool and redeploy the application that contains the workflow, then the workflow instance is no longer recoverable.

The Data Integration Service tries to recover the previous workflow state if the service restarts after an unexpected shutdown. By default, the Data Integration Service does not recover a workflow instance that stopped during a Command task, Mapping task, or Notification task. In addition, the Data Integration Service cannot recover a workflow instance by default if you cancel the workflow instance or cancel a running task in the workflow instance. You can configure the recovery options on the workflow to enable the Data Integration Service to recover a workflow instance in such cases.

When you configure the workflow options, you can configure the workflow for manual recovery or automatic recovery. If you configure automatic recovery, the Data Integration Service restarts the workflow from the point of interruption without any human interaction. If you configure manual recovery, you can restart the workflow.

When a workflow instance recovers or when you recover a workflow instance, the Data Integration Service restarts the task. The service continues processing the subsequent workflow objects. If a workflow instance recovers after a task is interrupted, the Monitor adds an entry for the task instance that runs in the recovered workflow. For example, if a workflow recovers three times and restarts a Mapping task each time, the Monitor contains three entries for the Mapping task.

Recovery Properties

The read-only recovery properties display for each workflow instance. You configure the recovery properties for the workflow definition in the Developer tool. You cannot change the values of the properties for the workflow instance.

The following table describes the read-only recovery properties for a workflow instance:

Property	Description
Recovery Enabled	Indicates that the workflow is enabled for recovery.
Automatically Recover Workflows	Indicates that the Data Integration Service process tries to automatically recover workflow instances that were interrupted. The workflow recovery starts after the Data Integration Service process restarts.

Recovering a Workflow

You can recover interrupted workflow instances that are enabled for recovery.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **Workflows**.

A list of workflow instances appears in the contents panel.

4. Select the interrupted workflow instance that you want to recover.
5. Click **Actions** > **Recover Selected Workflow**.

   Monitor the state of the workflow recovery run in the contents panel.

Workflow Logs

The Data Integration Service generates log events when you run a workflow. Log events include information about workflow errors, task progress, and the setting of workflow variables. Log events also include the analyses of the links that the Data Integration Service evaluates in a sequence flow.

If a workflow instance includes a Mapping task, the Data Integration Service generates a separate log file for the mapping. The mapping log file includes any errors encountered during the mapping run and load summary and transformation statistics.

You can view the workflow and mapping logs from the Monitor tab.

When you recover an interrupted workflow instance, the Data Integration Service appends log events to the current workflow log. When the recovered workflow instance includes a Mapping task that is restarted, the Data Integration Service creates a mapping log.

If the workflow runs on a grid, the recovery of the workflow instance might run on a different node than the original workflow instance run. If the recovery runs on a different node and the log directory is not in a shared location, the Data Integration Service creates a log file with the same name on the current node.

Workflow Log Information

The information in the workflow log file represents the sequence of events that occur when the workflow runs.

The Data Integration Service writes information to the workflow log when the following types of event occur:

- The Data Integration Service starts to run a task or another object in the workflow.
- A task or another object in the workflow is in progress.
- The Data Integration Service finishes running a task or another object in the workflow.
- The Data Integration Service sets or updates a workflow variable.
- The Data Integration Service evaluates the links in a sequence flow and identifies the correct path for the workflow process.
- The workflow encounters a workflow error.

Viewing Logs for a Workflow

You can download the log for a workflow instance to view the workflow instance details.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service.
4. Expand an application and select **Workflows**.

   A list of workflow instances appears in the contents panel.
5. Select a workflow instance.

6. Click **Actions** > **View Logs for Selected Object**.

   A dialog box appears with the option to open or save the log file.

Viewing Logs for a Mapping Run in a Workflow

You can download the log for a mapping run in a workflow to view the mapping details.

1. Click the **Execution Statistics** view.
2. In the Domain Navigator, expand a Data Integration Service.
3. Expand an application and select **Workflows**.

   A list of workflow instances appears in the contents panel.
4. Expand a workflow instance.
5. Expand a Mapping task, and then select the mapping run by the task.
6. Click **Actions** > **View Logs for Selected Object**.

   A dialog box appears with the option to open or save the log file.

Monitoring a Folder of Objects

You can view properties and statistics about objects in a folder that appears in the Navigator of the **Execution Statistics** view. You can select one of the following folders: Jobs, Deployed Mapping Jobs, Logical Data Objects, SQL Data Services, Web Services, or Workflows.

You can apply a filter to limit the number of objects that appear in the contents panel. You can create custom filters based on a time range. Custom filters allow you to select particular dates and times for job start times, end times, and elapsed times. Custom filters also allow you to filter results based on multiple filter criteria.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, select the folder.

   The contents panel shows a list of objects contained in the folder.
4. Right-click the header of the table to add or remove columns.
5. Select **Receive New Notifications** to dynamically display new jobs, operations, requests, or workflows in the **Monitor** tab.
6. Enter filter criteria to reduce the number of objects that appear in the contents panel.
7. Select the object in the contents panel to view details about the object in the details panel.

   The details panel shows more information about the object selected in the contents panel.
8. To view jobs that started around the same time as the selected job, click **Actions** > **View Context**.

   The selected job and other jobs that started around the same time appear in the **Context View** tab. You can also view the context of connections, deployed mappings, requests, and workflows.
9. Click the **Close** button to close the **Context View** tab.

Viewing the Context of an Object

View the context of an object to view other objects of the same type that started around the same time as the selected object. You might view the context of an object to troubleshoot a problem or to get a high-level

understanding of what is happening at a particular period of time. You can view the context of jobs, deployed mappings, connections, requests, and workflows.

For example, you notice that your deployed mapping failed. When you view the context of the deployed mapping, an unfiltered list of deployed mappings appears in a separate working view, showing you all deployed mappings that started around the same time as your deployed mapping. You notice that the other deployed mappings also failed. You determine that the cause of the problem is that the Data Integration Service was unavailable.

1. In the Administrator tool, click the **Monitor** tab.
2. Click the **Execution Statistics** view.
3. In the Domain Navigator, expand a Data Integration Service and select the category of objects.

   For example, select **Jobs**.
4. In the contents panel, select the object for which you want to view the context.

   For example, select a job.
5. Click **Actions** > **View Context**.

Configuring the Date and Time Custom Filter

You can apply a custom filter on a Start Time or End Time column in the contents panel of the **Monitor** tab to filter results.

1. Select Custom as the filter option for the Start Time or End Time column.

   The **Custom Filter: Date and Time** dialog box appears.
2. Enter the date range using the specified date and time formats.
3. Click **OK**.

Configuring the Elapsed Time Custom Filter

You can apply a custom filter on an Elapsed Time column in the contents panel of the **Monitor** tab to filter results.

1. Select Custom as the filter option for the Elapsed Time column.

   The **Custom Filter: Elapsed Time** dialog box appears.
2. Enter the time range.
3. Click **OK**.

Configuring the Multi-Select Custom Filter

You can apply a custom filter on columns in the contents panel of the **Monitor** tab to filter results based on multiple selections.

1. Select Custom as the filter option for the column.

   The **Custom Filter: Multi-Select** dialog box appears.
2. Select one or more filters.
3. Click **OK**.

CHAPTER 12

Log Management

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Log Management Overview

To perform the logging function, the Service Manager runs a Log Manager and a Log Agent.

The Log Manager runs on the master gateway node. It collects and processes log events for Service Manager domain operations, application services, and user activity. The log events contain operational and error messages for a domain. The Service Manager and the application services send log events to the Log Manager. When the Log Manager receives log events, it generates log event files. You can view service log events in the Administrator tool based on criteria that you provide.

The Log Agent runs on all nodes in the domain. You can view log events for the last workflow run with the Log Events window in the Workflow Monitor.

The Log Agent also collects and processes log events for jobs that the Data Integration Service runs. These include profile jobs, scorecard jobs, preview jobs, mapping jobs, and SQL data services. You can view log events for these jobs on the Monitoring tab.

Log event files are binary files that the Administrator tool Logs Viewer uses to display log events. When you view log events in the Administrator tool, the Log Manager uses the log event files to display the log events for the domain, application services, and user activity.

Domain logs include domain, application service, and user activity logs. You can view them in the Administrator tool. System logs are for use only by Informatica Support to address open support issues.

You can use the Administrator tool to perform the following tasks with the Log Manager:

- Configure the log location. Configure the node that runs the Log Manager, the directory path for log event files, purge options, and time zone for log events.
- Configure log management. Configure the Log Manager to purge logs or purge logs manually. Save log events to XML, text, or binary files. Configure the time zone for the time stamp in the log event files.
- View log events. View domain function, application service, and user activity log events on the Logs tab. Filter log events by domain, application service type, and user.

Log Manager Architecture

The Service Manager on the master gateway node controls the Log Manager. The Log Manager starts when you start the Informatica services. After the Log Manager starts, it listens for log events from the Service Manager and application services. When the Log Manager receives log events, it generates log event files.

The Log Manager creates the following types of log files:

- Log event files. Stores log events in binary format. The Log Manager creates log event files to display log events in the Logs tab. When you view events in the Administrator tool, the Log Manager retrieves the log events from the event nodes.

  The Log Manager stores the files by date and by node. Set the directory path with the infasetup tools defineDomain command -ld option.
- Guaranteed Message Delivery files. Stores domain, application service, and user activity log events. The Service Manager writes the log events to temporary Guaranteed Message Delivery files and sends the log events to the Log Manager.

  If the Log Manager becomes unavailable, the Guaranteed Message Delivery files stay in the default log directory on the node where the service runs. By default, the directory path is `<Informatica_installation_directory>/logs/<Node_Name>`. When the Log Manager becomes available, the Service Manager for the node reads the log events in the temporary files, sends the log events to the Log Manager, and deletes the temporary files.

Data Integration Service Job Log Events

Logs for the jobs that the Data Integration Service runs are stored in a separate location from the domain, application service, and user activity logs. The Data Integration Service writes job log events to text files on the node where it runs.

The Data Integration Service and the Log Manager perform the following tasks to process job log events for the Data Integration Service:

1. When the Data Integration Service runs a job, it writes log events to text files on the node. The Data Integration Service sends information about the logs to the Log Manager.
2. The Log Manager stores log information in the Model repository database. The Model repository database stores information such as the path to the log file location, the node that contains the log, and the Data Integration Service that created the log.
3. When you view a job log on the Monitor tab of the Administrator tool, the Log Manager retrieves the information from the Model repository database. The Log Manager uses the information to determine the location of the logs.
4. The Log manager dispatches a Log Agent to retrieve the log events on each node to display the log.

Log Manager Recovery

When a service generates log events, it sends them to the Log Manager on the master gateway node. When you have the high availability option and the master gateway node becomes unavailable, the application services send log events to the Log Manager on a new master gateway node.

The Service Manager, the application services, and the Log Manager perform the following tasks:

1. An application service process writes log events to a Guaranteed Message Delivery file.
2. The application service process sends the log events to the Service Manager on the gateway node for the domain.
3. The Log Manager processes the log events and writes log event files. The application service process deletes the temporary file.
4. If the Log Manager is unavailable, the Guaranteed Message Delivery files stay on the node running the service process. The Service Manager for the node sends the log events in the Guaranteed Message Delivery files when the Log Manager becomes available, and the Log Manager writes log event files.

Troubleshooting the Log Manager

Domain and application services write log events to Service Manager log files when the Log Manager cannot process log events. The Service Manager log files are located in the default logs directory. The Service Manager log files include catalina.out, localhost_<date>.txt, and node.log. Services write log events to different log files depending on the type of error.

Use the Service Manager log files to troubleshoot issues when the Log Manager cannot process log events. You will also need to use these files to troubleshoot issues when you contact Informatica Global Customer Support.

Note: You can troubleshoot an Informatica installation by reviewing the log files generated during installation. You can use the installation summary log file to find out which components failed during installation.

Log Location

The Service Manager on the master gateway node writes log event files to the log file directory. When you configure a node to serve as a gateway, you must configure the directory where the Service Manager on this node writes the log event files. Each gateway node must have access to the directory path.

You configure the log location in the Properties view for the domain. Configure a directory location that is accessible to the gateway node during installation or when you define the domain. Store the logs on a shared disk when you have more than one gateway node. If the Log Manager is unable to write to the directory path, it writes log events to node.log on the master gateway node.

When you configure the log location, the Administrator tool validates the directory as you update the configuration. If the directory is invalid, the update fails. The Log Manager verifies that the log directory has read/write permissions on startup. Log files might contain inconsistencies if the log directory is not shared in a highly available environment.

You can change the directory path for domain logs in the Administrator tool or with the log service directory parameter, -ld. You can use the -ld parameter with any of the following commands:

- infacmd isp SwitchToGatewayNode
- infasetup DefineDomain

- infasetup DefineGatewayNode
- infasetup UpdateGatewayNode

System Logs

System logs contain information that Informatica Support views to help solve issues that you raise with Support. Ordinarily, you have no need to view these logs.

By default, the directory path is `<Informatica_installation_directory>/logs/<Node_Name>/`.. You can change the default directory path for logs with the System Log Directory parameter, -sld. You can use the -sld parameter with any of the following commands:

- infasetup DefineDomain
- infasetup DefineGatewayNode
- infasetup DefineWorkerNode
- infasetup UpdateGatewayNode
- infasetup UpdateWorkerNode

When you create a custom location, you can use a local location or a location that all domain nodes share. The Service Manager adds the node name to the path and creates separate log directories for each node.

When you update the gateway node or worker node with a new default location for system logs, existing logs remain intact. The server creates future logs at the new location, and abandons logs at the old location.

If you specify a node name when you change the default path, the Service Manager adds it to the path. For example, if you specify C:/logs/node1/ as the system log directory, the Service Manager creates logs in directories under C:/logs/node1/node1/.

If you have multiple Informatica domains, you must configure a different directory path for the Log Manager in each domain. Multiple domains cannot use the same shared directory path.

Note: When you change the directory path, you must restart Informatica services on the node you changed.

Log Management Configuration

The Service Manager and the application services continually send log events to the Log Manager. As a result, the directory location for the logs can grow to contain a large number of log events.

You can purge logs events periodically to manage the amount of log events stored by the Log Manager. You can export logs before you purge them to keep a backup of the log events.

Purging Log Events

You can automatically or manually purge log events. The Service Manager purges log events from the log directory according to the purge properties you configure in the Log Management dialog box. You can manually purge log events to override the automatic purge properties.

Purging Log Events Automatically

The Service Manager purges log events from the log directory according to the purge properties. The default value for preserving logs is 30 days and the default maximum size for log event files is 200 MB.

When the number of days or the size of the log directory exceeds the limit, the Log Manager deletes the log event files, starting with the oldest log events. The Log Manager periodically verifies the purge options and purges log events. The Log Manager does not purge the current day log event files and folder.

Purging Log Events Manually

You can purge log events for the domain, application services, or user activity. When you purge log events, the Log Manager removes the log event files from the log directory. The Log Manager does not remove log event files currently being written to the logs.

Optionally, you can use the *infacmd* PurgeLog command to purge log events.

The following table lists the purge log options:

Option	Description
Log Type	Type of log events to purge. You can purge domain, service, user activity or all log events.
Service Type	When you purge application service log events, you can purge log events for a particular application service type or all application service types.
Purge Entries	Date range of log events you want to purge. You can select the following options: - All Entries. Purges all log events. - Before Date. Purges log events that occurred before this date. Use the yyyy-mm-dd format when you enter a date. Optionally, you can use the calendar to choose the date. To use the calendar, click the date field.

Time Zone

When the Log Manager creates log event files, it generates a time stamp based on the time zone for each log event. The log event time stamp includes date, time, and time zone information. When the Log Manager creates log folders, it labels folders according to a time stamp. When you export or purge log event files, the Log Manager uses this property to calculate which log event files to purge or export. Set the time zone to the location of the machine that stores the log event files.

Verify that you do not lose log event files when you configure the time zone for the Log Manager. If the application service that sends log events to the Log Manager is in a different time zone than the master gateway node, you may lose log event files you did not intend to delete. Configure the same time zone for each gateway node.

Note: When you change the time zone, you must restart Informatica Services on the node that you changed.

Configuring Log Management Properties

Configure the log management properties in the **Log Management** dialog box in Informatica Administrator.

1. In the Administrator console, click the **Logs** tab.
2. Select **Log Actions** > **Log Management**.
3. Enter the number of days for the Log Manager to preserve log events.
4. Enter the maximum disk size for the directory that contains the log event files.

5. Enter the time zone in the following format:

   ```
   GMT(+|-)<hours>:<minutes>
   ```

   For example: GMT+08:00
6. Click **OK**.

Using the Logs Tab

You can view domain, application service, and user activity log events in the Logs tab of the Administrator tool. When you view log events in the Logs tab, the Log Manager displays the generated log event files in the log directory. When an error message appears in the Administrator tool, the error provides a link to the Logs tab.

You can use the Logs tab to perform the following tasks:

- View log events and the Administrator tool operational errors. View log events for the domain, an application service, or user activity.
- Filter log event results. After you display the log events, you can display log events that match filter criteria.
- Configure columns. Configure the columns you want the Logs tab to display.
- Save log events. You can save log events in XML, text, and binary format.
- Purge log events. You can manually purge log events.
- Copy log event rows. You can copy log event rows.

Viewing Log Events

To view log events in the Logs tab of the Administrator tool, select the Domain, Service, or User Activity view. Next, configure the filter options. You can filter log events based on attributes such as log type, domain function category, application service type, application service name, user, message code, activity code, timestamp, and severity level. The available options depend on whether you choose to view domain, application service, or user activity log events.

To view more information about a log event, click the log event in the search results.

You can view logs to get more information about errors that you receive while working in the Administrator tool.

1. In the Administrator Tool, click the Logs tab.
2. In the contents panel, select Domain, Service, or User Activity view.
3. Configure the filter criteria to view a specific type of log event.

   The following table lists the query options:

Log Type	Option	Description
Domain	Category	Category of domain service you want to view.
Service	Service Type	Application service you want to view.

Log Type	Option	Description
Service	Service Name	Name of the application service for which you want to view log events. You can choose a single application service name or all application services.
Domain, Service	Severity	The Log Manager returns log events with this severity level.
User Activity	User	User name for the Administrator tool user.
User Activity	Security Domain	Security domain to which the user belongs.
Domain, Service, User Activity	Timestamp	Date range for the log events that you want to view. You can choose the following options: - Blank. View all log events. - Within Last Day - Within Last Month - Custom. Specify the start and end date. Default is Within Last Day.
Domain, Service	Thread	Filter criteria for text that appears in the thread data. You can use wildcards (*) in this text field.
Domain, Service	Message Code	Filter criteria for text that appears in the message code. You can also use wildcards (*) in this text field.
Domain, Service	Message	Filter criteria for text that appears in the message. You can also use wildcards (*) in this text field.
Domain, Service	Node	Name of the node for which you want to view log events.
Domain, Service	Process	Process identification number for the Windows or UNIX service process that generated the log event. You can use the process identification number to identify log events from a process when an application service runs multiple processes on the same node.
User Activity	Activity Code	Filter criteria for text that appears in the activity code. You can also use wildcards (*) in this text field.
User Activity	Activity	Filter criteria for text that appears in the activity. You can also use wildcards (*) in this text field.

4. Click the Filter button.

   The Log Manager retrieves the log events and displays them in the Logs tab with the most recent log events first.

5. Click the Reset Filter button to view a different set of log events.

   Tip: To search for logs related to an error or fatal log event, note the timestamp of the log event. Then, reset the filter and use a custom filter to search for log events during the timestamp of the event.

Configuring Log Columns

You can configure the Logs tab to display the following columns:

- Category
- Service Type
- Service Name
- Severity
- User
- Security Domain
- Timestamp
- Thread
- Message Code
- Message
- Node
- Process
- Activity Code
- Activity

Note: The columns appear based on the query options that you choose. For example, when you display a service type, the service name appears in the Logs tab.

1. In the Administrator Tool, click the **Logs** tab.
2. Select the **Domain**, **Service**, or **User Activity** view.
3. To add a column, right-click a column name, select **Columns**, and then the name of the column you want to add.
4. To remove a column, right-click a column name, select **Columns**, and then clear the checkmark next to the name of the column you want to remove.
5. To move a column, select the column name, and then drag it to the location where you want it to appear.

   The Log Manager updates the Logs tab columns with your selections.

Saving Log Events

You can save the log events that you filter and view in the Log Viewer. When you save log events, the Log Manager saves whatever logs that you are viewing based on the filter criteria. To save log events to a file, click Save Logs on the Log Actions menu.

The Log Manager does not delete the log events when you save them. The Administrator Tool prompts you to save or open the saved log events file.

Optionally, you can use the *infacmd* isp GetLog command to retrieve log events.

The format you choose to save log events to depends on how you plan to use the exported log events file:

- XML file. Use XML format if you want to analyze the log events in an external tool that uses XML or if you want to use XML tools, such as XSLT.
- Text file. Use a text file if you want to analyze the log events in a text editor.
- Binary file. Use binary format to back up the log events in binary format. You might need to use this format to send log events to Informatica Global Customer Support.

Exporting Log Events

You can export the log events to an XML, text, or binary file. To export log events to a file, click Export Logs on the Log Actions menu.

When you export log events, you can choose which logs you want to save. When you choose Service logs, you can export logs for a particular service type. You can choose the sort order of the log events in the export file.

The Log Manager does not delete the log events when you export them. The Administrator tool prompts you to save or open the exported log events file.

Optionally, you can use the *infacmd* GetLog command to retrieve log events.

The format you choose to export log events depends on how you plan to use the exported log events file:

- XML file. Use XML format if you want to analyze the log events in an external tool that uses XML or if you want to use XML tools, such as XSLT.
- Text file. Use a text file if you want to analyze the log events in a text editor.
- Binary file. Use binary format to back up the log events in binary format. You might need to use this format to send log events to Informatica Global Customer Support.

The following table describes the export log options for each log type:

Option	Log Type	Description
Type	Domain, Service, User Activity	Type of logs you want to export.
Service Type	Service	Type of application service for which to export log events. You can also export log events for all service types.
Export Entries	Domain, Service, User Activity	Date range of log events you want to export. You can select the following options: - All Entries. Exports all log events. - Before Date. Exports log events that occurred before this date. Use the yyyy-mm-dd format when you enter a date. Optionally, you can use the calendar to choose the date. To use the calendar, click the date field.
Export logs in descending chronological order	Domain, Service, User Activity	Exports log events starting with the most recent log events.

XML Format

When you export log events to an XML file, the Log Manager exports each log event as a separate element in the XML file. The following example shows an excerpt from a log events XML file:

```
<log xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:common="http://
www.informatica.com/pcsf/common" xmlns:metadata="http://www.informatica.com/pcsf/
metadata" xmlns:domainservice="http://www.informatica.com/pcsf/domainservice"
xmlns:logservice="http://www.informatica.com/pcsf/logservice" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance">
<logEvent xsi:type="logservice:LogEvent" objVersion="1.0.0" timestamp="1129098642698"
severity="3" messageCode="AUTHEN_USER_LOGIN_SUCCEEDED" message="User Admin successfully
logged in." user="Admin" stacktrace="" service="authenticationservice"
serviceType="PCSF" clientNode="sapphire" pid="0" threadName="http-8080-Processor24"
```

```
context="" />
<logEvent xsi:type="logservice:LogEvent" objVersion="1.0.0" timestamp="1129098517000"
severity="3" messageCode="LM_36854" message="Connected to node [garnet] on outbound
connection [id = 2]." user="" stacktrace="" service="Copper" serviceType="IS"
clientNode="sapphire" pid="4484" threadName="4528" context="" />
```

Text Format

When you export log events to a text file, the Log Manager exports the log events in Information and Content Exchange (ICE) Protocol. The following example shows an excerpt from a log events text file:

```
2006-02-27 12:29:41 : INFO : (2628 | 2768) : (IS | Copper) : sapphire : LM_36522 :
Started process [pid = 2852] for task instance Session task instance
[s_DP_m_DP_AP_T_DISTRIBUTORS4]:Executor - Master.
2006-02-27 12:29:41 : INFO : (2628 | 2760) : (IS | Copper) : sapphire : CMN_1053 :
Starting process [Session task instance [s_DP_m_DP_AP_T_DISTRIBUTORS4]:Executor -
Master].
2006-02-27 12:29:36 : INFO : (2628 | 2760) : (IS | Copper) : sapphire : LM_36522 :
Started process [pid = 2632] for task instance Session task instance
[s_DP_m_DP_AP_T_DISTRIBUTORS4]:Preparer.
2006-02-27 12:29:35 : INFO : (2628 | 2760) : (IS | Copper) : sapphire : CMN_1053 :
Starting process [Session task instance [s_DP_m_DP_AP_T_DISTRIBUTORS4]:Preparer].
```

Binary Format

When you export log events to a binary file, the Log Manager exports the log events to a file that Informatica Global Customer Support can import. You cannot view the file unless you convert it to text. You can use the *infacmd* ConvertLogFile command to convert binary log files to text files, XML files, or readable text on the screen.

Viewing Administrator Tool Log Errors

If you receive an error while starting, updating, or removing services in the Administrator tool, an error message in the contents panel of the service provides a link to the Logs tab. Click the link in the error message to access detail information about the error in the Logs tab.

Log Events

The Service Manager and application services send log events to the Log Manager. The Log Manager generates log events for each service type.

Log events include a timestamp, in milliseconds, and a thread name that identifies the event.

You can view the following log event types on the Logs tab:

- Domain log events. Log events generated from the Service Manager functions.
- Analyst Service log events. Log events about each Analyst Service running in the domain.
- Data Integration Service log events. Log events about each Data Integration Service running in the domain.
- Model Repository log events. Log events about each Model Repository Service running in the domain.
- User activity log events. Log events about domain and security management tasks that a user completes.

Log Event Components

The Log Manager uses a common format to store and display log events. You can use the components of the log events to troubleshoot Informatica.

Each log event contains the following components:

- Service type, category, or user. The Logs tab categorizes events by domain category, service type, or user. If you view application service logs, the Logs tab displays the application service names. When you view domain logs, the Logs tab displays the domain categories in the log. When you view user activity logs, the Logs tab displays the users in the log.
- Message or activity. Message or activity text for the log event. Use the message text to get more information about the log events for domain and application services. Use the activity text to get more information about log events for user activity. Some log events contain embedded log event in the message texts. For example, the following log events contains an embedded log event:

  ```
  Client application [PmDTM], connection [59]: recv failed.
  ```

  In this log event, the following log event is the embedded log event:

  ```
  [PmDTM], connection [59]: recv failed.
  ```

  When the Log Manager displays the log event, the Log Manager displays the severity level for the embedded log event.
- Security domain. When you view user activity logs, the Logs tab displays the security domain for each user.
- Message or activity code. Log event code. If the message type is error or fatal, click on the message code to open the Informatica Knowledge Base search for the message. You must configure the support portal credentials in the user account to do the search.
- Process. The process identification number for the Windows or UNIX service process that generated the log event. You can use the process identification number to identify log events from a process when an application service runs multiple processes on the same node.
- Node. Name of the node running the process that generated the log event.
- Thread. Identification number or name of a thread started by a service process.
- Time stamp. Date, time, and time zone of the log event that occurred.
- Severity. The severity level for the log event. When you view log events, you can configure the Logs tab to display log events for a specific severity level.

Domain Log Events

Domain log events are log events generated from the domain functions the Service Manager performs.

Use the domain log events to view information about the domain and troubleshoot issues. You can use the domain log events to troubleshoot issues related to the startup and initialization of nodes and application services for the domain.

Domain log events include log events from the following functions:

- Authorization. Log events that occur when the Service Manager authorizes user requests for services. Requests can come from the Administrator tool.
- Container Management. Log events that occur when the Service Manager manages containers on nodes with the compute role.
- Domain Configuration. Log events that occur when the Service Manager manages the domain configuration metadata.
- Licensing. Log events that occur when the Service Manager registers license information.

- License Usage. Log events that occur when the Service Manager verifies license information from application services.
- Log Manager. Log events from the Log Manager. The Log Manager runs on the master gateway node. It collects and processes log events for Service Manager domain operations and application services.
- Log Agent. Log events from the Log Agent.
- Monitoring. Log events about Domain Functions.
- Node Configuration. Log events that occur as the Service Manager manages node configuration metadata in the domain.
- User Management. Log events that occur when the Service Manager manages users, groups, roles, and privileges.
- Service Manager. Log events from the Service Manager and signal exceptions from DTM processes. The Service Manager manages all domain operations. If the error severity level of a node is set to Debug, when a service starts the log events include the environment variables used by the service.

Analyst Service Log Events

Analyst Service log events contain the following information:

- Managing projects. Log events about managing projects in the Informatica Analyst, such as creating objects, folders, and projects. Log events about creating profiles, scorecards, and reference tables.
- Running jobs. Log events about running profiles and scorecards. Logs about previewing data.
- User permissions. Log events about managing user permissions on projects.

Data Integration Service Log Events

Data Integration Service logs contain logs about the following events:

- Configuration. Log events about system or service configuration changes, application deployment or removal, and logs about the associated profiling warehouse.
- Data Integration Service processes. Log events about application deployment, data object cache refresh, and user requests to run mappings, jobs, or workflows.
- Service failures. Log events about failures that cause the Data Integration Service to be unavailable, such as Model repository connection failures or the service's failure to start.

Model Repository Service Log Events

Model repository Service log events contain the following information:

- Model repository connections. Log events for connections to the repository from Informatica Developer, Informatica Analyst, and the Data Integration Service.
- Model Repository Service. Log events about the Model Repository Service, including enabling, disabling, starting, and stopping the service.
- Repository operations. Log events for repository operations such as creating and deleting repository content, and adding deployed applications.
- User permissions. Log events about managing user permissions on the repository.

User Activity Log Events

User activity log events describe all domain and security management tasks that a user completes.

Use the user activity log events to determine when a user created, updated, or removed services, nodes, users, groups, or roles.

The Service Manager writes user activity log events when the Service Manager needs to authorize a user to perform one of the following domain actions:

- Enable or disable a service process.
- Start, stop, enable, or disable a service.
- Add, update, or shut down a node.
- Modify the domain properties.
- Move a folder in the domain.

The Service Manager also writes user activity log events each time a user adds, updates, or removes a user, group, operating system profile, or role.

The user activity log displays information about the user who performed the security action or who failed to log in. When the Service Manager logs an unsuccessful login attempt, the logs display an error message that contains the following information:

- User ID whose login attempt failed
- Security domain that rejected the login attempt
- Application and version that the user attempted to log in to. For example: `application [Informatica Administrator] version [10.5.2]`
- Client IP address from which the login attempt originated
- Reason for the rejection. For example: `[<error code>] The user [<user ID>] in security domain [Native] does not exist in the domain.]`

The user activity logs also displays information about security audit trails and log events for changes to users, groups, and permissions.

The Service Manager writes a user activity log event each time a user account is locked or unlocked. The Service Manager also writes a user activity log event each time a user tries to log in to the domain with a client application.

To include security audit trails in the user activity log events, you must enable the SecurityAuditTrail property for the PowerCenter Repository Service in the Administrator tool.

When you import one or more repository objects, you can generate audit logs.

The audit logs contain the following information about the .xml file imported:

- Host name and IP address of the client machine from which the .xml file was imported
- Full local path of the .xml import file
- The file name
- The file size in bytes
- Logged in user name
- Number of objects imported
- Time stamp of the import operation

Log Events

The Service Manager and application services send log events to the Log Manager. The Log Manager generates log events for each service type.

Log events include a timestamp, in milliseconds, and a thread name that identifies the event.

You can view the following log event types on the Logs tab:

- Domain log events. Log events generated from the Service Manager functions.
- Analyst Service log events. Log events about each Analyst Service running in the domain.
- Data Integration Service log events. Log events about each Data Integration Service running in the domain.
- Model Repository log events. Log events about each Model Repository Service running in the domain.
- User activity log events. Log events about domain and security management tasks that a user completes.

Log Aggregator

You can aggregate the log files of an application service that stops responding or shuts down unexpectedly. You might need to analyze multiple log files to figure out issues with an application service.

You can use the log aggregator to aggregate all the log files associated with an application service and compress required log files into a .zip file. You can download the .zip file and analyze the log files, or upload the .zip file to Informatica Global Customer Support for analysis.

You cannot store the history of aggregated logs. You must download or send the file to Informatica Global Customer Support after you aggregate the log files.

You can aggregate the hang and crash logs of the following application services:

- Analyst Service
- Data Integration Service
- Model Repository Service

In addition to the application service logs, the log aggregator captures debug information for the nodes in the domain. The log aggregator aggregates the log files of the associated application services when you aggregate the log files of an application service. For example, when you aggregate the log files of an Analyst Service, the log aggregator aggregates the log files of the Data Integration Service and the Model Repository Service associated with the Analyst Service.

The log collection directory in the master gateway node stores the application service logs when you aggregate the logs. All the node processes in the domain must have read/write access on the log collection directory. If the node processes cannot access the log collection directory, the aggregated logs do not appear in the aggregated logs listgrid. The core dump directory stores the core dump files of the nodes in the domain. Configure the log collection directory in the master gateway node and the core dump directory for each node in the domain.

When you process the aggregated logs you can choose the collectors from which you want to collect log information. The collectors are application services and nodes associated with the application service.

Aggregating Application Service Logs

You can aggregate log files associated with hang or crash scenarios of an application service.

1. Click the **Logs** tab in the Administrator tool.
2. Click the **Log Aggregator** tab.
3. Select the application service for which you want to aggregate the logs.
4. Select the scenario for which you want to aggregate the logs.

   You can choose between application service crash and hang scenarios.
5. Select the time interval to aggregate the logs.

   You can choose to aggregate the logs from the previous 6 hours to 3 days.
6. Click **Next**.
7. Select the collectors from which you want to aggregate the logs.

   The log aggregator displays the log files and the collectors based on the node to which they belong.
8. Click **Finish**.

   The list of logs associated with the scenario appears in the right pane. You can download the aggregated logs or send the logs to the Informatica Global Customer Support.

Processing Aggregated Application Service Logs

After you aggregate the application service logs, you must download the aggregated zip file or send the logs to Informatica Global Customer Support.

Aggregate the application service logs based on your requirement.

1. Select the logs that you want to process.
2. Click **Actions** > **Compress Logs**.

   The **Compressed Scenario Output** dialog box appears.
3. On the **Compressed Output** tab, click **Download** to download the aggregated log files as a zip file.
4. Optionally, click the **Send to Support** tab.
5. Enter the user name, password, and the TFTP directory of the Informatica My Support Portal.
6. Click **Send** to send the aggregated log files to Informatica Global Customer Support.

CHAPTER 13

Domain Reports

This chapter includes the following topics:

- Domain Reports Overview, 169
- License Management Report, 169
- Web Services Report, 176

Domain Reports Overview

You can run the following domain reports from the Reports tab in the Administrator tool:

- License Management Report. Monitors the number of software options purchased for a license and the number of times a license exceeds usage limits. The License Management Report displays the license usage information such as CPU and repository usage and the node configuration details.
- Web Services Report. Monitors activities of the web services running on a Web Services Hub. The Web Services Report displays run-time information such as the number of successful or failed requests and average service time. You can also view historical statistics for a specific period of time.

Note: If the master gateway node runs on a UNIX machine and the UNIX machine does not have a graphics display server, you must install X Virtual Frame Buffer on the UNIX machine to view the report charts in the License Report or the Web Services Report. If you have multiple gateway nodes running on UNIX machines, install X Virtual Frame Buffer on each UNIX machine.

License Management Report

You can monitor the list of software options purchased with a license and the number of times a license exceeds usage limits. The License Management Report displays the general properties, CPU and repository usage, user details, hardware and node configuration details, and the options purchased for each license.

You can save the License Management Report as a PDF on your local machine. You can also email a PDF version of the report to someone.

Run the License Management Report to monitor the following license usage information:

- Licensing details. Shows general properties for every license assigned in the domain.
- CPU usage. Shows the number of logical CPUs used to run application services in the domain. The License Management Report counts logical CPUs instead of physical CPUs for license enforcement. If the

number of logical CPUs exceeds the number of authorized CPUs, then the License Management Report shows that the domain exceeded the CPU limit.

- Repository usage.
- User information. Shows information about users in the domain.
- Hardware configuration. Shows details about the machines used in the domain.
- Node configuration. Shows details about each node in the domain.
- Licensed options. Shows a list of Informatica options for each license.

Licensing Properties

The Licensing section of the License Management Report shows information about each license in the domain.

The following table describes the licensing information in the License Management Report:

Property	Description
Name	Name of the license.
Version	Version of Informatica platform.
Expiration Date	Date when the license expires.
Serial Number	Serial number of the license. The serial number identifies the customer or project. The original and incremental keys for a license have the same serial number.
Deployment Level	Level of deployment. Values are Development and Production.
Operating System / BitMode	Operating system and bitmode for the license. Indicates whether the license is installed on a 32-bit or 64-bit operating system.
CPU	Maximum number of authorized logical CPUs.
AT Named Users	Maximum number of users who are assigned the License Access for Informatica Analyst privilege.
Product Bitmode	Bitmode of the server binaries that are installed. Values are 32-bit or 64-bit.

CPU Summary

The CPU Summary section of the License Management Report shows the maximum number of logical CPUs used to run application services in the domain. Use the CPU summary information to determine if the CPU usage exceeded the license limits. If the number of logical CPUs is greater than the total number of CPUs authorized by the license, the License Management Report indicates that the CPU limit is exceeded.

The License Management Report determines the number of logical CPUs based on the number of processors, cores, and threads. Use the following formula to calculate the number of logical CPUs:

`N*C*T`, where

N is the number of processors.

C is the number of cores in each processor.

T is the number of threads in each core.

For example, a machine contains 4 processors. Each processor has 2 cores. The machine contains 8 (4*2) physical cores. Hyperthreading is enabled, where each core contains 3 threads. The number of logical CPUs is 24 (4*2*3).

Note: Although the License Management Report includes threads in the calculation of logical CPUs, Informatica license compliance is based on the number of physical cores, not threads. To be compliant, the number of physical cores must be less than or equal to the maximum number of licensed CPUs. If the License Management Report shows that you have exceeded the license limit but the number of physical cores is less than or equal to the maximum number of licensed CPUs, you can ignore the message. If you have a concern about license compliance, contact your Informatica account manager.

The following table describes the CPU summary information in the License Management Report:

Property	Description
Domain	Name of the domain on which the report runs.
Current Usage	Maximum number of logical CPUs used concurrently on the day the report runs.
Peak Usage	Maximum number of logical CPUs used concurrently during the last 12 months.
Peak Usage Date	Date when the maximum number of logical CPUs were used concurrently during the last 12 months.
Days Exceeded License Limit	Number of days that the CPU usage exceeded the license limits. The domain exceeds the CPU license limit when the number of concurrent logical CPUs exceeds the number of authorized CPUs.

CPU Detail

The CPU Detail section of the License Management Report provides CPU usage information for each host in the domain. The CPU Detail section shows the maximum number of logical CPUs used each day in a selected time period.

The report counts the number of logical CPUs on each host that runs application services in the domain. The report groups logical CPU totals by node.

The following table describes the CPU detail information in the License Management Report:

Property	Description
Host Name	Host name of the machine.
Current Usage	Maximum number of logical CPUs that the host used concurrently on the day the report runs.
Peak Usage	Maximum number of logical CPUs that the host used concurrently during the last 12 months.
Peak Usage Date	Date in the last 12 months when the host concurrently used the maximum number of logical CPUs.
Assigned Licenses	Name of all licenses assigned to services that run on the node.

Repository Summary

The Repository Summary section of the License Management Report provides repository usage information for the domain. Use the repository summary information to determine if the repository usage exceeded the license limits.

The following table describes the repository summary information in the License Management Report:

Property	Description
Current Usage	Maximum number of repositories used concurrently in the domain on the day the report runs.
Peak Usage	Maximum number of repositories used concurrently in the domain during the last 12 months.
Peak Usage Date	Date in the last 12 months when the maximum number of repositories were used concurrently.
Days Exceeded License Limit	Number of days that the repository usage exceeded the license limits.

User Summary

The User Summary section of the License Management Report provides information about Analyst tool users in the domain.

The following table describes the user summary information in the License Management Report:

Property	Description
User Type	Type of user in the domain.
Current Named Users	Maximum number of users who are assigned the License Access for Informatica Analyst privilege on the day the report runs.
Peaked Name Users	Maximum number of users who are assigned the License Access for Informatica Analyst privilege during the last 12 months.
Peak Named Users Date	Date during the last 12 months when the maximum number of concurrent users were assigned the License Access for Informatica Analyst privilege.

User Detail

The User Detail section of the License Management Report provides information about each Analyst tool user in the domain.

The following table describes the user detail information in the License Management Report:

Property	Description
User Type	Type of user in the domain.
User Name	User name.

Property	Description
Days Logged In	Number of days the user logged in to the Analyst tool and performed profiling during the last 12 months.
Peak Unique IP Addresses in a Day	Maximum number of machines that the user was logged in to and performed profiling on during a single day of the last 12 months.
Average Unique IP Addresses	Daily average number of machines that the user was logged in to and running profiling on during the last 12 months.
Peak IP Address Date	Date when the user logged in to and performed profiling on the maximum number of machines during a single day of the last 12 months.
Peak Daily Sessions	Maximum number of times in a single day of the last 12 months that the user logged in to any Analyst tool and performed profiling.
Average Daily Sessions	Average number of times per day in the last 12 months that the user logged in to any Analyst tool and performed profiling.
Peak Session Date	Date in the last 12 months when the user had the most daily sessions in the Analyst tool.

Hardware Configuration

The Hardware Configuration section of the License Management Report provides details about machines used in the domain.

The following table describes the hardware configuration information in the License Management Report:

Property	Description
Host Name	Host name of the machine.
Logical CPUs	Number of logical CPUs used to run application services in the domain.
Sockets	Number of sockets on the machine.
Consumed cores	Number of cores on the machine.
Cores per socket	Number of cores for each socket on the machine.
CPU Model	Model of the CPU.
Hyperthreading Enabled	Indicates whether hyperthreading is enabled.
Virtual Machine	Indicates whether the machine is a virtual machine.

Node Configuration

The Node Configuration section of the License Management Report provides details about each node in the domain.

The following table describes the node configuration information in the License Management Report:

Property	Description
Node Name	Name of the node or nodes assigned to a machine for a license.
Host Name	Host name of the machine.
IP Address	IP address of the node.
Operating System	Operating system of the machine on which the node runs.
Status	Status of the node.
Gateway	Indicates whether the node is a gateway node.
Service Type	Type of the application service configured to run on the node.
Service Name	Name of the application service configured to run on the node.
Service Status	Status of the application service.
Assigned License	License assigned to the application service.

Licensed Options

The Licensed Options section of the License Management Report provides details about each option for every license assigned to the domain.

The following table describes the licensed option information in the License Management Report:

Property	Description
License Name	Name of the license.
Description	Name of the license option.
Status	Status of the license option.
Issued On	Date when the license option was issued.
Expires On	Date when the license option expires.

Running the License Management Report

Run the License Management Report from the **Reports** tab in the Administrator tool.

1. Click the **Reports** tab in the Administrator tool.
2. Click the **License Management Report** view.

   The License Management Report appears.
3. Click **Save** to save the License Management Report as a PDF.

   If a License Management Report contains multibyte characters, you must configure the Service Manager to use a Unicode font.

4. Click **Email** to send a copy of the License Management Report in an email.

   The **Send License Management Report** page appears.

Configuring a Unicode Font for the Report

Before you can save a License Management Report that contains multibyte characters or non-English characters, configure the Service Manager to use a Unicode font when generating the PDF file.

1. Install a Unicode font on the master gateway node.
2. Use a text editor to create a file named AcUtil.properties.
3. Add the following properties to the file:

   ```
   PDF.Font.Default=Unicode_font_name
   PDF.Font.MultibyteList=Unicode_font_name
   ```

   Unicode_font_name is the name of the Unicode font installed on the master gateway node.

   You might also need to add the following property if the font file is not available in the locale:

   ```
   Unicode_font_name_path=Unicode_font_file_location
   ```

   For example:

   ```
   PDF.Font.Default=Arial Unicode MS
   PDF.Font.MultibyteList=Arial Unicode MS
   Arial Unicode MS_path=/usr/lib/X11/fonts/TrueType
   ```

4. Save the AcUtil.properties file to the following location:

   InformaticaInstallationDir\services\AdministratorConsole\administrator

5. Use a text editor to open the licenseUtility.css file in the following location:

   InformaticaInstallationDir\services\AdministratorConsole\administrator\css

6. Append the Unicode font name to the value of each font-family property.

   For example:

   ```
   font-family: Arial Unicode MS, Verdana, Arial, Helvetica, sans-serif;
   ```

7. Restart Informatica services on each node in the domain.

Sending the License Management Report in an Email

You must configure the SMTP settings for the domain before you can send the License Management Report in an email.

The domain administrator can send the License Management Report in an email from Send License Management Report page in the Administrator tool.

1. Enter the following information:

Property	Description
To Email	Email address to which you send the License Management Report.
Subject	Subject of the email.
Customer Name	Name of the organization that purchased the license.

Property	Description
Request ID	Request ID that identifies the project for which the license was purchased.
Contact Name	Name of the contact person in the organization.
Contact Phone Number	Phone number of the contact person.
Contact Email	Email address of the contact person at the customer site.

2. Click OK.

   The Administrator tool sends the License Management Report in an email.

Web Services Report

To analyze the performance of web services running on a Web Services Hub, you can run a report for the Web Services Hub or for a web service running on the Web Services Hub.

The Web Services Report provides run-time and historical information on the web service requests handled by the Web Services Hub. The report displays aggregated information for all web services in the Web Services Hub and information for each web service running on the Web Services Hub. The Web Services Report also provides historical information.

Understanding the Web Services Report

You can run the Web Services Report for a time interval that you choose. The Web Services Hub collects information on web services activities and caches 24 hours of information for use in the Web Services Report. It also writes the information to a history file.

Time Interval

By default, the Web Services Report displays activity information for a five-minute interval. You can select one of the following time intervals to display activity information for a web service or Web Services Hub:

- 5 seconds
- 1 minute
- 5 minutes
- 1 hour
- 24 hours

The Web Services Report displays activity information for the interval ending at the time you run the report. For example, if you run the Web Services Report at 8:05 a.m. for an interval of one hour, the Web Services Report displays the Web Services Hub activity from 7:05 a.m. and 8:05 a.m.

Caching

The Web Services Hub caches 24 hours of activity data. The cache is reinitialized every time the Web Services Hub is restarted. The Web Services Report displays statistics from the cache for the time interval that you run the report.

History File

The Web Services Hub writes the cached activity data to a history file. The Web Services Hub stores data in the history file for the number of days that you set in the MaxStatsHistory property of the Web Services Hub. For example, if the value of the MaxStatsHistory property is 5, the Web Services Hub keeps five days of data in the history file.

Contents of the Web Services Report

The Web Services Report view contains information about the web services in the domain. When you select a web services hub in the Navigator, you can view the following information about the web services it contains:

- Properties view. Displays General Properties, Web Services Hub Summary, and Historical Statistics for the web services hub.
- Web Services view. Lists the web services in the web services hub. When you select a web service, you can view Properties, Top IP Addresses, and Historical Statistics for the web service.

General Properties and Web Services Hub Summary

To view the general properties and summary information for the Web Services Hub, select the Properties view in the content panel.

The following table describes the general properties:

Property	Description
Name	Name of the Web Services Hub.
Description	Short description of the Web Services Hub.
Service type	Type of Service. For a Web Services Hub, the service type is ServiceWSHubService.

The following table describes the Web Services Hub Summary properties:

Property	Description
# of Successful Message	Number of requests that the Web Services Hub processed successfully.
# of Fault Responses	Number of fault responses generated by web services in the Web Services Hub. The fault responses could be due to any error.
Total Messages	Total number of requests that the Web Services Hub received.
Last Server Restart Tme	Date and time when the Web Services Hub was last started.
Avg. # of Service Partitions	Average number of partitions allocated for all web services in the Web Services Hub.
% of Partitions in Use	Percentage of web service partitions that are in use for all web services in the Web Services Hub.
Avg. # of Run Instances	Average number of instances running for all web services in the Web Services Hub.

Web Services Historical Statistics

To view historical statistics for the web services in the Web Services Hub, select the Properties view in the content panel. The detail panel displays data from the Web Services Hub history file for the date that you specify.

The following table describes the historical statistics:

Property	Description
Time	Time of the event.
Web Service	Name of the web service for which the information is displayed. When you click the name of a web service, the Web Services Report displays the Service Statistics window.
Successful Requests	Number of requests successfully processed by the web service.
Fault Responses	Number of fault responses sent by the web service.
Avg. Service Time	Average time it takes to process a service request received by the web service.
Max Service Time	The largest amount of time taken by the web service to process a request.
Min Service Time	The smallest amount of time taken by the web service to process a request.
Avg. Service Partitions	Average number of session partitions allocated for the web service.
Percent Partitions in Use	Percentage of partitions in use by the web service.
Avg Run Instances	Average number of instances running for the web service.

Web Services Run-time Statistics

To view run-time statistics for each web service in the Web Services Hub, select the Web Services view in the content panel. The Web Services view lists the statistics for each web service.

The report provides the following information for each web service for the selected time interval:

Property	Description
Service name	Name of the web service for which the information is displayed.
Successful Requests	Number of requests received by the web service that the Web Services Hub processed successfully.
Fault Responses	Number of fault responses generated by the web services in the Web Services Hub.
Avg. Service Time	Average time it takes to process a service request received by the web service.
Avg. Service Partitions	Average number of session partitions allocated for the web service.
Avg. Run Instances	Average number of instances of the web service running during the interval.

Web Service Properties

To view the properties of a web service, select the web service in the Web Services view of the content panel. In the details panel, the Properties view displays the properties for the web service.

The report provides the following information for the selected web service:

Property	Description
# of Successful Requests	Number of requests received by the web service that the Web Services Hub processed successfully.
# of Fault Responses	Number of fault responses generated by the web services in the Web Services Hub.
Total Messages	Total number of requests that the Web Services Hub received.
Last Server Restart Time	Date and time when the Web Services Hub was last started
Last Service Time	Number of seconds it took to process the most recent service request
Average Service Time	Average time it takes to process a service request received by the web service.
Avg.# of Service Partitions	Average number of session partitions allocated for the web service.
Avg. # of Run Instances	Average number of instances of the web service running during the interval.

Web Service Top IP Addresses

To view the top IP addresses for a web service, select a web service in the Web Services view of the content panel and select the Top IP Addresses view in the details panel. The Top IP Addresses displays the most active IP addresses for the web service, listed in the order of longest to shortest service times.

The report provides the following information for each of the most active IP addresses:

Property	Description
Top 10 Client IP Addresses	The list of client IP addresses and the longest time taken by the web service to process a request from the client. The client IP addresses are listed in the order of longest to shortest service times. Use the *Click here* link to display the list of IP addresses and service times.

Web Service Historical Statistics

To view a table of historical statistics for a web service, select a web service in the Web Services view of the content panel and select the Table view in the details panel. The details panel displays a table of historical statistics for the web service.

The table provides the following information for the selected web service:

Property	Description
Time	Time of the event.
Web Service	Name of the web service for which the information is displayed.
Successful Requests	Number of requests successfully processed by the web service.
Fault Responses	Number of requests received for the web service that could not be processed and generated fault responses.
Avg. Service Time	Average time it takes to process a service request received by the web service.
Min. Service Time	The smallest amount of time taken by the web service to process a request.
Max. Service Time	The largest amount of time taken by the web service to process a request.
Avg. Service Partitions	Average number of session partitions allocated for the web service.
Percent Partitions in Use	Percentage of partitions in use by the web service.
Avg. Run Instances	Average number of instances running for the web service.

Running the Web Services Report

Run the Web Services Report from the Reports tab in the Administrator tool.

Before you run the Web Services Report for a Web Services Hub, verify that the Web Services Hub is enabled. You cannot run the Web Services Report for a disabled Web Services Hub.

1. In the Administrator tool, click the Reports tab.
2. Click Web Services.
3. In the Navigator, select the Web Services Hub for which to run the report.

   In the content panel, the Properties view displays the properties of the Web Services Hub. The details view displays historical statistics for the services in the Web Services Hub.
4. To specify a date for historical statistics, click the date filter icon in the details panel, and select the date.
5. To view information about each service, select the Web Services view in the content panel.

   The Web Services view displays summary statistics for each service for the Web Services Hub.
6. To view additional information about a service, select the service from the list.

   In the details panel, the Properties view displays the properties for the service.
7. To view top IP addresses for the service, select the Top IP Addresses view in the details panel.
8. To view table attributes for the service, select the Table view in the detail panel.

Running the Web Services Report for a Secure Web Services Hub

To run a Web Services Hub on HTTPS, you must have an SSL certificate file for authentication of message transfers. When you create a Web Services Hub to run on HTTPS, you must specify the location of the keystore file that contains the certificate for the Web Services Hub. To run the Web Services Report in the

Administrator tool for a secure Web Services Hub, you must import the SSL certificate into the Java certificate file. The Java certificate file is named *cacerts* and is located in the /lib/security directory of the Java directory. The Administrator tool uses the cacerts certificate file to determine whether to trust an SSL certificate.

In a domain that contains multiple nodes, the node where you generate the SSL certificate affects how you access the Web Services Report for a secure Web Services Hub.

Use the following rules and guidelines to run the Web Services Report for a secure Web Services Hub in a domain with multiple nodes:

- For each secure Web Services Hub running in a domain, generate an SSL certificate and import it to a Java certificate file.
- The Administrator tool searches for SSL certificates in the certificate file of a gateway node. The SSL certificate for a Web Services Hub running on worker node must be generated on a gateway node and imported into the certificate file of the same gateway node.
- To view the Web Services Report for a secure Web Services Hub, log in to the Administrator tool from the gateway node that has the certificate file containing the SSL certificate of the Web Services Hub for which you want to view reports.
- If a secure Web Services Hub runs on a worker node, the SSL certificate must be generated and imported into the certificate file of the gateway node. If a secure Web Services Hub runs on a gateway and a worker node, the SSL certificate of both nodes must be generated and imported into the certificate file of the gateway node. To view reports for the secure Web Services Hub, log in to the Administrator tool from the gateway node.
- If the domain has two gateway nodes and a secure Web Services Hub runs on each gateway node, access to the Web Services Reports depends on where the SSL certificate is located.

  For example, gateway node GWN01 runs Web Services Hub WSH01 and gateway node GWN02 runs Web Services Hub WSH02. You can view the reports for the Web Services Hubs based on the location of the SSL certificates:

  - If the SSL certificate for WSH01 is in the certificate file of GWN01 but not GWN02, you can view the reports for WSH01 if you log in to the Administrator tool through GWN01. You cannot view the reports for WSH01 if you log in to the Administrator tool through GWN02. If GWN01 fails, you cannot view reports for WSH01.
  - If the SSL certificate for WSH01 is in the certificate files of GWN01 and GWN02, you can view the reports for WSH01 if you log in to the Administrator tool through GWN01 or GWN02. If GWN01 fails, you can view the reports for WSH01 if you log in to the Administrator tool through GWN02.
- To ensure successful failover when a gateway node fails, generate and import the SSL certificates of all Web Services Hubs in the domain into the certificates files of all gateway nodes in the domain.

CHAPTER 14

Node Diagnostics

This chapter includes the following topics:

- Node Diagnostics Overview, 182
- Informatica Network Login, 183
- Generating Node Diagnostics, 184
- Downloading Node Diagnostics, 185
- Uploading Node Diagnostics, 185
- Analyzing Node Diagnostics, 186

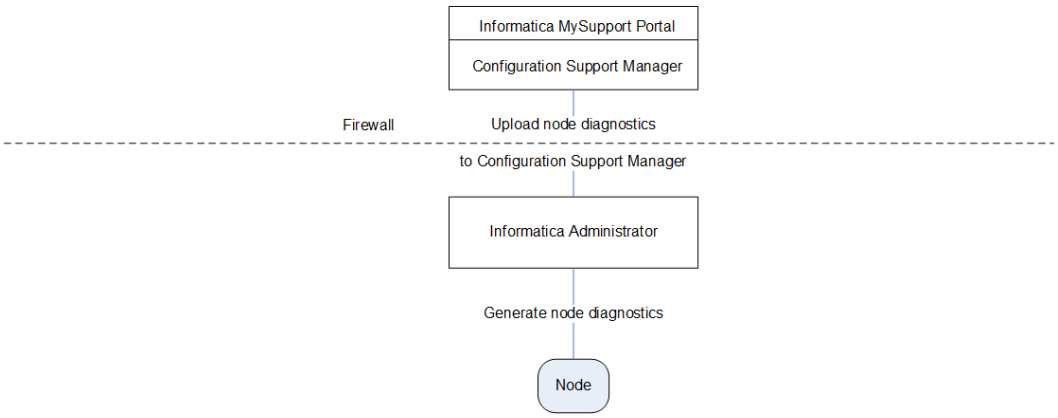
Node Diagnostics Overview

The Configuration Support Manager is a web-based application that you can use to track Informatica updates and diagnose issues in your environment.

You can discover comprehensive information about your technical environment and diagnose issues before they become critical.

Generate node diagnostics from the Informatica Administrator and upload them to the Configuration Support Manager in the Informatica MySupport Portal. Then, check the node diagnostics against business rules and recommendations in the Configuration Support Manager.

The following image shows the operational flow to generate and upload node diagnostics:



Complete the following tasks to generate and upload node diagnostics:

1. Log in to the Informatica MySupport Portal.

2. Generate node diagnostics. The Service Manager analyzes the services of the node and generates node diagnostics including information such as operating system details, CPU details, database details, and patches.
3. Optionally, download node diagnostics to your local drive.
4. Upload node diagnostics to the Configuration Support Manager, a diagnostic web application outside the firewall. The Configuration Support Manager is a part of the Informatica MySupport Portal. The Service Manager connects to the Configuration Support Manager through the HTTPS protocol and uploads the node diagnostics.
5. Review the node diagnostics in the Configuration Support Manager to find troubleshooting information for your environment.

Informatica Network Login

You must log in to the Informatica Network to upload node diagnostics to the Configuration Support Manager. The login credentials are not specific to a user. The same credentials are applicable for all users who have access to the Administrator tool. Register at http://communities.informatica.com if you do not have the customer portal login details. You need to enter the customer portal login details and save these details. Alternatively, you can enter the customer portal details each time you upload node diagnostics to the Configuration Support Manager. You can generate node diagnostics without entering the login details.

To maintain login security, you must log out of the Configuration Support Manager and the Node Diagnostics Upload page of the Administrator tool.

- To log out of the Configuration Support Manager, click the logout link.
- To log out of the Upload page, click **Close Window**.

Note: If you close these windows through the web browser close button, you remain logged in to the Configuration Support Manager. Other users can access the Configuration Support Manager without valid credentials.

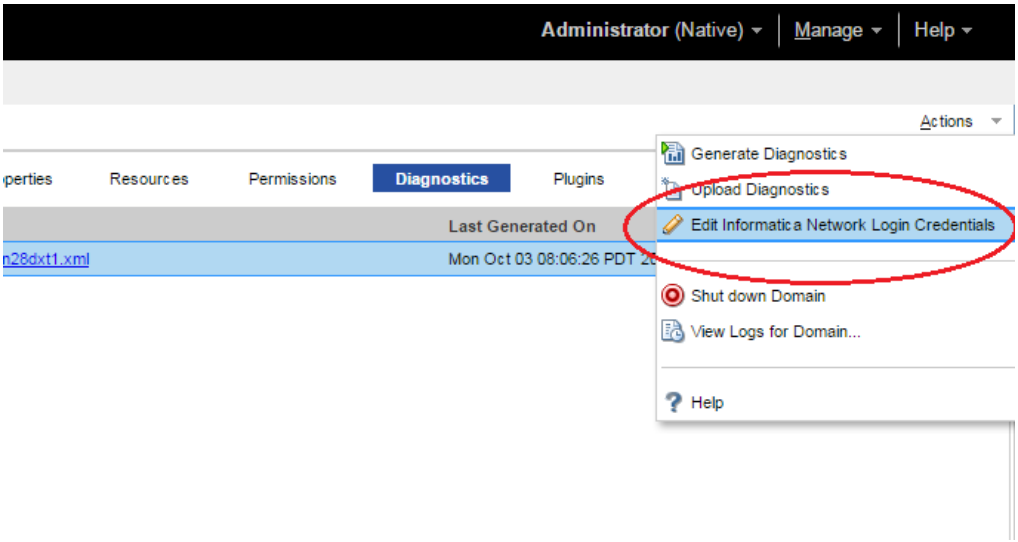
Logging In to the Informatica Network

Before you generate and upload node diagnostics, you must log in to the Informatica Network.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the contents panel, click **Diagnostics**.

   A list of all the nodes in the domain appears.

4. Click the **Actions** menu in the upper right corner of the page and select **Edit Informatica Network Login Credentials**:



   The **Edit Informatica Network Login Credentials** dialog box appears.

5. Enter the following customer portal login details:

Field	Description
Email Address	Email address with which you registered your customer portal account.
Password	Password for your customer portal account.
Project ID	Unique ID assigned to your support project.

6. Click **OK**.

Generating Node Diagnostics

When you generate node diagnostics, the Administrator tool generates node diagnostics in an XML file.

The XML file contains details about services, logs, environment variables, operating system parameters, system information, and database clients. Worker node diagnostics only contain node metadata; they do not include domain metadata.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the contents panel, click **Diagnostics**.

   A list of all nodes in the domain appears.

4. Select the node.
5. Click **Generate Diagnostics File**.
6. Click **Yes** to confirm that you want to generate node diagnostics.

Note: You can also generate diagnostics from the **Actions** menu on the **Diagnostics** tab.

The csmagent<host name>.xml file, which contains the node diagnostics, is generated at `INFA_HOME/server/csm/output`. The node diagnostics and the time stamp of the generated file appear.

7. To run diagnostics for your environment, upload the csmagent<host name>.xml file to the Configuration Support Manager.

   Alternatively, you can download the XML file to your local drive.

After you generate node diagnostics for the first time, you can regenerate or upload them.

Downloading Node Diagnostics

After you generate node diagnostics, you can download the file to your local drive.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the contents panel, click **Diagnostics**.

   A list of all nodes in the domain appears.
4. Click the diagnostics file name of the node.

   The file opens in another browser window.
5. Click **File** > **Save As**. Then, specify a location to save the file.
6. Click **Save**.

   The XML file is saved to your local drive.

Uploading Node Diagnostics

You can upload node diagnostics to the Configuration Support Manager through the Administrator tool. You must enter the customer portal login details before you upload node diagnostics.

When you upload node diagnostics, you can update or create a configuration in the Configuration Support Manager. Create a configuration the first time you upload the node diagnostics. Update a configuration to view the latest diagnostics of the configuration. To compare current and previous node configurations of an existing configuration, upload the current node diagnostics as a new configuration.

Note: If you do not have access to the Internet, you can download the file and upload it at a later time. You can also send the file to the Informatica Global Customer Support in an email to troubleshoot or to upload.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the domain.
3. In the contents panel, click **Diagnostics**.

   A list of all nodes in the domain appears.
4. Select the node.
5. Generate node diagnostics.
6. Click **Upload Diagnostics File to CSM**.

You can upload the node diagnostics as a new configuration or as an update to an existing configuration.

7. To upload a new configuration, go to step 10.

   To update a configuration, select **Update an existing configuration**.
8. Select the configuration you want to update from the list of configurations.
9. Go to step 12.
10. Select **Upload as a new configuration**.
11. Enter the following configuration details:

Field	Description
Name	Configuration name.
Description	Configuration description.
Type	Type of the node, which is one of the following types: - Production - Development - Test/QA

12. Click **Upload Now**.

    After you upload the node diagnostics, go to the Configuration Support Manager to analyze the node diagnostics.
13. Click **Close Window**.

    Note: If you close the window by using the close button in the browser, the user authentication session does not end and you cannot upload node diagnostics to the Configuration Support Manager with another set of customer portal login credentials.

Analyzing Node Diagnostics

Use the Configuration Support Manager to analyze node diagnostics.

Use the Configuration Support Manager to complete the following tasks:

- Diagnose issues before they become critical.
- Identify bug fixes.
- Identify recommendations that can reduce risk of unplanned outage.
- View details of your technical environment.
- Manage your configurations efficiently.
- Subscribe to proactive alerts through email and RSS.
- Run advanced diagnostics with compare configuration.

Identify Bug Fixes

You can use the Configuration Support Manager to resolve issues encountered during operations. To expedite resolution of support issues, you can generate and upload node diagnostics to the Configuration

Support Manager. You can analyze node diagnostics in the Configuration Support Manager and find a solution to your issue.

For example, when you run a Sorter session that processes a large volume of data, you notice that there is some data loss. You generate node diagnostics and upload them to the Configuration Support Manager. When you review the diagnostics for bug fix alerts, you see that a bug fix, EBF178626, is available for this. You apply EBF178626, and run the session again. All data is successfully loaded.

Identify Recommendations

You can use the Configuration Support Manager to avoid issues in your environment. You can troubleshoot issues that arise after you make changes to the node properties by comparing different node diagnostics in the Configuration Support Manager. You can also use the Configuration Support Manager to identify recommendations or updates that may help you improve the performance of the node.

For example, you upgrade the node memory to handle a higher volume of data. You generate node diagnostics and upload them to the Configuration Support Manager. When you review the diagnostics for operating system warnings, you find the recommendation to increase the total swap memory of the node to twice that of the node memory for optimal performance. You increase swap space as suggested in the Configuration Support Manager and avoid performance degradation.

Tip: Regularly upload node diagnostics to the Configuration Support Manager and review node diagnostics to maintain your environment efficiently.

CHAPTER 15

Security Management

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Informatica Security Overview

You can secure the Informatica domain to protect from threats from inside and outside the network on which the domain runs.

Security for the Informatica domain includes the following types of security:

Infrastructure Security

Infrastructure security protects the Informatica domain against unauthorized access to or modification of services and resources in the Informatica domain. Infrastructure security includes the following aspects:

- Protection of data transmitted and stored within the Informatica domain
- Authentication of users and services connecting to the Informatica domain
- Security of connections for external components, including client applications and relational databases for repositories, sources, and targets.

Operational Security

Operational security controls access to the data and services in the Informatica domain. Operational security includes the following aspects:

- Setting restrictions to user access to data and metadata based on the role of the user in the organization
- Setting restrictions to user ability to perform operations within the Informatica domain based on the user role in the organization

Informatica stores the domain configuration information and the list of users authorized to access the domain in the domain configuration repository. The domain configuration repository also contains the groups, roles, privileges, and permissions that are assigned to each user in the Informatica domain.

Informatica organizes the list of users by security domains. A security domain contains a collection of user accounts. A domain can have multiple security domains.

Infrastructure Security

Infrastructure security includes user and service authentication, secure communication within the domain, and secure data storage.

Authentication

The Service Manager authenticates the services that run in the domain and the users who log in to the Informatica client tools.

You can configure the Informatica domain to use the following types of authentication:

Native Authentication

Native authentication is a mode of authentication available only for user accounts in the Informatica domain. When the Informatica domain uses native authentication, the Service Manager stores user credentials and privileges in the domain configuration repository and performs all user authentication within the Informatica domain.

If the Informatica domain uses native authentication, by default, the domain has a Native security domain and all user accounts belong to the Native security domain.

Informatica uses user name and passwords to authenticate users and services in the Informatica domain.

Lightweight Directory Access Protocol (LDAP) Authentication

LDAP is a software protocol for accessing users and resources on a network. If the Informatica domain uses LDAP authentication, the user accounts and credentials are stored in the LDAP directory service.

The user privileges and permissions are stored in the domain configuration repository. You must periodically synchronize the user accounts in the domain configuration repository with the user accounts in the LDAP directory service.

Informatica uses user name and passwords to authenticate informatica users and services in the Informatica domain.

Kerberos Authentication

Kerberos is a network authentication protocol which uses tickets to authenticate users and services in a network. When the Informatica domain uses Kerberos authentication, the user accounts and credentials are stored in the Kerberos principal database, which can be an LDAP directory service. The user privileges and permissions are stored in the domain configuration repository. You must periodically synchronize the user accounts in the domain configuration repository with the user accounts in the Kerberos principal database.

Informatica uses the Kerberos tickets to authenticate Informatica users and services in the Informatica domain.

Secure Domain Communication

The Informatica domain has various options to secure the data and metadata that are transmitted between the Service Manager and services in the domain and the client applications. Informatica uses the TCP/IP and HTTP protocols to communicate between components in the domain and uses SSL certificates to secure the communication between services and the Service Manager in the domain.

The SSL/TLS protocol uses public key cryptography to encrypt and decrypt network traffic. The public key used to encrypt and decrypt traffic is stored in an SSL certificate that can be self-signed or signed. A self-signed certificate is signed by the creator of the certificate. Because the identity of the signer is not verified, a self-signed certificate is less secure than a signed certificate. A signed certificate is an SSL certificate that has the identity of the person who requested the certificate verified by a certificate authority (CA). Informatica recommends CA signed certificates for a higher level of security.

A keystore contains private keys and certificates. It is used to provide a credential. A truststore contains the certificate of trusted SSL/TLS servers. It is used to verify a credential.

To secure connections in the domain, Informatica requires keystores and truststores in PEM and JKS formats. You can use the following programs to create the required files:

keytool

Use keytool to create an SSL certificate or a Certificate Signing Request (CSR) as well as keystores and truststores in JKS format.

For more information about keytool, see the documentation on the following website:

http://docs.oracle.com/javase/7/docs/technotes/tools/windows/keytool.html

OpenSSL

You can use OpenSSL to create an SSL certificate or CSR as well as convert a keystore in JKS format to PEM format.

For more information about OpenSSL, see the documentation on the following website:

https://www.openssl.org/docs/

The type of connection that you secure determines the files required.

Secure Data Storage

Informatica encrypts sensitive data, such as passwords and secure connection parameters, before it stores the data in the domain configuration repository. Informatica also saves sensitive files, such as configuration files, in a secure directory.

Operational Security

You can assign privileges, roles, and permissions to users or groups of users to manage the level of access users and groups can have and the scope of the actions that users and groups can perform in the domain.

You can use the following methods to manage user and group access in the domain:

Privileges

Privileges determine the actions that users can perform in the Informatica client tools. You can assign a set of privileges to a user to restrict access to the services available in the domain. You can also assign privileges to a group to allow all users in the group the same access to services.

Roles

A role is a set of privileges that you can assign to users or groups. You can use roles to more easily manage assignments of privileges to users. You can create a role with limited privileges and assign it to users and groups that have restricted access to domain services. Or you can create roles with related privileges to assign to users and groups that require the same level of access.

Permissions

Permissions define the level of access that users have to an object. A user who has the privilege to perform a certain action might require permission to perform the action on a particular object. For example, to manage an application service, a user must have the privilege to manage services and permission on the specific application service.

Default Administrator Group

The Informatica domain has a system-defined Administrator group that includes all privileges and permissions for a service. Any user account that you add to the Administrator group has privileges and permissions on all services and objects in the domain. When you install Informatica services, the installer creates a user account that belongs to the Administrator group. You can use the default Administrator account to initially log in to the Administrator tool.

Domain Configuration Repository

The domain configuration repository contains information about the domain configuration and user privileges and permissions.

If the Informatica domain uses native user authentication, the domain configuration repository also contains the user credentials. If the domain uses LDAP authentication, the domain configuration repository does not contain the user credentials. All LDAP user credentials are stored outside the Informatica domain, in the LDAP directory service.

When you create the Informatica domain during installation, the installer creates a domain configuration repository in a relational database. You must specify the database in which to create the domain configuration repository. You can create the repository on a database secured with the SSL protocol.

Security Domain

A security domain is a collection of user accounts and groups in the Informatica domain.

The Informatica domain can have the following types of security domains:

Native Security Domain

The Native security domain contains the users and groups created and managed in the Administrator tool. Informatica stores all credentials for user accounts in the Native security domain in the domain configuration repository. By default, the Native security domain is created during installation. After installation, you cannot create additional Native security domains or delete the Native security domain.

LDAP Security Domain

An LDAP security domain contains users and groups imported from an LDAP directory service. If the Informatica domain uses LDAP authentication, you can create an LDAP security domain and add users and groups that you import from the LDAP directory service.

When you install Informatica services and create a domain that uses native or LDAP authentication, the installer creates the Native security domain but does not create an LDAP security domain. You can create LDAP security domains after installation.

User Authentication Overview

User authentication in the Informatica domain depends on the type of authentication that you configure when you install the Informatica services.

The Informatica domain can use the following types of authentication to authenticate users in the Informatica domain:

- Native user authentication
- LDAP user authentication

Native user accounts are stored in the Informatica domain and can only be used within the Informatica domain. LDAP user accounts are stored in an LDAP directory service and are shared by applications within the enterprise.

You can select the type of authentication to use in the Informatica domain during installation. You can use native authentication and LDAP authentication together in the Informatica domain. The Service Manager authenticates the users based on the security domain. If a user belongs to the native security domain, the Service Manager authenticates the user in the domain configuration repository. If the user belongs to an LDAP security domain, the Service Manager passes the user name and password to the LDAP server for authentication.

Native User Authentication

If the Informatica domain uses native authentication, the Service Manager stores all user account information and performs all user authentication within the Informatica domain. When a user logs in, the Service Manager uses the native security domain to authenticate the user name and password.

The native security domain is created at installation and cannot be deleted. An Informatica domain can have only one native security domain. You create and maintain user accounts in the native security domain in the Administrator tool. The Service Manager stores details about the user accounts, including the user credentials and privileges, in the domain configuration repository.

LDAP User Authentication

You can configure the Informatica domain to allow users in an LDAP directory service to log in to Informatica client applications. The Informatica domain can use LDAP user authentication in addition to native user authentication.

To enable the Informatica domain to use LDAP user authentication, you must set up a connection to an LDAP server and specify the users and groups from the LDAP directory service that can have access to the Informatica domain. You can use the Administrator tool to set up the connection to the LDAP server.

When you synchronize the LDAP security domains with the LDAP directory service, the Service Manager imports the list of LDAP user accounts with access to the Informatica domain into the LDAP security domains. When you assign privileges and permissions to users in LDAP security domains, the Service Manager stores the information in the domain configuration repository. The Service Manager does not store the user credentials in the domain configuration repository.

When a user logs in, the Service Manager passes the user name and password to the LDAP server for authentication.

Note: The Service Manager requires that LDAP users log in to a client application with a password even though an LDAP directory service may allow a blank password for anonymous login mode.

LDAP Security Domains Overview

An LDAP security domain contains a set of users and groups that are imported from an LDAP directory service. You must create an LDAP security domain if you use LDAP user authentication.

Configure the LDAP security domains to store the list of users from an LDAP directory service that you want to allow access to the Informatica domain and client applications. The LDAP security domain does not store user account credentials. When a user logs in to an Informatica client, the Service Manager verifies that the user account is in a security domain. If the user account belongs to an LDAP security domain, the Service Manager authenticates the user with the LDAP directory service.

After installation, you can add users and groups to the native security domain. If you have users in an LDAP directory service that you want to give access to Informatica client applications, you can set up LDAP security domains in addition to the native security domain. Configure a connection to the LDAP server and import the users and groups into the LDAP security domains.

After installation, you can configure a connection to the LDAP server and import users and groups from the LDAP directory service into the LDAP security domain.

Setting Up an LDAP Security Domain

You can create an LDAP security domain for user accounts that you import from an LDAP directory service. To organize different groups of users, you can create multiple LDAP security domains.

You create and manage LDAP users and groups in the LDAP directory service. Set up a connection to the LDAP server and use search filters to specify the users and groups that can have access to the Informatica domain. Then import the user accounts into LDAP security domains. If the LDAP server uses the SSL protocol, you must also specify the location of the SSL certificate.

You can import users from the following LDAP directory services:

- Microsoft Active Directory Service

  Note: If you use Kerberos authentication, you can import users only from a Microsoft Active Directory (AD) directory service.
- Sun Java System Directory Service
- Novell e-Directory Service
- IBM Tivoli Directory Service
- Open LDAP Directory Service

After you import users into an LDAP security domain, you can assign roles, privileges, and permissions to the users. You can assign LDAP user accounts to native groups to organize them based on their roles in the Informatica domain. You cannot use the Administrator tool to create, edit, or delete users and groups in an LDAP security domain.

Use the LDAP Configuration dialog box to set up the connection to the LDAP directory service and create the LDAP security domain. You can also use the LDAP Configuration dialog box to set up a synchronization schedule.

To set up the LDAP security domain, perform the following steps:

1. Set up the connection to the LDAP directory service.
2. Configure a security domain.
3. Schedule the synchronization times.

Step 1. Set Up the Connection to the LDAP Server

Configure the connection to the LDAP server that contains the directory service from which you want to import the user accounts for the Informatica domain.

When you configure the connection to the LDAP server, indicate that the Service Manager must ignore the case-sensitivity of the distinguished name attributes of the LDAP user accounts when it assigns users to groups in the Informatica domain. If the Service Manager does not ignore case sensitivity, the Service Manager might not assign all the users that belong to a group.

If you modify the LDAP connection properties to connect to a different LDAP directory service, ensure that the user and group filters in the LDAP security domains are correct for the new LDAP directory service. Verify that the filters include the users and groups that you want to use in the Informatica domain.

To set up a connection to the LDAP directory service, perform the following tasks:

1. In the Administrator tool, click the **Security** tab.
2. Click the **Actions** menu and select **LDAP Configuration**.
3. In the **LDAP Configuration** dialog box, click the **LDAP Connectivity** tab.

4. Configure the connection properties for the LDAP server.

   You might need to consult the LDAP administrator to get the information about the LDAP server.

   The following table describes the LDAP server configuration properties:

Property	Description
Server name	Name of the machine hosting the LDAP directory service.
Port	Listening port for the LDAP server. This is the port number to communicate with the LDAP directory service. Typically, the LDAP server port number is 389. If the LDAP server uses SSL, the LDAP server port number is 636. The maximum port number is 65535.
LDAP Directory Service	Type of LDAP directory service. Select from the following directory services: - Microsoft Active Directory Service - Sun Java System Directory Service - Novell e-Directory Service - IBM Tivoli Directory Service - Open LDAP Directory Service **Note:** If you use Kerberos authentication, you must select Microsoft Active Directory Service.
Name	Distinguished name (DN) for the principal user. The user name often consists of a common name (CN), an organization (O), and a country (C). The principal user name is an administrative user with access to the directory. Specify a user that has permission to read other user entries in the LDAP directory service. Leave blank for anonymous login. For more information, see the documentation for the LDAP directory service.
Password	Password for the principal user. Leave blank for anonymous login. Not available if you use Kerberos authentication.
Use SSL Certificate	Indicates that the LDAP server uses the Secure Socket Layer (SSL) protocol.
Trust LDAP Certificate	Determines whether the Service Manager can trust the SSL certificate of the LDAP server. If selected, the Service Manager connects to the LDAP server without verifying the SSL certificate. If not selected, the Service Manager verifies that the SSL certificate is signed by a certificate authority before connecting to the LDAP server. To enable the Service Manager to recognize a self-signed certificate as valid, specify the truststore file and password to use.
Not Case Sensitive	Indicates that the Service Manager must ignore case-sensitivity for distinguished name attributes when assigning users to groups. Enable this option.

Property	Description
Group Membership Attribute	Name of the attribute that contains group membership information for a user. This is the attribute in the LDAP group object that contains the DNs of the users or groups who are members of a group. For example, *member* or *memberof*.
Maximum Size	Maximum number of user accounts to import into a security domain. For example, if the value is set to 100, you can import a maximum of100 user accounts into the security domain. If the number of user to be imported exceeds the value for this property, the Service Manager generates an error message and does not import any user. Set this property to a higher value if you have many users to import. Default is 1000.

5. Click Test Connection to verify that the connection to the LDAP server is valid.

Step 2. Configure Security Domains (LDM)

Create a security domain for each set of user accounts and groups you want to import from the LDAP directory service. Set up search bases and filters to define the set of user accounts and groups to include in a security domain. The Service Manager uses the user search bases and filters to import user accounts and the group search bases and filters to import groups. The Service Manager imports groups and the list of users that belong to the groups. It imports the groups that are included in the group filter and the user accounts that are included in the user filter.

The names of users and groups to be imported from the LDAP directory service must conform to the same rules as the names of native users and groups. The Service Manager does not import LDAP users or groups if names do not conform to the rules of native user and group names.

Note: Unlike native user names, LDAP user names can be case-sensitive.

When you set up the LDAP directory service, you can use different attributes for the unique ID (UID). The Service Manager requires a particular UID to identify users in each LDAP directory service. Before you configure the security domain, verify that the LDAP directory service uses the required UID.

The following table lists the required UID for each LDAP directory service:

LDAP Directory Service	UID
IBMTivoliDirectory	uid
Microsoft Active Directory	sAMAccountName
NovellE	uid
OpenLDAP	uid
SunJavaSystemDirectory	uid

The Service Manager does not import the LDAP attribute that indicates that a user account is enabled or disabled. You must enable or disable an LDAP user account in the Administrator tool. The status of the user account in the LDAP directory service affects user authentication in application clients. For example, a user account is enabled in the Informatica domain but disabled in the LDAP directory service. If the LDAP directory service allows disabled user accounts to log in, then the user can log in to application clients. If the

LDAP directory service does not allow disabled user accounts to log in, then the user cannot log in to application clients.

Note: If you modify the LDAP connection properties to connect to a different LDAP server, the Service Manager does not delete the existing security domains. You must ensure that the LDAP security domains are correct for the new LDAP server. Modify the user and group filters in the security domains or create additional security domains so that the Service Manager correctly imports the users and groups that you want to use in the Informatica domain.

To configure an LDAP security domain, perform the following steps:

1. In the Administrator tool, click the **Security** tab.
2. Click the **Actions** menu and select **LDAP Configuration**.
3. In the **LDAP Configuration** dialog box, click the **Security Domains** tab.
4. Click **Add**.
5. Use LDAP query syntax to create filters to specify the users and groups to be included in the security domain you are creating.

   You might need to consult the LDAP administrator to get the information about the users and groups available in the LDAP directory service.

   The following table describes the filter properties that you can set for a security domain:

Property	Description
Security Domain	Name of the LDAP security domain. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or contain the following special characters: , + / < > @ ; \ % ? The name can contain an ASCII space character except for the first and last character. All other space characters are not allowed.
User search base	Distinguished name (DN) of the entry that serves as the starting point to search for user names in the LDAP directory service. The search finds an object in the directory according to the path in the distinguished name of the object. For example, in Microsoft Active Directory, the distinguished name of a user object might be cn=UserName,ou=OrganizationalUnit,dc=DomainName, where the series of relative distinguished names denoted by dc=DomainName identifies the DNS domain of the object.
User filter	An LDAP query string that specifies the criteria for searching for users in the directory service. The filter can specify attribute types, assertion values, and matching criteria. For example: `(objectclass=*)` searches all objects. `(&(objectClass=user)(!(cn=susan)))` searches all user objects except "susan." For more information about search filters, see the documentation for the LDAP directory service.
Group search base	Distinguished name (DN) of the entry that serves as the starting point to search for group names in the LDAP directory service.
Group filter	An LDAP query string that specifies the criteria for searching for groups in the directory service.

6. Click **Preview** to view a subset of the list of users and groups that fall within the filter parameters.

   If the preview does not display the correct set of users and groups, modify the user and group filters and search bases to get the correct users and groups.
7. To add another LDAP security domain, repeat steps *1* through *6*.

8. To immediately synchronize the users and groups in the security domains with the users and groups in the LDAP directory service, click **Synchronize Now**.

   The Service Manager synchronizes the users in all the LDAP security domains with the users in the LDAP directory service. The time it takes for the synchronization process to complete depends on the number of users and groups to be imported.

9. Click **OK** to save the security domains.

Step 3. Schedule the Synchronization Times

You can set up a schedule for the Service Manager to periodically synchronize the list of users and groups in the LDAP security domain with the list of users and groups in the LDAP directory service.

Important: Before you start the synchronization process, verify that the /etc/hosts file contains an entry for the host name of the LDAP server. If the Service Manager cannot resolve the host name for the LDAP server, the user synchronization can fail.

During synchronization, the Service Manager imports users and groups from the LDAP directory service. The Service Manager deletes any user or group from the LDAP security domain that is no longer included in the search filters used for the import.

By default, the Service Manager does not have a scheduled time to synchronize with the LDAP directory service. To ensure that the list of users and groups in the LDAP security domains is accurate, you can schedule the times during the day when the Service Manager synchronizes the LDAP security domains. The Service Manager synchronizes the LDAP security domains with the LDAP directory service every day at the times you set.

Note: During synchronization, the Service Manager locks the user account that it synchronizes. When the user account is locked, the Service Manager cannot authenticate the user account. Users might not be able to log in to application clients. If users are logged in to application clients when synchronization starts, the users might not be able to perform tasks. The duration of the synchronization process depends on the number of users and groups to be synchronized. To avoid usage disruption, synchronize the security domains during times when most users are not logged in. To synchronize more than 100 users or groups, enable paging on the LDAP directory service before you run the synchronization. If you do not enable paging on the LDAP directory service, the synchronization can fail.

To set up a schedule to synchronize the LDAP security domains with the LDAP directory service, perform the following steps:

1. In the Administrator tool, click the **Security** tab.
2. Click the **Actions** menu and select **LDAP Configuration**.
3. In the **LDAP Configuration** dialog box, click the **Schedule** tab.
4. Click the **Add** button (+) to add a time.

   The synchronization schedule uses a 24-hour time format.

   You can add as many synchronization times in the day as you require. If the list of users and groups in the LDAP directory service changes often, you can schedule the Service Manager to synchronize multiple times a day.

5. To immediately synchronize the users and groups in the security domains with the users and groups in the LDAP directory service, click **Synchronize Now**.
6. Click **OK** to save the synchronization schedule.

   Note: If you restart the Informatica domain before the Service Manager synchronizes with the LDAP directory service, the synchronization times that you added are lost.

Using Nested Groups in the LDAP Directory Service

An LDAP security domain can contain nested LDAP groups. The Service Manager can import nested groups that are created in the following manner:

- Create the groups under the same organizational units (OU).
- Set the relationship between the groups.

For example, you want to create a nested grouping where GroupB is a member of GroupA and GroupD is a member of GroupC.

1. Create GroupA, GroupB, GroupC, and GroupD within the same OU.
2. Edit GroupA, and add GroupB as a member.
3. Edit GroupC, and add GroupD as a member.

You cannot import nested LDAP groups into an LDAP security domain that are created in a different way.

Using a Self-Signed SSL Certificate

You can connect to an LDAP server that uses an SSL certificate signed by a certificate authority (CA). By default, the Service Manager does not connect to an LDAP server that uses a self-signed certificate.

To use a self-signed certificate, import the self-signed certificate into a truststore file and use the INFA_JAVA_OPTS environment variable to specify the truststore file and password:

```
setenv INFA_JAVA_OPTS -Djavax.net.ssl.trustStore=<TrustStoreFile>
-Djavax.net.ssl.trustStorePassword=<TrustStorePassword>
```

On Windows, configure INFA_JAVA_OPTS as a system variable.

Restart the node for the change to take effect. The Service Manager uses the truststore file to verify the SSL certificate.

keytool is a key and certificate management utility that allows you to generate and administer keys and certificates for use with the SSL security protocol. You can use keytool to create a truststore file or to import a certificate to an existing truststore file. You can find the keytool utility in the following directory:

```
<PowerCenterClientDir>\CMD_Utilities\PC\java\bin
```

For more information about using keytool, see the documentation on the following web site: http://docs.oracle.com/javase/7/docs/technotes/tools/windows/keytool.html.

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Deleting an LDAP Security Domain

To permanently prohibit users in an LDAP security domain from accessing application clients, you can delete the LDAP security domain. When you delete an LDAP security domain, the Service Manager deletes all user accounts and groups in the LDAP security domain from the domain configuration database.

1. In the LDAP Configuration dialog box, click the Security Domains tab.

   The LDAP Configuration dialog box displays the list of security domains.
2. To ensure that you are deleting the correct security domain, click the security domain name to view the filter used to import the users and groups and verify that it is the security domain you want to delete.

3. Click the Delete button next to a security domain to delete the security domain.
4. Click OK to confirm that you want to delete the security domain.

Privileges and Roles Overview

You manage user security with privileges and roles.

Privileges

Privileges determine the actions that users can perform in application clients. Informatica includes the following privileges:

- Domain privileges. Determine actions that users can perform on the Informatica domain using the Administrator tool and the infacmd and pmrep command line programs.
- Analyst Service privilege. Determines actions that users can perform using Informatica Analyst.
- Content Management Service privilege. Determines actions that users can perform using reference tables in the Informatica Developer tool and the Informatica Analyst tool.
- Data Integration Service privilege. Determines actions on applications that users can perform using the Administrator tool and the infacmd command line program. This privilege also determines whether users can drill down and export profile results.
- Metadata Manager Service privileges. Determine actions that users can perform using Metadata Manager.
- Model Repository Service privilege. Determines actions on projects that users can perform using Informatica Analyst and Informatica Developer.
- PowerCenter Repository Service privileges. Determine PowerCenter repository actions that users can perform using the Repository Manager, Designer, Workflow Manager, Workflow Monitor, and the pmrep and pmcmd command line programs.
- PowerExchange application service privileges. Determine actions that users can perform on the PowerExchange Listener Service and PowerExchange Logger Service using the infacmd pwx commands.
- Reporting Service privileges. Determine reporting actions that users can perform using Data Analyzer.
- Reporting and Dashboards Service privileges. Determine actions that users can perform using Jaspersoft.
- Scheduler Service privileges. Determine actions that users can perform using the Scheduler Service.
- Test Data Manager Service privileges. Determine data discovery, data masking, data subset, and test data generation tasks that users can perform using the Test Data Manager.

You assign privileges to users and groups for application services. You can assign different privileges to a user for each application service of the same service type.

You assign privileges to users and groups on the **Security tab** of the Administrator tool.

The Administrator tool organizes privileges into levels. A privilege is listed below the privilege that it includes. Some privileges include other privileges. When you assign a privilege to users and groups, the Administrator tool also assigns any included privileges.

Privilege Groups

The domain and application service privileges are organized into privilege groups. A privilege group is an organization of privileges that define common user actions. For example, the domain privileges include the following privilege groups:

- Tools. Includes privileges to log in to the Administrator tool.
- Security Administration. Includes privileges to manage users, groups, roles, and privileges.
- Domain Administration. Includes privileges to manage the domain, folders, nodes, grids, licenses, and application services.

Tip: When you assign privileges to users and user groups, you can select a privilege group to assign all privileges in the group.

Roles

A role is a collection of privileges that you assign to a user or group. Each user within an organization has a specific role, whether the user is a developer, administrator, basic user, or advanced user.

For example, the PowerCenter Developer role includes all the PowerCenter Repository Service privileges or actions that a developer performs.

You assign a role to users and groups for the domain and for application services in the domain.

Tip: If you organize users into groups and then assign roles and permissions to the groups, you can simplify user administration tasks. For example, if a user changes positions within the organization, move the user to another group. If a new user joins the organization, add the user to a group. The users inherit the roles and permissions assigned to the group. You do not need to reassign privileges, roles, and permissions. For more information, see the Informatica How-To Library article
Using Groups and Roles to Manage Informatica Access Control.

Domain Privileges

Domain privileges determine the actions that users can perform using the Administrator tool and the infacmd and pmrep command line programs.

The following table describes each domain privilege group:

Privilege Group	Description
Security Administration	Includes privileges to manage users, groups, roles, and privileges.
Domain Administration	Includes privileges to manage the domain, folders, nodes, grids, licenses, application services, and connections.
Monitoring	Includes privileges to configure monitoring statistics and reports, view monitoring for integration objects, and access monitoring.
Tools	Includes privileges to log in to the Administrator tool.
Cloud Administration	Includes privileges to add Informatica Cloud organizations in the Administrator tool and view them.

Analyst Service Privilege

The Analyst Service privilege determines actions that licensed users can perform on projects using the Analyst tool.

The following table lists the privileges and permissions required to manage projects and objects in projects:

Privilege	Permission	Description
Run Profiles and Scorecards	Read on projects. Execute on relational data source connection.	User is able to run profiles and scorecards for licensed users in the Analyst tool.
Access Mapping Specifications	Read on projects.	User is able to access mapping specifications for licensed users in the Analyst tool.
Load Mapping Specification Results	Write on projects.	User is able to load the results of a mapping specification for licensed users to a table or flat file. **Note:** Selecting this privilege also grants the **Access Mapping Specifications** privilege by default.
Manage Glossaries	-	User is able to manage the business glossary.
View Glossaries	-	User is able to view published Business Glossary assets in the Library workspace. This is equivalent to providing read permission for glossaries and Glossary assets in the Glossary Security workspace.
Workspace Access	-	User is able to access the following workspaces in the Analyst tool: - **Design** workspace. - **Discovery** workspace. - **Glossary** workspace. - **Scorecards** workspace. **Note:** Selecting this privilege also grants access to projects in the Analyst tool. If the user does not have this privilege, the user must have either the **Design Workspace**, **Discovery Workspace**, **Glossary Workspace**, or **Scorecards Workspace** privilege to access projects.
Design Workspace	-	User is able to access the **Design** workspace.
Discovery Workspace	-	User is able to access the **Discovery** workspace.
Glossary Workspace	-	User is able to access the **Glossary** workspace.
Scorecards Workspace	-	User is able to access the **Scorecards** workspace.

Content Management Service Privileges

The Content Management Service privileges determine actions that licensed users can perform on reference tables.

The following table lists the privileges and permissions required to manage reference tables:

Privilege	Permission	Description
Create Reference Tables	Write on project	- Create a reference table in the Analyst and Developer tool. - Create a reference table with infacmd rtm import. - Import a reference table object to the Model repository. - Copy a reference table in the Analyst and Developer tool. - Create a reference table from profile data. **Note:** The Create privilege also grants the Edit privilege by default.
Edit Reference Table Data and Metadata	Read on project	- Edit reference table data values in the Developer tool and Analyst tool. - Add profile data to a reference table. - Add or delete columns in a reference table. Change reference table metadata such as column names, descriptions, and default values.

Data Integration Service Privilege

The Data Integration Service privileges determine actions that users can perform on applications using the Administrator tool and the infacmd command line program. They also determine whether users can drill down and export profile results using the Analyst tool and the Developer tool.

The following table lists the required permissions and the actions that users can perform with the privilege in the Application Administration privilege group:

Privilege Name	Permission On	Description
Manage Applications	Data Integration Service	User is able to perform the following actions: - Back up and restore an application to a file. - Deploy an application to a Data Integration Service and resolve name conflicts. - Start an application after deployment. - Find an application. - Start or stop objects in an application. - Configure application properties.

The following table lists the required permissions and the actions that users can perform with the privilege in the Profiling Administration privilege group:

Privilege Name	Permission On	Description
Drilldown and Export Results	Read on project Execute on relational data source connection is also required to drill down on live data	User is able to perform the following actions: - Drill down profiling results. - Export profiling results.

Managing Roles

A role is a collection of privileges that you can assign to users and groups. You can assign the following types of roles:

- System-defined. Roles that you cannot edit or delete.
- Custom. Roles that you can create, edit, and delete.

A role includes privileges for the domain or an application service type. You assign roles to users or groups for the domain or for each application service in the domain. For example, you can create a Developer role that includes privileges for the PowerCenter Repository Service. A domain can contain multiple PowerCenter Repository Services. You can assign the Developer role to a user for the Development PowerCenter Repository Service. You can assign a different role to that user for the Production PowerCenter Repository Service.

When you select a role in the Roles section of the Navigator, you can view all users and groups that have been directly assigned the role for the domain and application services. You can view the role assignments by users and groups or by services. To navigate to a user or group listed in the Assignments section, right-click the user or group and select Navigate to Item.

You can search for system-defined and custom roles.

System-Defined Roles

A system-defined role is a role that you cannot edit or delete. The Administrator role is a system-defined role.

When you assign the Administrator role to a user or group for the domain, Analyst Service, Data Integration Service, Metadata Manager Service, Model Repository Service, PowerCenter Repository Service, or Reporting Service, the user or group is granted all privileges for the service. The Administrator role bypasses permission checking. Users with the Administrator role can access all objects managed by the service.

Administrator Role

When you assign the Administrator role to a user or group for the domain or the Data Integration Service, the user or group can complete some tasks that are determined by the Administrator role, not by privileges or permissions.

You can assign a user or group all privileges for the domain or the Data Integration Service and then grant the user or group full permissions on all domains. However, this user or group cannot complete the tasks determined by the Administrator role.

For example, a user assigned the Administrator role for the domain can configure domain properties in the Administrator tool. A user assigned all domain privileges and permission on the domain cannot configure domain properties.

The following table lists the tasks determined by the Administrator role for the domain or the Data Integration Service:

Service	Tasks
Domain	- Configure domain properties. - Create operating system profiles. - Delete operating system profiles. - Grant permission on the domain and operating system profiles. - Manage and purge log events. - Receive domain alerts. - Run the License Report. - View user activity log events. - Shut down the domain. - Access the service upgrade wizard.
Data Integration Service	- Upgrade the Data Integration Service using the Actions menu.

Custom Roles

A custom role is a role that you can edit or delete.

By default, the Administrator tool includes the following custom roles:

- Analyst Service custom role
- Metadata Manager Service custom roles
- Operator custom role
- PowerCenter Repository Service custom roles
- Reporting Service custom roles
- Test Data Manager Service custom roles

You can edit the privileges for these roles, or delete the roles. You can also create your own custom roles.

Creating Custom Roles

When you create a custom role, you assign privileges to the role for the domain or for an application service type. A role can include privileges for one or more services.

1. In the Administrator tool, click the Security tab.
2. On the Security Actions menu, click Create Role.

   The Create Role dialog box appears.

3. Enter the following properties for the role:

Property	Description
Name	Name of the role. The role name is case insensitive and cannot exceed 128 characters. It cannot include a tab, newline character, or the following special characters: , + " \ < > ; / * % ? The name can include an ASCII space character except for the first and last character. All other space characters are not allowed.
Description	Description of the role. The description cannot exceed 765 characters or include a tab, newline character, or the following special characters: < > "

4. Click the Privileges tab.
5. Expand the domain or an application service type.
6. Select the privileges to assign to the role for the domain or application service type.
7. Click OK.

Editing Properties for Custom Roles

When you edit a custom role, you can change the description of the role. You cannot change the name of the role.

1. In the Administrator tool, click the Security tab.
2. In the Roles section of the Navigator, select a role.
3. Click Edit.
4. Change the description of the role and click OK.

Editing Privileges Assigned to Custom Roles

You can change the privileges assigned to a custom role for the domain and for each application service type.

1. In the Administrator tool, click the Security tab.
2. In the Roles section of the Navigator, select a role.
3. Click the Privileges tab.
4. Click Edit.

   The Edit Roles and Privileges dialog box appears.
5. Expand the domain or an application service type.
6. To assign privileges to the role, select the privileges for the domain or application service type.
7. To remove privileges from the role, clear the privileges for the domain or application service type.
8. Repeat the steps to change the privileges for each service type.
9. Click OK.

Deleting Custom Roles

When you delete a custom role, the custom role and all privileges that it included are removed from any user or group assigned the role.

To delete a custom role, right-click the role in the Roles section of the Navigator and select Delete Role. Confirm that you want to delete the role.

Assigning Privileges and Roles to Users and Groups

You determine the actions that users can perform by assigning the following items to users and groups:

- Privileges. A privilege determines the actions that users can perform in application clients.
- Roles. A role is a collection of privileges. When you assign a role to a user or group, you assign the collection of privileges belonging to the role.

Use the following rules and guidelines when you assign privileges and roles to users and groups:

- You assign privileges and roles to users and groups for the domain and for each application service that is running in the domain.
- You can assign different privileges and roles to a user or group for each application service of the same service type.
- A role can include privileges for the domain and multiple application service types. When you assign the role to a user or group for one application service, privileges for that application service type are assigned to the user or group.

If you change the privileges or roles assigned to a user, the changed privileges or roles take effect the next time that the user logs in.

Note: You cannot edit the privileges or roles assigned to the default Administrator user account.

Inherited Privileges

A user or group can inherit privileges from the following objects:

- Group. When you assign privileges to a group, all subgroups and users belonging to the group inherit the privileges.
- Role. When you assign a role to a user, the user inherits the privileges belonging to the role. When you assign a role to a group, the group and all subgroups and users belonging to the group inherit the privileges belonging to the role. The subgroups and users do not inherit the role.

You cannot revoke privileges inherited from a group or role. You can assign additional privileges to a user or group that are not inherited from a group or role.

The Privileges tab for a user or group displays all the roles and privileges assigned to the user or group for the domain and for each application service. Expand the domain or application service to view the roles and privileges assigned for the domain or service. Click the following items to display additional information about the assigned roles and privileges:

- Name of an assigned role. Displays the role details on the details panel.
- Information icon for an assigned role. Highlights all privileges inherited with that role.

Privileges that are inherited from a role or group display an inheritance icon. The tooltip for an inherited privilege displays which role or group the user inherited the privilege from.

Assigning Privileges and Roles to a User or Group by Navigation

1. In the Administrator tool, click the Security tab.
2. In the Navigator, select a user or group.
3. Click the Privileges tab.
4. Click Edit.

   The Edit Roles and Privileges dialog box appears.

5. To assign roles, expand the domain or an application service on the Roles tab.
6. To grant roles, select the roles to assign to the user or group for the domain or application service.

   You can select any role that includes privileges for the selected domain or application service type.
7. To revoke roles, clear the roles assigned to the user or group.
8. Repeat steps *1* through *7* to assign roles for another service.
9. To assign privileges, click the Privileges tab.
10. Expand the domain or an application service.
11. To grant privileges, select the privileges to assign to the user or group for the domain or application service.
12. To revoke privileges, clear the privileges assigned to the user or group.

    You cannot revoke privileges inherited from a role or group.
13. Repeat steps *9* through *12* to assign privileges for another service.
14. Click OK.

Configuring Privileges in Catalog Service for Data Asset Analytics

You can use Data Asset Analytics with Enterprise Data Catalog to view analytical insights about the catalog in the form of reports. Analytical insights include information about users configured to access the catalog, assets and resources, asset usage, and enrichment and collaboration details associated with assets.

Before accessing Data Asset Analytics using the **Analytics** tab in Enterprise Data Catalog, make sure that you configure the required privileges for the configured users to access the tab.

Perform the following steps to configure the required permissions for the configured users to access Data Asset Analytics:

1. Log in to Informatica Administrator.
2. Click the **Security** tab.
3. Click the **Users** tab.

   The **Users** panel appears.
4. From the **Users** panel, expand the directory that includes the configured user. For example, if you have used native authentication to authenticate users, expand the **Native** directory to view the list of configured users.
5. Select the user for whom you want to configure access permissions for Data Asset Analytics.

   The **<user name>** page appears.
6. Click the **Privileges** tab.
7. Click **Edit** under the user name.

   The **Edit Roles and Privileges** dialog box appears with the **Roles** and the **Privileges** tabs.
8. Click the **Privileges** tab in the dialog box.
9. Expand the Catalog Service that you had configured.
10. Scroll down to the **Data Asset Analytics Privileges** option.

11. Select from the following privileges that you want to assign to the user, based on your requirements:

Privilege	Description
Report Management : View and Download Raw Data	The privilege to view, configure, and create analytical details in the **Data Asset Analytics** tab and download the reports in CSV format. As a user with this privilege, you cannot access the **Application Configuration** section of Data Asset Analytics to configure the **Application** and **Server Settings** tab. The reports that you create as a user with this privilege can only be accessed by you.
Dashboard Management : Visualize	The privilege to view the analytical details in the **Data Asset Analytics** tab. As a user with this privilege, you can only view the analytical charts.
Admin Management : Configuration	The administrator privilege to perform all operations in the **Data Asset Analytics** tab. The reports that you create as a user with this privilege can only be accessed by you.

12. If you have configured any custom roles and want the role to be assigned to the user, make sure that you enable the required custom role from the **Custom Role** drop-down list located in the **Roles** tab for the user. A custom role defines a set of permissions and privileges that you have configured. Instead of assigning specific permissions for each user configured, you can assign the custom role to all users who require similar permissions.
13. Click **OK**.

Viewing Users with Privileges for a Service

You can view all users that have privileges for the domain or an application service.

1. In the Administrator tool, click the Security tab.
2. On the Security Actions menu, click Service User Privileges.

   The Services dialog box appears.
3. Select the domain or an application service.

   The details panel displays all users that have privileges for the domain or application service.
4. Right-click a user name and click Navigate to Item to navigate to the user.

Troubleshooting Privileges and Roles

I cannot assign privileges or roles to users for an existing Metadata Manager Service, PowerCenter Repository Service, or Reporting Service.

You cannot assign privileges and roles to users and groups for an existing Metadata Manager Service, PowerCenter Repository Service, or Reporting Service in the following situations:

- The application service is disabled.
- The PowerCenter Repository Service is running in exclusive mode.

I cannot assign privileges to a user for an enabled Reporting Service.

Data Analyzer uses the user account name and security domain name in the format `UserName@SecurityDomain` to determine the length of the user login name. You cannot assign privileges or roles to a user for a Reporting Service when the combination of the user name, @ symbol, and security domain name exceeds 128 characters.

I removed a privilege from a group. Why do some users in the group still have that privilege?

You can use any of the following methods to assign privileges to a user:

- Assign a privilege directly to a user.
- Assign a role to a user.
- Assign a privilege or role to a group that the user belongs to.

If you remove a privilege from a group, users that belong to that group can be directly assigned the privilege or can inherit the privilege from an assigned role.

I am assigned all domain privileges and permission on all domain objects, but I cannot complete all tasks in the Administrator tool.

Some of the Administrator tool tasks are determined by the Administrator role, not by privileges or permissions. You can be assigned all privileges for the domain and granted full permissions on all domain objects. However, you cannot complete the tasks determined by the Administrator role.

I am assigned the Administrator role for an application service, but I cannot configure the application service in the Administrator tool.

When you have the Administrator role for an application service, you are an application client administrator. An application client administrator has full permissions and privileges in an application client.

However, an application client administrator does not have permissions or privileges on the Informatica domain. An application client administrator cannot log in to the Administrator tool to manage the service for the application client for which it has administrator privileges.

To manage an application service in the Administrator tool, you must have the appropriate domain privileges and permissions.

I am assigned the Administrator role for the PowerCenter Repository Service, but I cannot use the Repository Manager to perform an advanced purge of objects or to create reusable metadata extensions.

You must have the Manage Services domain privilege and permission on the PowerCenter Repository Service in the Administrator tool to perform the following actions in the Repository Manager:

- Perform an advanced purge of object versions at the PowerCenter repository level.
- Create, edit, and delete reusable metadata extensions.

My privileges indicate that I should be able to edit objects in an application client, but I cannot edit any metadata.

You might not have the required object permissions in the application client. Even if you have the privilege to perform certain actions, you may also require permission to perform the action on a particular object.

I cannot use pmrep to connect to a new PowerCenter Repository Service running in exclusive mode.

The Service Manager might not have synchronized the list of users and groups in the PowerCenter repository with the list in the domain configuration database. To synchronize the list of users and groups, restart the PowerCenter Repository Service.

I am assigned all privileges in the Folders privilege group for the PowerCenter Repository Service and have read, write, and execute permission on a folder. However, I cannot configure the permissions for the folder.

Only the folder owner or a user assigned the Administrator role for the PowerCenter Repository Service can complete the following folder management tasks:

- Assign operating system profiles to folders if the PowerCenter Integration Service uses operating system profiles. Requires permission on the operating system profile.
- Change the folder owner.
- Configure folder permissions.
- Delete the folder.
- Designate the folder to be shared.
- Edit the folder name and description.

I am assigned the Administrator role for the Metadata Manager Service, but I cannot create or restore the Metadata Manager repository.

To create or restore the Metadata Manager repository, you must be in the default Administrator group. Users in the default Administrator group have more privileges than users that are assigned the Administrator role for an application service.

I am assigned the Load Resources privilege for the Metadata Manager Service, but I get an "insufficient privileges" error when I try to load Business Glossary resources.

To load Business Glossary resources, the Load Resource, Manage Resource, and View Model privileges are required. You also need write permission on any business glossary resource that you want to load.

Domain Object Permissions

You configure privileges and permissions to manage user security within the domain. Permissions define the level of access a user has to a domain object. To log in to the Administrator tool, a user must have permission on at least one domain object. If a user has permission on an object, but does not have the domain privilege that grants the ability to modify the object type, then the user can only view the object.

For example, if a user has permission on a node, but does not have the Manage Nodes and Grids privilege, the user can view the node properties, but cannot configure, shut down, or remove the node.

You can configure permissions on the following types of domain objects:

Domain Object Type	Description of Permission
Domain	Enables Administrator tool users to access all objects in the domain. When users have permission on a domain, they inherit permission on all objects in the domain.
Folder	Enables Administrator tool users to access all objects in the folder in the Administrator tool. When users have permission on a folder, they inherit permission on all objects in the folder.
Node	Enables Administrator tool users to view and edit the node properties. Without permission, a user cannot use the node when defining an application service or creating a grid.
Grid	Enables Administrator tool users to view and edit the grid properties. Without permission, a user cannot assign the grid to a Data Integration Service or PowerCenter Integration Service.
License	Enables Administrator tool users to view and edit the license properties. Without permission, a user cannot use the license when creating an application service.
Application Service	Enables Administrator tool users to view and edit the application service properties.
Operating System Profile	Enables PowerCenter users to run workflows associated with the operating system profile. If the user that runs a workflow does not have permission on the operating system profile assigned to the workflow, the workflow fails.

You can use the following methods to manage domain object permissions:

- Manage permissions by domain object. Use the Permissions view of a domain object to assign and edit permissions on the object for multiple users or groups.
- Manage permissions by user or group. Use the Manage Permissions dialog box to assign and edit permissions on domain objects for a specific user or group.

Note: You configure permissions on an operating system profile differently than you configure permissions on other domain objects.

Permissions by Domain Object

Use the **Permissions** view of a domain object to assign, view, and edit permissions on the domain object for multiple users or groups.

Assigning Permissions on a Domain Object

When you assign permissions on a domain object, you grant users and groups access to the object.

1. On the Manage tab, select the **Services and Nodes** view.
2. In the Navigator, select the domain object.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Click **Actions** > **Assign Permission**.

   The **Assign Permissions** dialog box displays all users or groups that do not have permission on the object.

6. Enter the filter conditions to search for users and groups, and click the **Filter** button.
7. Select a user or group, and click **Next**.
8. Select **Allow**, and click **Finish**.

Viewing Permission Details on a Domain Object

When you view permission details, you can view the origin of effective permissions.

1. On the Manage tab, select the **Services and Nodes** view.
2. In the Navigator, select the domain object.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Enter the filter conditions to search for users and groups, and click the **Filter** button.
6. Select a user or group and click **Actions** > **View Permission Details**.

   The **Permission Details** dialog box appears. The dialog box displays direct permissions assigned to the user or group, direct permissions assigned to parent groups, and permissions inherited from parent objects. In addition, permission details display whether the user or group is assigned the Administrator role which bypasses permission checking.
7. Click **Close**.
8. Or, click **Edit Permissions** to edit direct permissions.

Editing Permissions on a Domain Object

You can edit direct permissions on a domain object for a user or group. You cannot revoke inherited permissions or your own permissions.

Note: If you revoke direct permission on an object, the user or group might still inherit permission from a parent group or object.

1. On the Manage tab, select the **Services and Nodes** view.
2. In the Navigator, select the domain object.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Enter the filter conditions to search for users and groups, and click the **Filter** button.
6. Select a user or group and click **Actions** > **Edit Direct Permissions**.

   The **Edit Direct Permissions** dialog box appears.
7. To assign permission on the object, select **Allow**.
8. To revoke permission on the object, select **Revoke**.

   You can view whether the permission is directly assigned or inherited by clicking **View Permission Details**.
9. Click **OK**.

Permissions by User or Group

Use the **Manage Permissions** dialog box to view, assign, and edit domain object permissions for a specific user or group.

Viewing Permission Details for a User or Group

When you view permission details, you can view the origin of effective permissions.

1. In the header of Infomatica Administrator, click **Manage** > **Permissions**.

   The **Manage Permissions** dialog box appears.
2. Click the **Groups** or **Users** tab.
3. Enter a string to search for users and groups, and click the **Filter** button.
4. Select a user or group.
5. Select a domain object and click the **View Permission Details** button.

   The **Permission Details** dialog box appears. The dialog box displays direct permissions assigned to the user or group, direct permissions assigned to parent groups, and permissions inherited from parent objects. In addition, permission details display whether the user or group is assigned the Administrator role which bypasses permission checking.
6. Click **Close**.
7. Or, click **Edit Permissions** to edit direct permissions.

Assigning and Editing Permissions for a User or Group

When you edit domain object permissions for a user or group, you can assign permissions and edit existing direct permissions. You cannot revoke inherited permissions or your own permissions.

Note: If you revoke direct permission on an object, the user or group might still inherit permission from a parent group or object.

1. In the header of Infomatica Administrator, click **Manage** > **Permissions**.

   The **Manage Permissions** dialog box appears.
2. Click the **Groups** or **Users** tab.
3. Enter a string to search for users and groups and click the **Filter** button.
4. Select a user or group.
5. Select a domain object and click the **Edit Direct Permissions** button.

   The **Edit Direct Permissions** dialog box appears.
6. To assign permission on the object, select **Allow**.
7. To revoke permission on the object, select **Revoke**.

   You can view whether the permission is directly assigned or inherited by clicking **View Permission Details**.
8. Click **OK**.
9. Click **Close**.

Operating System Profile Permissions

Use the **Configure Operating System Profiles** dialog box to assign, view, and edit permissions on operating system profiles.

Assigning Permissions on an Operating System Profile

When you assign permissions on an operating system profile, PowerCenter users can run workflows assigned to the operating system profile.

1. On the **Security** tab, click **Actions** > **Configure Operating System Profiles**.

   The **Configure Operating System Profiles** dialog box appears.
2. Select the operating system profile, and click the **Permissions** tab.
3. Select the **Groups** or **Users** view, and click the **Assign Permission** button.

   The **Assign Permissions** dialog box displays all users or groups that do not have permission on the operating system profile.
4. Enter the filter conditions to search for users and groups, and click the **Filter** button.
5. Select a user or group, and click **Next**.
6. Select **Allow**, and click **Finish**.

Viewing Permission Details on an Operating System Profile

When you view permission details, you can view the origin of effective permissions.

1. On the **Security** tab, click **Actions** > **Configure Operating System Profiles**.

   The **Configure Operating System Profiles** dialog box appears.
2. Select the operating system profile, and click the **Permissions** tab.
3. Select the **Groups** or **Users** view.
4. Enter the filter conditions to search for users and groups, and click the **Filter** button.
5. Select a user or group and click **Actions** > **View Permission Details**.

   The **Permission Details** dialog box appears. The dialog box displays direct permissions assigned to the user or group, direct permissions assigned to parent groups, and permissions inherited from parent objects. In addition, permission details display whether the user or group is assigned the Administrator role which bypasses permission checking.
6. Click **Close**.
7. Or, click **Edit Permissions** to edit direct permissions.

Editing Permissions on an Operating System Profile

You can edit direct permissions on an operating system profile for a user or group. You cannot revoke inherited permissions or your own permissions.

Note: If you revoke direct permission on an object, the user or group might still inherit permission from a parent group or object.

1. On the **Security** tab, click **Actions** > **Configure Operating System Profiles**.

   The **Configure Operating System Profiles** dialog box appears.
2. Select the operating system profile, and click the **Permissions** tab.
3. Select the **Groups** or **Users** view.

4. Enter the filter conditions to search for users and groups, and click the **Filter** button.
5. Select a user or group and click **Actions** > **Edit Direct Permissions**.

   The **Edit Direct Permissions** dialog box appears.
6. To assign permission on the operating system profile, select **Allow**.
7. To revoke permission on the operating system profile, select **Revoke**.

   You can view whether the permission is directly assigned or inherited by clicking **View Permission Details**.
8. Click **OK**.

Connection Permissions

Permissions control the level of access that a user or group has on the connection.

You can configure permissions on a connection in the Analyst tool, Developer tool, or Administrator tool.

Any connection permission that is assigned to a user or group in one tool also applies in other tools. For example, you grant GroupA permission on ConnectionA in the Developer tool. GroupA has permission on ConnectionA in the Analyst tool and Administrator tool also.

Any connection permission that is assigned to a user or group in one tool also applies in other tools. For example, you grant GroupA permission on ConnectionA in the Developer tool. GroupA has permission on ConnectionA in the Administrator tool also.

The following Informatica components use the connection permissions:

- Administrator tool. Enforces read, write, and execute permissions on connections.
- Analyst tool. Enforces read, write, and execute permissions on connections.
- Informatica command line interface. Enforces read, write, and grant permissions on connections.
- Developer tool. Enforces read, write, and execute permissions on connections.
  For SQL data services, the Developer tool does not enforce connection permissions. Instead, it enforces column-level and pass-through security to restrict access to data.
- Data Integration Service. Enforces execute permissions when a user tries to preview data or run a mapping, scorecard, or profile.

Note: You cannot assign permissions on the following connections: profiling warehouse, data object cache database, or Model repository.

Types of Connection Permissions

You can assign different permission types to users to perform the following actions:

Action	Permission Types
View all connection metadata, except passwords, such as connection name, type, description, connection strings, and user names.	Read
Edit all connection metadata, including passwords. Delete the connection. Users with Write permission inherit Read permission.	Write

Action	Permission Types
Access the physical data in the underlying data source defined by the connection. Users can preview data, run a mapping, run a mapping in a workflow Mapping task, run a scorecard, or run a profile that uses the connection.	Execute
Grant and revoke permissions on connections.	Grant

Default Connection Permissions

The domain administrator has all permissions on all connections. The user that creates a connection has read, write, execute, and grant permission on the connection. By default, all users have permission to perform the following actions on connections:

- View basic connection metadata, such as connection name, type, and description.
- Use the connection in mappings in the Developer tool.
- Create profiles in the Analyst tool on objects in the connection.

Assigning Permissions on a Connection

When you assign permissions on a connection, you define the level of access a user or group has to the connection.

1. On the Manage tab, select the **Connections** view.
2. In the Navigator, select the connection.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Click **Actions** > **Assign Permission**.

   The **Assign Permissions** dialog box displays all users or groups that do not have permission on the connection.
6. Enter the filter conditions to search for users and groups, and click the **Filter** button.
7. Select a user or group, and click **Next**.
8. Select **Allow** for each permission type that you want to assign.
9. Click **Finish**.

Viewing Permission Details on a Connection

When you view permission details, you can view the origin of effective permissions.

1. On the Manage tab, select the **Connections** view.
2. In the Navigator, select the connection.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Enter the filter conditions to search for users and groups, and click the **Filter** button.
6. Select a user or group and click **Actions** > **View Permission Details**.

The **View Permission Details** dialog box appears. The dialog box displays direct permissions assigned to the user or group and direct permissions assigned to parent groups. In addition, permission details display whether the user or group is assigned the Administrator role which bypasses the permission check.

7. Click **Close**.
8. Or, click **Edit Permissions** to edit direct permissions.

Editing Permissions on a Connection

You can edit direct permissions on a connection for a user or group. You cannot revoke inherited permissions or your own permissions.

Note: If you revoke direct permission on an object, the user or group might still inherit permission from a parent group or object.

1. On the Manage tab, select the **Connections** view.
2. In the Navigator, select the connection.
3. In the contents panel, select the **Permissions** view.
4. Click the **Groups** or **Users** tab.
5. Enter the filter conditions to search for users and groups, and click the **Filter** button.
6. Select a user or group and click **Actions** > **Edit Direct Permissions**.

   The **Edit Direct Permissions** dialog box appears.
7. Choose to allow or revoke permissions.
   - Select **Allow** to assign a permission.
   - Clear **Allow** to revoke a single permission.
   - Select **Revoke** to revoke all permissions.

   You can view whether the permission is directly assigned or inherited by clicking **View Permission Details**.
8. Click **OK**.

APPENDIX A

Code Pages

This appendix includes the following topics:

- Supported Code Pages for Application Services, 219
- Supported Code Pages for Sources and Targets, 219

Supported Code Pages for Application Services

Informatica supports code pages for internationalization. Informatica uses International Components for Unicode (ICU) for its globalization support. For a list of code page aliases in ICU, see http://demo.icu-project.org/icu-bin/convexp.

When you assign an application service code page in the Administrator tool, you select the code page description.

You must use UTF-8 compatible code pages for the domain, Model Repository Service, and for each Data Integration Service process.

Supported Code Pages for Sources and Targets

Informatica supports code pages for internationalization. Informatica uses International Components for Unicode (ICU) for its globalization support. For a list of code page aliases in ICU, see http://demo.icu-project.org/icu-bin/convexp.

When you assign a code page using the *pmrep* CreateConnection command or define a code page in a parameter file, you enter the code page name.

The following table lists the name, description, and ID for supported code pages for sources and targets:

Name	Description	ID
Adobe-Standard-Encoding	Adobe Standard Encoding	10073
BOCU-1	Binary Ordered Compression for Unicode (BOCU-1)	10010
CESU-8	ICompatibility Encoding Scheme for UTF-16 (CESU-8)	10011

Name	Description	ID
cp1006	ISO Urdu	10075
cp1098	PC Farsi	10076
cp1124	ISO Cyrillic Ukraine	10077
cp1125	PC Cyrillic Ukraine	10078
cp1131	PC Cyrillic Belarus	10080
cp1381	PC Chinese GB (S-Ch Data mixed)	10082
cp850	PC Latin1	10036
cp851	PC DOS Greek (without euro)	10037
cp856	PC Hebrew (old)	10040
cp857	PC Latin5 (without euro update)	10041
cp858	PC Latin1 (with euro update)	10042
cp860	PC Portugal	10043
cp861	PC Iceland	10044
cp862	PC Hebrew (without euro update)	10045
cp863	PC Canadian French	10046
cp864	PC Arabic (without euro update)	10047
cp865	PC Nordic	10048
cp866	PC Russian (without euro update)	10049
cp868	PC Urdu	10051
cp869	PC Greek (without euro update)	10052
cp922	IPC Estonian (without euro update)	10056
cp949c	PC Korea - KS	10028
ebcdic-xml-us	EBCDIC US (with euro) - Extension for XML4C(Xerces)	10180
EUC-KR	EUC Korean	10029
GB_2312-80	Simplified Chinese (GB2312-80)	10025
gb18030	GB 18030 MBCS codepage	1392
GB2312	Chinese EUC	10024

Name	Description	ID
HKSCS	Hong Kong Supplementary Character Set	9200
hp-roman8	HP Latin1	10072
HZ-GB-2312	Simplified Chinese (HZ GB2312)	10092
IBM037	IBM EBCDIC US English	2028
IBM-1025	EBCDIC Cyrillic	10127
IBM1026	EBCDIC Turkey	10128
IBM1047	IBM EBCDIC US English IBM1047	1047
IBM-1047-s390	EBCDIC IBM-1047 for S/390 (lf and nl swapped)	10167
IBM-1097	EBCDIC Farsi	10129
IBM-1112	EBCDIC Baltic	10130
IBM-1122	EBCDIC Estonia	10131
IBM-1123	EBCDIC Cyrillic Ukraine	10132
IBM-1129	ISO Vietnamese	10079
IBM-1130	EBCDIC Vietnamese	10133
IBM-1132	EBCDIC Lao	10134
IBM-1133	ISO Lao	10081
IBM-1137	EBCDIC Devanagari	10163
IBM-1140	EBCDIC US (with euro update)	10135
IBM-1140-s390	EBCDIC IBM-1140 for S/390 (lf and nl swapped)	10168
IBM-1141	EBCDIC Germany, Austria (with euro update)	10136
IBM-1142	EBCDIC Denmark, Norway (with euro update)	10137
IBM-1142-s390	EBCDIC IBM-1142 for S/390 (lf and nl swapped)	10169
IBM-1143	EBCDIC Finland, Sweden (with euro update)	10138
IBM-1143-s390	EBCDIC IBM-1143 for S/390 (lf and nl swapped)	10170
IBM-1144	EBCDIC Italy (with euro update)	10139
IBM-1144-s390	EBCDIC IBM-1144 for S/390 (lf and nl swapped)	10171
IBM-1145	EBCDIC Spain, Latin America (with euro update)	10140

Name	Description	ID
IBM-1145-s390	EBCDIC IBM-1145 for S/390 (If and nl swapped)	10172
IBM-1146	EBCDIC UK, Ireland (with euro update)	10141
IBM-1146-s390	EBCDIC IBM-1146 for S/390 (If and nl swapped)	10173
IBM-1147	EBCDIC French (with euro update)	10142
IBM-1147-s390	EBCDIC IBM-1147 for S/390 (If and nl swapped)	10174
IBM-1147-s390	EBCDIC IBM-1147 for S/390 (If and nl swapped)	10174
IBM-1148	EBCDIC International Latin1 (with euro update)	10143
IBM-1148-s390	EBCDIC IBM-1148 for S/390 (If and nl swapped)	10175
IBM-1149	EBCDIC Iceland (with euro update)	10144
IBM-1149-s390	IEBCDIC IBM-1149 for S/390 (If and nl swapped)	10176
IBM-1153	EBCDIC Latin2 (with euro update)	10145
IBM-1153-s390	EBCDIC IBM-1153 for S/390 (If and nl swapped)	10177
IBM-1154	EBCDIC Cyrillic Multilingual (with euro update)	10146
IBM-1155	EBCDIC Turkey (with euro update)	10147
IBM-1156	EBCDIC Baltic Multilingual (with euro update)	10148
IBM-1157	EBCDIC Estonia (with euro update)	10149
IBM-1158	EBCDIC Cyrillic Ukraine (with euro update)	10150
IBM1159	IBM EBCDIC Taiwan, Traditional Chinese	11001
IBM-1160	EBCDIC Thai (with euro update)	10151
IBM-1162	Thai (with euro update)	10033
IBM-1164	EBCDIC Vietnamese (with euro update)	10152
IBM-1250	MS Windows Latin2 (without euro update)	10058
IBM-1251	MS Windows Cyrillic (without euro update)	10059
IBM-1255	MS Windows Hebrew (without euro update)	10060
IBM-1256	MS Windows Arabic (without euro update)	10062
IBM-1257	MS Windows Baltic (without euro update)	10064
IBM-1258	MS Windows Vietnamese (without euro update)	10066

Name	Description	ID
IBM-12712	EBCDIC Hebrew (updated with euro and new sheqel, control characters)	10161
IBM-12712-s390	EBCDIC IBM-12712 for S/390 (lf and nl swapped)	10178
IBM-1277	Adobe Latin1 Encoding	10074
IBM13121	IBM EBCDIC Korean Extended CP13121	11002
IBM13124	IBM EBCDIC Simplified Chinese CP13124	11003
IBM-1363	PC Korean KSC MBCS Extended (with \ <-> Won mapping)	10032
IBM-1364	EBCDIC Korean Extended (SBCS IBM-13121 combined with DBCS IBM-4930)	10153
IBM-1371	EBCDIC Taiwan Extended (SBCS IBM-1159 combined with DBCS IBM-9027)	10154
IBM-1373	Taiwan Big-5 (with euro update)	10019
IBM-1375	MS Taiwan Big-5 with HKSCS extensions	10022
IBM-1386	PC Chinese GBK (IBM-1386)	10023
IBM-1388	EBCDIC Chinese GB (S-Ch DBCS-Host Data)	10155
IBM-1390	EBCDIC Japanese Katakana (with euro)	10156
IBM-1399	EBCDIC Japanese Latin-Kanji (with euro)	10157
IBM-16684	EBCDIC Japanese Extended (DBCS IBM-1390 combined with DBCS IBM-1399)	10158
IBM-16804	EBCDIC Arabic (with euro update)	10162
IBM-16804-s390	EBCDIC IBM-16804 for S/390 (lf and nl swapped)	10179
IBM-25546	ISO-2022 encoding for Korean (extension 1)	10089
IBM273	IBM EBCDIC German	2030
IBM277	EBCDIC Denmark, Norway	10115
IBM278	EBCDIC Finland, Sweden	10116
IBM280	IBM EBCDIC Italian	2035
IBM284	EBCDIC Spain, Latin America	10117
IBM285	IBM EBCDIC UK English	2038
IBM290	EBCDIC Japanese Katakana SBCS	10118

Name	Description	ID
IBM297	IBM EBCDIC French	2040
IBM-33722	Japanese EUC (with \ <-> Yen mapping)	10017
IBM367	IBM367	10012
IBM-37-s390	EBCDIC IBM-37 for S/390 (lf and nl swapped)	10166
IBM420	EBCDIC Arabic	10119
IBM424	EBCDIC Hebrew (updated with new sheqel, control characters)	10120
IBM437	PC United States	10035
IBM-4899	EBCDIC Hebrew (with euro)	10159
IBM-4909	ISO Greek (with euro update)	10057
IBM4933	IBM Simplified Chinese CP4933	11004
IBM-4971	EBCDIC Greek (with euro update)	10160
IBM500	IBM EBCDIC International Latin-1	2044
IBM-5050	Japanese EUC (Packed Format)	10018
IBM-5123	EBCDIC Japanese Latin (with euro update)	10164
IBM-5351	MS Windows Hebrew (older version)	10061
IBM-5352	MS Windows Arabic (older version)	10063
IBM-5353	MS Windows Baltic (older version)	10065
IBM-803	EBCDIC Hebrew	10121
IBM833	IBM EBCDIC Korean CP833	833
IBM834	IBM EBCDIC Korean CP834	834
IBM835	IBM Taiwan, Traditional Chinese CP835	11005
IBM836	IBM EBCDIC Simplified Chinese Extended	11006
IBM837	IBM Simplified Chinese CP837	11007
IBM-838	EBCDIC Thai	10122
IBM-8482	EBCDIC Japanese Katakana SBCS (with euro update)	10165
IBM852	PC Latin2 (without euro update)	10038
IBM855	PC Cyrillic (without euro update)	10039

Name	Description	ID
IBM-867	PC Hebrew (with euro update)	10050
IBM870	EBCDIC Latin2	10123
IBM871	EBCDIC Iceland	10124
IBM-874	PC Thai (without euro update)	10034
IBM-875	EBCDIC Greek	10125
IBM-901	PC Baltic (with euro update)	10054
IBM-902	PC Estonian (with euro update)	10055
IBM918	EBCDIC Urdu	10126
IBM930	IBM EBCDIC Japanese	930
IBM933	IBM EBCDIC Korean CP933	933
IBM935	IBM EBCDIC Simplified Chinese	935
IBM937	IBM EBCDIC Traditional Chinese	937
IBM939	IBM EBCDIC Japanese CP939	939
IBM-942	PC Japanese SJIS-78 syntax (IBM-942)	10015
IBM-943	PC Japanese SJIS-90 (IBM-943)	10016
IBM-949	PC Korea - KS (default)	10027
IBM-950	Taiwan Big-5 (without euro update)	10020
IBM-964	EUC Taiwan	10026
IBM-971	EUC Korean (DBCS-only)	10030
IMAP-mailbox-name	IMAP Mailbox Name	10008
is-960	Israeli Standard 960 (7-bit Hebrew encoding)	11000
ISO-2022-CN	ISO-2022 encoding for Chinese	10090
ISO-2022-CN-EXT	ISO-2022 encoding for Chinese (extension 1)	10091
ISO-2022-JP	ISO-2022 encoding for Japanese	10083
ISO-2022-JP-2	ISO-2022 encoding for Japanese (extension 2)	10085
ISO-2022-KR	ISO-2022 encoding for Korean	10088
ISO-8859-10	ISO 8859-10 Latin 6 (Nordic)	13

Name	Description	ID
ISO-8859-13	ISO 8859-13 PC Baltic (without euro update)	10014
ISO-8859-15	ISO 8859-15 Latin 9 (Western European)	201
ISO-8859-2	ISO 8859-2 Eastern European	5
ISO-8859-3	ISO 8859-3 Southeast European	6
ISO-8859-4	ISO 8859-4 Baltic	7
ISO-8859-5	ISO 8859-5 Cyrillic	8
ISO-8859-6	ISO 8859-6 Arabic	9
ISO-8859-7	ISO 8859-7 Greek	10
ISO-8859-8	ISO 8859-8 Hebrew	11
ISO-8859-9	ISO 8859-9 Latin 5 (Turkish)	12
JapanEUC	Japanese Extended UNIX Code (including JIS X 0212)	18
JEF	Japanese EBCDIC Fujitsu	9000
JEF-K	Japanese EBCDIC-Kana Fujitsu	9005
JIPSE	NEC ACOS JIPSE Japanese	9002
JIPSE-K	NEC ACOS JIPSE-Kana Japanese	9007
JIS_Encoding	ISO-2022 encoding for Japanese (extension 1)	10084
JIS_X0201	ISO-2022 encoding for Japanese (JIS_X0201)	10093
JIS7	ISO-2022 encoding for Japanese (extension 3)	10086
JIS8	ISO-2022 encoding for Japanese (extension 4)	10087
JP-EBCDIC	EBCDIC Japanese	9010
JP-EBCDIK	EBCDIK Japanese	9011
KEIS	HITACHI KEIS Japanese	9001
KEIS-K	HITACHI KEIS-Kana Japanese	9006
KOI8-R	IRussian Internet	10053
KSC_5601	PC Korean KSC MBCS Extended (KSC_5601)	10031
Latin1	ISO 8859-1 Western European	4
LMBCS-1	Lotus MBCS encoding for PC Latin1	10103

Name	Description	ID
LMBCS-11	Lotus MBCS encoding for MS-DOS Thai	10110
LMBCS-16	Lotus MBCS encoding for Windows Japanese	10111
LMBCS-17	Lotus MBCS encoding for Windows Korean	10112
LMBCS-18	Lotus MBCS encoding for Windows Chinese (Traditional)	10113
LMBCS-19	Lotus MBCS encoding for Windows Chinese (Simplified)	10114
LMBCS-2	Lotus MBCS encoding for PC DOS Greek	10104
LMBCS-3	Lotus MBCS encoding for Windows Hebrew	10105
LMBCS-4	Lotus MBCS encoding for Windows Arabic	10106
LMBCS-5	Lotus MBCS encoding for Windows Cyrillic	10107
LMBCS-6	Lotus MBCS encoding for PC Latin2	10108
LMBCS-8	Lotus MBCS encoding for Windows Turkish	10109
macintosh	Apple Latin 1	10067
MELCOM	MITSUBISHI MELCOM Japanese	9004
MELCOM-K	MITSUBISHI MELCOM-Kana Japanese	9009
MS1250	MS Windows Latin 2 (Central Europe)	2250
MS1251	MS Windows Cyrillic (Slavic)	2251
MS1252	MS Windows Latin 1 (ANSI), superset of Latin1	2252
MS1253	MS Windows Greek	2253
MS1254	MS Windows Latin 5 (Turkish), superset of ISO 8859-9	2254
MS1255	MS Windows Hebrew	2255
MS1256	MS Windows Arabic	2256
MS1257	MS Windows Baltic Rim	2257
MS1258	MS Windows Vietnamese	2258
MS1361	MS Windows Korean (Johab)	1361
MS874	MS-DOS Thai, superset of TIS 620	874
MS932	MS Windows Japanese, Shift-JIS	2024

Name	Description	ID
MS936	MS Windows Simplified Chinese, superset of GB 2312-80, EUC encoding	936
MS949	MS Windows Korean, superset of KS C 5601-1992	949
MS950	MS Windows Traditional Chinese, superset of Big 5	950
SCSU	Standard Compression Scheme for Unicode (SCSU)	10009
UNISYS	UNISYS Japanese	9003
UNISYS-K	UNISYS-Kana Japanese	9008
US-ASCII	7-bit ASCII	1
UTF-16_OppositeEndian	UTF-16 encoding of Unicode (Opposite Platform Endian)	10004
UTF-16_PlatformEndian	UTF-16 encoding of Unicode (Platform Endian)	10003
UTF-16BE	UTF-16 encoding of Unicode (Big Endian)	1200
UTF-16LE	UTF-16 encoding of Unicode (Lower Endian)	1201
UTF-32_OppositeEndian	UTF-32 encoding of Unicode (Opposite Platform Endian)	10006
UTF-32_PlatformEndian	UTF-32 encoding of Unicode (Platform Endian)	10005
UTF-32BE	UTF-32 encoding of Unicode (Big Endian)	10001
UTF-32LE	UTF-32 encoding of Unicode (Lower Endian)	10002
UTF-7	UTF-7 encoding of Unicode	10007
UTF-8	UTF-8 encoding of Unicode	106
windows-57002	Indian Script Code for Information Interchange - Devanagari	10094
windows-57003	Indian Script Code for Information Interchange - Bengali	10095
windows-57004	Indian Script Code for Information Interchange - Tamil	10099
windows-57005	Indian Script Code for Information Interchange - Telugu	10100
windows-57007	Indian Script Code for Information Interchange - Oriya	10098
windows-57008	Indian Script Code for Information Interchange - Kannada	10101
windows-57009	Indian Script Code for Information Interchange - Malayalam	10102
windows-57010	Indian Script Code for Information Interchange - Gujarati	10097
windows-57011	Indian Script Code for Information Interchange - Gurumukhi	10096

Name	Description	ID
x-mac-centraleurroman	Apple Central Europe	10070
x-mac-cyrillic	Apple Cyrillic	10069
x-mac-greek	Apple Greek	10068
x-mac-turkish	Apple Turkish	10071

Restrictions for Code Pages for Sources and Targets

Consider the following restrictions when you assign a source or target code page:

- Select IBM EBCDIC as your source database connection code page only if you access EBCDIC data, such as data from a mainframe extract file.
- The following code pages are not supported for ODBC connections:
  - UTF-16 encoding of Unicode (Opposite Platform Endian)
  - UTF-16 encoding of Unicode (Platform Endian)
  - UTF-16 encoding of Unicode (Big Endian)
  - UTF-16 encoding of Unicode (Lower Endian)

APPENDIX B

Informatica Platform Connectivity

This appendix includes the following topics:

- Connectivity Overview, 230
- Domain Connectivity, 231
- Native Connectivity, 233
- ODBC Connectivity, 234
- JDBC Connectivity, 235

Connectivity Overview

The Informatica platform uses the following types of connectivity to communicate among clients, services, and other components in the domain:

TCP/IP network protocol

Application services and the Service Managers in a domain use TCP/IP network protocol to communicate with other nodes and services. The clients also use TCP/IP to communicate with application services. You can configure the host name and port number for TCP/IP communication on a node when you install the Informatica services. You can configure the port numbers used for services on a node during installation or in Informatica Administrator.

Native drivers

The Data Integration Service uses native drivers to communicate with databases. Native drivers are packaged with the database server and client software. Install and configure the native database client software on the machines where the services run.

ODBC

The ODBC drivers are installed with the Informatica services and the Informatica clients. The integration services use ODBC drivers to communicate with databases.

JDBC

The Model Repository Service uses JDBC to connect to the Model repository database. The Reporting Service uses JDBC to connect to the Data Analyzer repository and data sources.

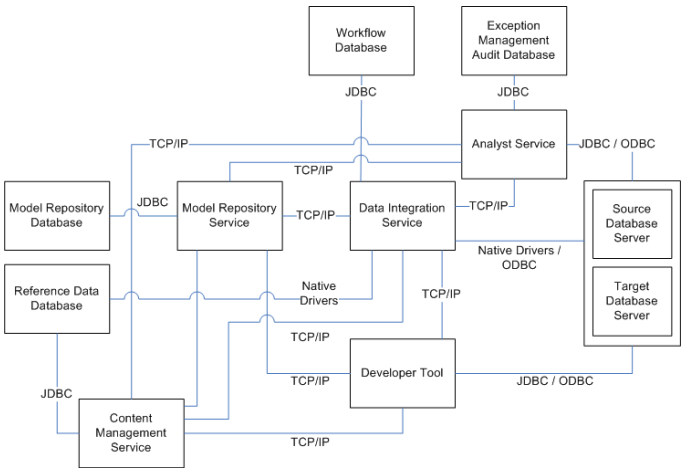
The gateway nodes in the Informatica domain use JDBC to connect to the domain configuration repository.

Domain Connectivity

Services on a node in an Informatica domain use TCP/IP to connect to services on other nodes. Because services can run on multiple nodes in the domain, services rely on the Service Manager to route requests. The Service Manager on the master gateway node handles requests for services and responds with the address of the requested service.

Nodes communicate through TCP/IP on the port you select for a node when you install Informatica Services. When you create a node, you select a port number for the node. The Service Manager listens for incoming TCP/IP connections on that port.

The following figure shows an overview of the connectivity for components in the platform:



The platform uses connection objects to define connectivity information for source and target databases. The connection objects can use native or ODBC connectivity. The Data Integration Service uses connection objects to connect to sources and targets.

The services and clients connect in the following ways:

Model Repository Service

The Model Repository Service uses JDBC to read or write data and metadata in the Model repository. It uses TCP/IP to communicate with the Data Integration Service and the clients.

Data Integration Service

The Data Integration Service uses ODBC or native drivers to connect and read data from a source database and write data to a target database. It uses TCP/IP to communicate with the Model Repository Service, Content Management Service, and client applications.

Informatica Developer

The Developer tool uses TCP/IP to send data transformation requests to the Data Integration Service. It uses TCP/IP to communicate with the Content Management Service to manage reference tables, probabilistic model files, and to retrieve configuration and status information for identity population files and address validation reference data files. When you preview mappings or data objects in the Developer tool, it uses JDBC or ODBC drivers to connect to the source or target database to fetch the metadata required for preview.

Informatica Analyst

The Analyst Service uses TCP/IP to send requests to the Data Integration Service. It uses TCP/IP to communicate with the Content Management Service to manage reference tables. When an Analyst tool user previews profiles or objects, the Analyst Service fetches the metadata that the preview requires

from the source or target database. The Analyst Service uses JDBC or ODBC drivers to connect to the source or target database.

If you use ODBC to connect to the source or target database, install the ODBC driver on the node where the Analyst Service runs.

The Analyst Service can also connect to an exception management audit database. The exception management audit database is a centralized audit trail for the work that Analyst tool users perform on Human task instances. The Analyst Service uses JDBC drivers to connect to the exception management audit database.

Content Management Service

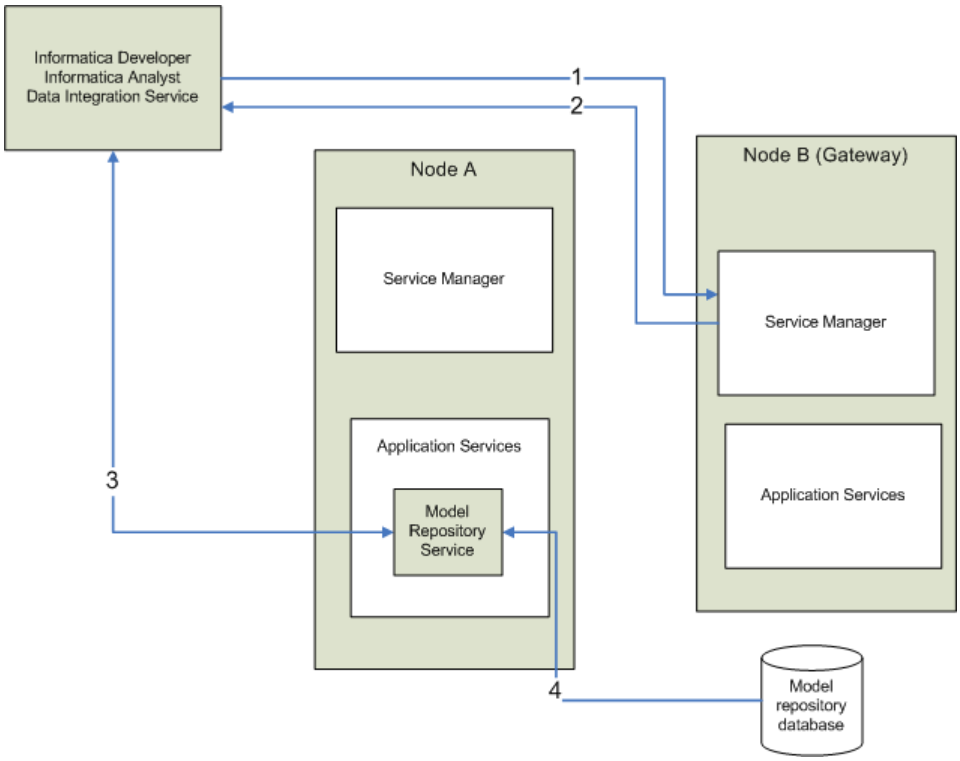
The Content Management Service manages the locations and other properties for reference data. The Content Management Service uses TCP/IP to communicate with the Data Integration Service to read and write data in reference tables. The Content Management Service uses JDBC to communicate directly with the reference data warehouse when it creates reference tables.

If multiple instances of a Content Management Service exist in an Informatica domain, the master Content Management Service updates the Data Integration Service. The master Content Management Service uses TCP/IP to communicate with the Domain Service to identify the Model Repository Service and the Data Integration Service to use.

Model Repository Connectivity

The Model Repository Service connects to the Model repository using JDBC drivers. Informatica Developer, Informatica Analyst, Informatica Administrator, and the Data Integration Service communicate with the Model Repository Service over TCP/IP. Informatica Developer, Informatica Analyst, and Data Integration Service are Model repository clients.

The following figure shows how a Model repository client connects to the Model repository database:



1. A Model repository client sends a repository connection request to the master gateway node, which is the entry point to the domain.
2. The Service Manager sends back the host name and port number of the node running the Model Repository Service. In the diagram, the Model Repository Service is running on node A.
3. The repository client establishes a TCP/IP connection with the Model Repository Service process on node A.
4. The Model Repository Service process communicates with the Model repository database over JDBC. The Model Repository Service process stores objects in or retrieves objects from the Model repository database based on requests from the Model repository client.

Note: The Model repository tables have an open architecture. Although you can view the repository tables, never manually edit them through other utilities. Informatica is not responsible for corrupted data that is caused by customer alteration of the repository tables or data within those tables.

Native Connectivity

To establish native connectivity between an application service and a database, you must install the database client software on the machine where the service runs.

The following table describes the syntax for the native connection string for each supported database system:

Database	Connect String Syntax	Example
IBM DB2	*dbname*	mydatabase
Microsoft SQL Server	*servername@dbname*	sqlserver@mydatabase
Oracle	*dbname*.world (same as TNSNAMES entry)	oracle.world
Sybase ASE	*servername@dbname*	sambrown@mydatabase **Note:** Sybase ASE servername is the name of the Adaptive Server from the interfaces file.
Teradata	*ODBC_data_source_name* or *ODBC_data_source_name@db_name* or *ODBC_data_source_name@db_user_name*	TeradataODBC TeradataODBC@mydatabase TeradataODBC@sambrown **Note:** Use Teradata ODBC drivers to connect to source and target databases.

ODBC Connectivity

Open Database Connectivity (ODBC) provides a common way to communicate with different database systems.

To use ODBC connectivity, you must install the following components on the machine hosting the Informatica service or client tool:

- **Database client software.** Install the client software for the database system. This installs the client libraries needed to connect to the database.

  Note: Some ODBC drivers contain wire protocols and do not require the database client software.

- **ODBC drivers.** The DataDirect closed 32-bit or 64-bit ODBC drivers are installed when you install the Informatica services. The DataDirect closed 32-bit ODBC drivers are installed when you install the Informatica clients. The database server can also include an ODBC driver.

After you install the necessary components you must configure an ODBC data source for each database that you want to connect to. A data source contains information that you need to locate and access the database, such as database name, user name, and database password. On Windows, you use the ODBC Data Source Administrator to create a data source name. On UNIX, you add data source entries to the odbc.ini file found in the system $ODBCHOME directory.

When you create an ODBC data source, you must also specify the driver that the ODBC driver manager sends database calls to.

The following table shows the recommended ODBC drivers to use with each database:

Database	ODBC Driver	Requires Database Client Software
Informix	DataDirect Informix Wire Protocol	No
Microsoft Access	Microsoft Access driver	No
Microsoft Excel	Microsoft Excel driver	No
Microsoft SQL Server	DataDirect SQL Server Wire Protocol	No
Netezza	Netezza SQL	Yes
Teradata	Teradata ODBC driver	Yes
SAP HANA	SAP HANA ODBC driver	Yes

JDBC Connectivity

JDBC (Java Database Connectivity) is a Java API that provides connectivity to relational databases. Java-based applications can use JDBC drivers to connect to databases.

The following services and clients use JDBC to connect to databases:

JDBC drivers are installed with the Informatica services and the Informatica clients.

APPENDIX C

Configure the Web Browser

This appendix includes the following topic:

- Configure the Web Browser, 236

Configure the Web Browser

You can run the Administrator tool in the Microsoft Internet Explorer, Microsoft Edge, Google Chrome, or Safari web browser.

To use the Administrator tool, configure the following options in the browser:

Scripting and ActiveX

Enable the following controls on Microsoft Internet Explorer:

- Active scripting
- Allow programmatic clipboard access
- Run ActiveX controls and plug-ins
- Script ActiveX controls marked safe for scripting

To configure the controls, click **Tools** > **Internet options** > **Security** > **Custom level**.

TLS 1.0

If you configure HTTPS for Informatica Administrator on a domain that runs on AIX, Internet Explorer requires TLS 1.0. To enable TLS 1.0, click **Tools** > **Internet Options** > **Advanced**. The TLS 1.0 setting is listed below the Security heading.

Trusted sites

In Microsoft Internet Explorer, Microsoft Edge, and Google Chrome, add the URL of the Informatica web application to the list of trusted sites. In Safari, add the certificate of the Informatica web application to the keychain. If you are using Chrome version 86.0.42x or later, you must also set the `AuthServerWhitelist` and `AuthNegotiateDelegateWhitelist` policies.

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